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Design of an Atmospheric Boundary Layer Wind Tunnel

Rinka van Dommelen
Master Thesis

MASTER OF SCIENCE OF BUILDING PHYSICS AND SERVICES
AT THE DEPARTMENT OF BUILT ENVIRONMENT
OF THE EINDHOVEN UNIVERSITY OF TECHNOLOGY.

Thesis committee:
Prof. dr. ir. B.J.E. Blocken
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Eindhoven University of Technology
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Eindhoven, The Netherlands
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Preface

This thesis is the final of three research studies for my Master Building Physics and Services. The research is about the design of a wind tunnel and an instrument I developed to optimize such designs.

This thesis is the final work of a very extended Master which I started in 2008. Due to an accident in the same year I lost the ability to read. This affected the progress of my studies dramatically. Thanks to the great support of especially my parents, my beloved sister Milon, and Patrick I was able to keep, slowly, making progress in my studies. The TU/e also contributed greatly in support; the supervisors over the years, as well as TU/e making it financially possible to finish my study.

For the work of this final thesis I would like to thank my daily supervisor Christof Gromke for investing a lot of energy in me as well as the subject of wind tunnel design. He kept me motivated and sharp on subjects as various physical principles which we discussed regularly. I also would like to thank my other supervisors Pierre Gousseau, who especially helped me to get familiar with the subject of wind tunnels, Jan Diepens who helped me to understand the workings of a fan, and of course my professor Bert Blocken who was always helpful with additional feedback even in busy times. Also Marton Balczó, from Budapest University of Technology and Economics, was very enthusiastic in sharing his wind tunnel knowledge. Many people took the effort of reading and commenting on (parts of) my thesis, including Erik Migschofsky, Steven van Egeraat, Geert Jansen, and Ineke Wijnheijmer, thank you!

I am glad to be about to finish my Masters, and thankful to all the friends who kept me going through the past years. Especially my partner Patrick who gave me all the support I could have wished for!

Rinka van Dommelen
Den Haag, 6 April, 2013

Abstract

The group *Urban Physics and Environmental Fluid Mechanics* of the TU/e studies air flows using computational models. These studies need to be validated in an atmospheric boundary layer (ABL) wind tunnel. Currently none of the Dutch universities have an ABL wind tunnel. The TU/e plans to build an ABL wind tunnel to facilitate this research. The aim of the research is to provide a cost efficient design for this new wind tunnel, potentially reusing parts of the previous wind tunnels.

For design of a wind tunnel, the fan power, the test section dimensions, and the test section velocity are important parameters. To optimize these parameters the Wind tunnel Design Instrument (WinDI) has been developed. WinDI calculates the total pressure loss for one or more elements, and can be used to optimize a wind tunnel for minimum pressure loss or maximum test section dimensions and velocity. The WinDI is usable for different types of wind tunnels and other duct systems such as ventilation systems.

Five design options are proposed which have been optimized by WinDI. Three designs use a combination of new parts and existing parts from former wind tunnels, two use only new parts. The combination of the existing closed wind tunnel parts with the existing centrifugal fan can fulfill the requirements with a test section cross section of $1\text{ m} \times 1.5\text{ m}$ and a velocity of 10 m/s . The designs with all new parts have a test section cross section of $1.5\text{ m} \times 2\text{ m}$ and velocity of 20 m/s . Both an open circuit and closed circuit configuration have been optimized.

In this research the following was found, which has been visualized by WinDI:

- The largest gain in pressure loss is to find in simultaneously optimizing for the settling chamber cross section, corners cross section, and diffuser length between these elements.
- Generally assumed for wind tunnel design is that pressure loss sensitive parts, such as corners and the settling chamber, have to be placed in an as low as possible velocity / large as possible cross section. Using WinDI it was found that these cross sections have an optimum.

- The equation for square diffusers from Barlow et al. (1999) contains an error in the coefficients. The difference in calculated pressure loss can be over 100%. The correct coefficients have been verified by the author and can be found in this thesis.

Samenvatting

(Dutch summary)

De groep Stedebouwfysica van de TU/e doet onderzoek naar stromingen in de bebouwde- en natuurlijke omgeving aan de hand van computermodellen. Deze onderzoeksresultaten dienen gevalideerd te worden in een ABL windtunnel. Momenteel heeft geen van de universiteiten in Nederland een ABL windtunnel, de TU/e heeft plannen zo'n windtunnel te realiseren. Het doel van dit onderzoek is verschillende ontwerpvoorstellen te doen voor een atmosferische grenslaagwindtunnel (ABL windtunnel), geoptimaliseerd op drukverlies.

Voor het ontwerp van een windtunnel zijn de volgende aspecten van belang: het minimaal ventilatorvermogen, de maximale testruimte, en de luchtsnelheid in de testruimte. Voor deze aspecten is het Windtunnel Design Instrument (WinDI) ontwikkeld. WinDI berekent het drukverlies over een of meerdere onderdelen en kan gebruikt worden voor optimalisatie van drukverlies voor verschillende soorten windtunnels en andere leidingsystemen.

Met gebruik van WinDI zijn vijf ontwerpen gemaakt voor ABL windtunnels. Drie ontwerpen hergebruiken onderdelen van bestaande windtunnels, twee gebruiken uitsluitend nieuwe onderdelen. Het hergebruik van de onderdelen van een gesloten circuit windtunnel in combinatie met het hergebruik van een centrifugale ventilator voldoet aan de onderzoekseisen met een testruimte van $1\text{ m} \times 1.5\text{ m}$ en een testluchtsnelheid van 10 m/s . De ontwerpen met uitsluitend nieuwe onderdelen hebben een testruimte van $1.5\text{ m} \times 2\text{ m}$ en een testluchtsnelheid van 20 m/s . Zowel een open- als een gesloten circuit windtunnel zijn voor deze testruimte geoptimaliseerd.

Tijdens het onderzoek zijn de volgende feiten aan het licht gekomen, die met behulp van WinDI inzichtelijk zijn gemaakt:

- Algemeen aangenomen voor het ontwerp van windtunnels is dat drukverlies gevoelige onderdelen, zoals een 'corner' en een 'settling chamber', in een zo groot mogelijke doorsnede geplaatst dienen te worden om het drukverlies te beperken. Echter met de optimalisatie van WinDI is gevonden dat deze doorsneden een optimum hebben.

- Het gezamenlijk optimaliseren van de 'corner' doorsnede, de 'settling chamber' doorsnede, en de 'diffuser' lengte daartussen, heeft het meeste invloed op het totale drukverlies.
- De drukverliesberekening voor 'diffusers' met een rechthoekige doorsnede van Barlow et al. (1999) bevat een fout in de coëfficiënten. Hierdoor kan het drukverlies in een diffuser meer dan twee maal zo groot uitvallen. Dit verslag bevat de aangepaste coëfficiënten.

List of Symbols

α	angle between the center-line and wall ($^{\circ}$)
β	porosity, open area ratio ($-$)
$\dot{m}$	mass rate (kg/s)
μ	efficiency ($-$)
ρ	density (kg/m^3)
ν	kinematic viscosity (m^2/s)
A	cross section (m^2)
Ar	area ratio ($-$)
c	chord length (m)
D	diameter (m)
D_{mesh}	mesh hart distance (m)
D_{wire}	wire diameter (m)
f	friction factor ($-$)
g	gravitational acceleration (m/s^2)
H	height (m)
h	gap size between chords (m)
K	pressure loss coefficient ($-$)
L	length (m)
P	power (W)

p	pressure (Pa)
p_{stat}	static pressure at reference level (Pa)
q	dynamic pressure (Pa)
Re	Reynolds number ($-$)
RF	reserve factor ($-$)
z	height above reference level (m)
v	velocity (m/s)

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Chapter 1

Introduction

Wind tunnels are build for aerodynamic studies to understand flow behaviour around objects. They consist of a combination of ducts and a fan to provide air flow in the test section. This thesis focuses on the type of an Atmospheric Boundary Layer (ABL) wind tunnel. An ABL wind tunnel simulates the airflow as it flows over the earth's surface and is used for urban and environmental experiments.

The first flow experiments arose around 1700 using small fans with a test object in front of it (Pope and Harper, 1966). The fans gradually expanded to wind tunnels by adding more parts, such as a closed test sections and flow straighteners, to the design. In the 20th century, they got into the shapes as they are known nowadays. Alongside wind tunnels, computers started to gain popularity in the 1970's and 80's. It was expected that computer simulations would soon replace the wind tunnel experiments (Barlow et al., 1999, Moonen et al., 2006a). However, up to this date the physics of turbulent flows is not yet fully understood. Therefore computational data are simplifications of the reality and wind tunnel studies are needed to validate models.

The design of a wind tunnel is in general based on empirical relations for the pressure loss equations. The equations are manually combined to represent a wind tunnel, making it difficult to see the influence of a changing parameter on the total pressure loss of a wind tunnel. This research focuses on the combination of pressure loss equations, resulting in the wind tunnel design instrument (WinDI). The core principle of WinDI is to minimize the required input parameters and optimize a wind tunnel on minimum pressure loss.

1.1 Problem statement

The group *Urban Physics and Environmental Fluid Mechanics* of the unit of *Building Physics and Services* (BPS) at the *Eindhoven University of Technology* (TU/e)

studies air flows using computational models. These studies need to be validated in an ABL wind tunnel. Currently none of the Dutch universities have a wind tunnel usable for validation of these ABL experiments. Wind tunnel data to compare with the simulated data have to be found elsewhere. This data cannot be checked or edited. The Dutch wind tunnels for building studies are the property of commercial organizations and their data are not always available for research. Rent of these tunnels is very expensive (around €5000/day), and long experiments (if any) are unaffordable. It would be a great advantage if the researches can validate their specific models in a wind tunnel at the TU/e.

For the design of a wind tunnel the pressure loss needs to be calculated. The pressure loss of a part combines parameters and all the parts need to be manually combined to represent a wind tunnel. The influence of a changing parameter on the total pressure loss is difficult to find out. The risk of manual errors is also present.

1.2 Motivation

An ABL wind tunnel for the unit BPS will be a huge advantage for all who work in flow research. Different study cases can then be validated in at the TU/e. Also an ABL wind tunnel at the TU/e can serve in research of (urban) aerodynamics, wind loads, and dynamics of structures. And the wind tunnel can be used for understanding flow dynamics, by visualizing flows using smoke. It will, apart from the research advantages, give the unit BPS more body and might interest more students for research in this field.

To keep construction cost low, parts from previous wind tunnels can be reused. One is the closed circuit wind tunnel “David” from the Department of Applied Physics with an axial fan, the other is an open circuit ABL wind tunnel from the Katholieke Universiteit Leuven with an centrifugal fan. The aim of the research is to provide a cost efficiently design of a new wind tunnel potentially re-using parts of the previous wind tunnels. To design a wind tunnel the influence of a changing parameter on the pressure loss need to be made clear. Therefore a minimum of input parameters is desired. It would also be useful if the equations do not need to be made manually. This is the motive for WinDI.

1.3 Research questions

For the cost efficient design of an ABL wind tunnel the following research questions have been formulated:

1. What are the various parts in wind tunnels?

- (a) What are their properties?
 - (b) What are their functions?
 - (c) What causes the pressure loss of each part?
 - (d) How can the pressure loss of each part be calculated?
2. What are the required parameters to calculate the fan power of a wind tunnel and how can this be calculated from a minimum number of input parameters?
 3. How can a wind tunnel design be optimized for minimum pressure loss?
 4. What are the design requirements for the ABL wind tunnel?
 - (a) What is the ideal velocity (range)?
 - (b) What are the required dimensions?
 - (c) What are the cost limitations?
 5. How can parts of the previous wind tunnels be transformed into an ABL wind tunnel?
 - (a) What are the various options and which parts are being reused?
 - (b) What are the compromises to be made?
 6. What are the life cycle cost indications for the ABL wind tunnel, and which cost are most demanding?
 7. What is the most efficient design for the new ABL wind tunnel?

1.4 Thesis structure

This thesis consist of two parts. The first part is concerned with general wind tunnel information and the second part with the development of a Wind tunnel Design Instrument (WinDI) and the design optimization for a wind tunnel. The first chapter, chapter 2 covers some elementary physical principals underlying the wind tunnel characteristics. The wind tunnel characteristics of the parts are presented in chapter 3, including a short wind tunnel introduction in paragraph 3.2.1. The second part of the thesis is about the design optimization for a new wind tunnel. In the chapter 4 the pressure losses in a wind tunnel (section 4.1) and the development of WinDI (section 4.2) will be explained. WinDI is used in chapter 5 to evaluate design options and make optimizations. Also requirements and a cost indication can found in this chapter. Conclusions and answers to the research questions are given in chapter 6.

Chapter 2

Elementary Physical principles

The air in a wind tunnel flows through different ducts that change in size and shape. Here the flow encounters different elementary physical principles, the most important are mass continuity and Bernoulli's law. In this chapter these physical principles in relation to wind tunnels are explained. More details can be found in (Bertin and Smith, 1998, Çengel and Boles, 2006, Kundu and Cohen, 2008, Munson et al., 2009, Nakayama and Boucher, 1998).

2.1 Pressure

Everything in the earth's atmosphere experiences pressure. This pressure consists of static pressure, 1.013 bar under standard conditions at sea level, and dynamic pressure. An air parcel flowing on a streamline always experiences the static pressure, which is omni directional, and the dynamic pressure depending on the flow direction. The total pressure is the sum of the static and dynamic pressure, see equation 2.1. The static pressure is a measure of the potential energy, the dynamic pressure a measure of the kinetic energy. For certain conditions will the total pressure be constant, see Bernoulli's law, equation 2.6.

$$p_{tot} = p_{stat} + q \quad (2.1)$$

where	p_{tot}	= total pressure	(Pa)
	p_{stat}	= static pressure	(Pa)
	q	= dynamic pressure	(Pa)

In a wind tunnel, the total pressure along the duct decreases in flow direction. The fan increases the total pressure (static and dynamic) with the amount lost through the wind tunnel (Hemphill, 2012). To have the same static pressure in the test section as the local static atmospheric pressure, an opening a so called

breather can be integrated in the test section. In this way the test section will be in direct contact with the local atmospheric pressure. As a result the pressure in the test section is equal to the local atmospheric pressure. The parts upstream of the test section are subjected to a greater than atmospheric pressure, the parts downstream the test section to less than atmospheric pressure.

The pressure can be given in either absolute pressure (relative to vacuum) or gauge pressure (relative to the local atmospheric pressure). In this thesis the pressure refers to the gauge pressure.

2.2 Mass continuity

In a steady state flow, with no change in time, the mass of fluid that enters a duct comes out. The cross section of a duct can be increased and decreased, the mass is never lost. This is the basic physical principle called mass continuity. The continuity equation states that in a steady state flow the mass flow does not change. For the case of a flow in a duct this can be written as is shown in equation 2.2.

$$\dot{m}_{in} = \dot{m}_{out} \quad (2.2a)$$

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2 \quad (2.2b)$$

where $\dot{m}$ = flow rate (kg/s)
 ρ = density (kg/m^3)
 A = cross section (m^2)
 v = velocity (m/s)

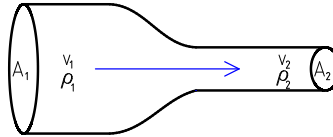


Figure 2.1: Continuity in a duct with changing cross section. A larger area and lower velocity at the inlet (1), and smaller area and higher velocity at the outlet (2).

For subsonic wind tunnels (Mach number lower than 0.3), as in this study, the flow can be seen as a steady state and incompressible ($\rho_1 = \rho_2$) (Munson et al., 2009). Therefore the continuity equation can be simplified to equation 2.3.

$$A_1 V_1 = A_2 V_2 \quad (2.3)$$

For a higher velocity in the test section the cross section is reduced, vice versa when a low velocity is preferred. The simplified mass continuity is the basis for the design of a wind tunnel.

2.3 Bernoulli's law

Bernoulli's law states for an incompressible and inviscid fluid, along a streamline, that the total pressure is constant (see equation 2.4).

$$p_{stat} + \rho g z + \frac{1}{2} \rho v^2 = constant \quad (2.4)$$

where	ρ	= density	(kg/m^3)
	p_{stat}	= static pressure at reference level	(kg/m^3)
	g	= gravitational acceleration	(m/s^2)
	z	= height above reference level	(m)
	v	= velocity	(m/s)

Bernoulli's law is valid under the conditions:

- along a streamline
- viscosity is negligible
- steady conditions
- incompressible fluid

The last two conditions are generally true for wind tunnels. In the main stream (outside the boundary layer) viscosity is also negligible. Therefore outside the boundary layer and on a streamline the Bernoulli equation can be used. In wind tunnels for environmental flow studies the difference in height, and therefore the gravitation term, is negligible. When the elevation is constant, the static and dynamic pressure are interchangeable; where the static pressure increases the velocity decreases and where the static pressure decreases the velocity increases, see figure 2.2. Locally Bernoulli's law can be simplified to:

$$p_{stat} + \frac{1}{2} \rho v^2 = constant \quad (2.5)$$

or

$$p_{stat} + q = constant \quad (2.6)$$

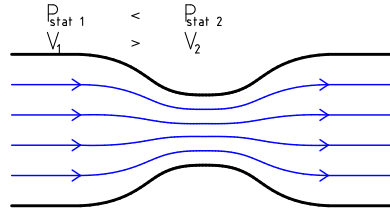


Figure 2.2: Bernoulli's law on a streamline for changing duct cross section.

Bernoulli's law is used to calculate the velocity in a flow when the static and total pressure are known. Both can be measured with a Pitot-static-tube. The Pitot-static-tube is a measurement device which measures the static and total pressure (see figure 2.3). The static pressure is measured at an opening parallel to the flow (in white), the total pressure is measured in the flow direction at the stagnation point. With the total and static pressure the dynamic pressure can be calculated via Bernoulli's law. This way the velocity can be obtained. The velocity is especially of interest for the test section condition.

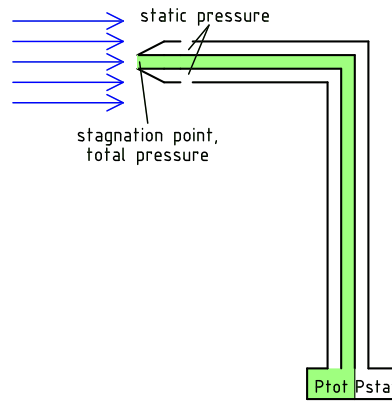


Figure 2.3: Pitot-static-tube. The static pressure is measured at an opening parallel to the flow (in white), the total pressure is measured in flow direction at the stagnation point (green).

2.4 Similarity

To have experiments which outcomes can be transferred to full-scale, the model in the wind tunnel and the outdoor reality aim to have a geometric and dynamic similarity. Geometrically similar objects have the same shape, i.e. a constant scaling factor for all principle directions (see figure 2.4). A dynamically similar flow around objects behaves in the same way. For most wind tunnel experiments of

urban and environmental flows, dynamic similarity only depends on the Reynolds number.

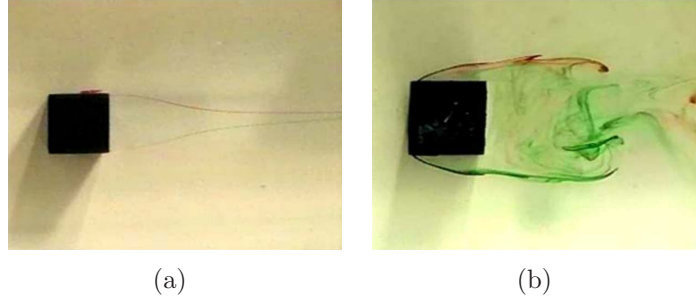


Figure 2.4: The objects are geometrically similar, they have the same shape. Their flow is dynamically different due to the different flow behaviour around the object. (©Stanford University 2007)

Osborn Reynolds (who named the Reynolds number) discovered that the kind of flow in a pipe depends on the flow characteristics: the velocity of the flow, the characteristic diameter, and the viscosity. The Reynolds number is the ratio of kinetic energy to viscous energy. The Reynolds number is also used to describe the flow around an object.

$$Re = \frac{vD}{\nu} \quad (2.7)$$

where	Re	= Reynolds number	(—)
	v	= velocity of the flow	(m/s)
	D	= characteristic diameter	(m)
	ν	= kinematic viscosity ($\nu = 1.51 \times 10^{-5}$ m/s for air at 20°C)	(m ² /s)

The larger the velocity and / or the larger the object, the larger the Reynolds number (Re). For low Reynolds numbers (< 1000) the flow is laminar, at large Reynolds numbers the flow is fully turbulent ($> 10^5$), they depend however on the geometry of the object. In general the larger the Reynolds number the more turbulent the flow. For experiments in ABL wind tunnels a scaled model is placed in the wind tunnel test section. The size of the model in the wind tunnel and the object in reality differs. The air characteristics (velocity and density) are equal to the full-scale situation ($Air_{windtunnel} = Air_{real}$). For example a building of 5 m wide in a calm flow of 0.3 m/s has a Reynolds number of 100.000. To have the same Reynolds number in the test section with an object scale of 1:500, the velocity needs to be 500 times as larger, in this case 150 m/s which is unrealistic.

As a result the Reynolds number of the model and the reality cannot be similar ($D_{real} > D_{wt} \rightarrow Re_{real} > Re_{wt}$).

For a dynamically similar flow the model and the prototype have to be in the same turbulence regimes. A flow around an object with a sharp edges becomes turbulent if the Reynolds number is above the critical value of 10.000 (Snyder, 1981). Above this critical value the flow behaviour around sharp edged objects is comparable. However at round edges the flow behaviour depends stronger on the Reynolds number. Here the critical value for dynamic similarity is higher ($Re_{criticalround} > 10^6$) (Kundu and Cohen, 2008). To be able to compare rounded objects in a wind tunnel the difference in Reynolds number can be compensated for by adding roughness to the models' surface and thereby increasing the turbulence in the flow near the surface.

To be above the critical values for sharp edges a cube in a wind tunnel flow of 20 m/s (wind force 8) should be larger than 8 mm. In a velocity of 5 m/s (wind force 4) the cubes must be larger than 3.2 cm.

2.5 Boundary layer and flow separation

The boundary layer influences the flow characteristics, i.a. flow separation. Flow separation disturbs the continuity of the flow. A boundary layer (with thickness δ) is the area where the flow experiences friction from the solid surface over which it flows, see figure 2.5. Here the velocity changes from zero at the objects' surface to the velocity in the free stream. The boundary layer increases along the length of the surface.

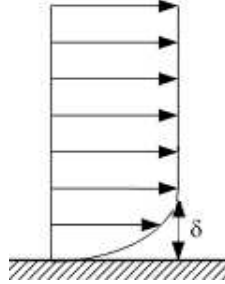


Figure 2.5: Boundary layer (δ) along a plate (Kundu and Cohen, 2008).

When the velocity in the boundary layer is slowed down too much compared to the free stream, the flow streamlines separate and create vortices, see figure 2.6. Separation causes a non continuous flow through a wind tunnel, the velocity will fluctuate with time (Chong et al., 2009). This needs to be avoided. For wind tunnels the risk of separation occurs when the wall surfaces make a corner, and at increasing and decreasing cross sections (Johl et al., 2004).

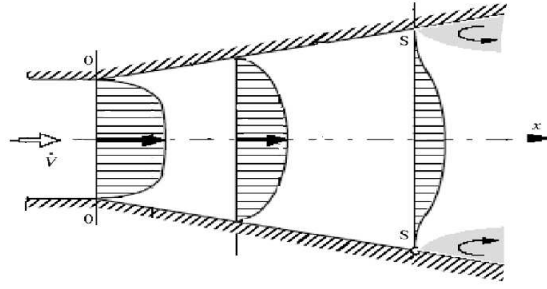
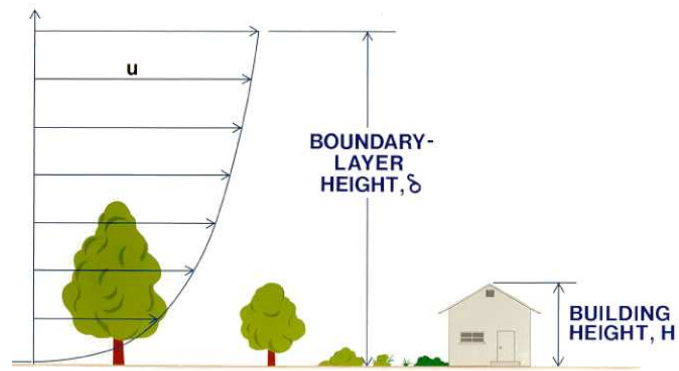
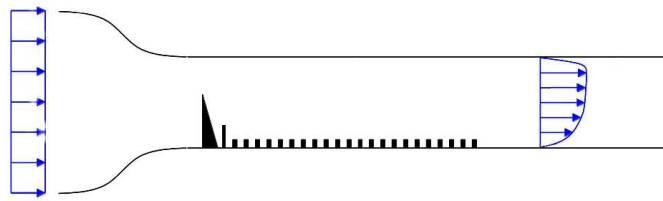


Figure 2.6: Separation in a diffuser. The flow in the boundary layer is slowed, at point S the flow streamlines separate (Kundu and Cohen, 2008).

The velocity over the earth's surface changes from zero at the ground, to free stream at around 500 m to 1000 m (see figure 2.7(a)). All buildings are situated in the earth's atmospheric boundary layer. In a wind tunnel for urban and environmental flows this atmospheric boundary layer is artificially created along a fetch, upstream of the test section, see chapter 3.2.9 and figure 2.7(b).



(a)



(b)

Figure 2.7: Generation of the atmospheric boundary layer. (a) Boundary layer of the flow over the earth (University of California, 2013), (b) generation of an atmospheric boundary layer flow profile in the fetch of a wind tunnel.

Chapter 3

Wind tunnel characteristics

This report focuses on the type of an Atmospheric Boundary Layer (ABL) wind tunnel. The main variations for these tunnels are open circuit and closed circuit wind tunnels which is explained in the first section. The workings of wind tunnel and the elements it comprises are explained in the second section.

3.1 Open versus closed circuit wind tunnels

In a closed circuit wind tunnel the air circulates through a closed system of connected parts. In an open wind tunnel, the inlet and exit are not directly coupled, though they might indirectly be coupled via possible air circulation in the room.

Closed circuit wind tunnels consist of more elements than an open circuit wind tunnel. It consists of parts that return the air and make the total dimension of a closed circuit larger. The recirculated air is less disturbed and has lower turbulence intensities. Also particles for visualization will not pollute the room. However, the main advantage of a closed circuit is that not all kinetic energy is lost at the end of the circuit. Therefore, in general, less power is needed to run a closed circuit wind tunnel. Resulting in a cheaper fan with lower operation cost (Basak et al., 2011).

Open circuit wind tunnels comprise fewer elements and have therefore lower construction costs. However, air that exits an open wind tunnel means that all the kinetic energy is lost. Though with careful design the power consumption of an open wind tunnel can be competitive to that of a closed circuit wind tunnel (Eckert et al., 1976). In open wind tunnels the returning air needs to be taken into account. The room where the air is returning can be considered an inefficient part of the wind tunnel and should at least not influence the inlet flow. An option is to create an inefficient design for the returning flow, rather than not have any control at all. This can be done by a separate section or room to guide the air

through.

Two types of open wind tunnel can be distinguished: blow down and sucking operation mode, see figure 3.1. The test section air in a sucking wind tunnel is not disturbed by the fan. A blow down configuration is mostly used for tests where the flow leaving the test section is greatly disturbed (Mehta, 1977). A disturbed flow through the fan influences the continuousness of the produced flow.

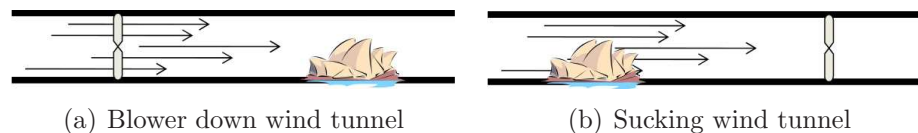


Figure 3.1: A sucking and blow down wind tunnel (Moonen, 2012).

3.2 Various elements in the previous closed circuit wind tunnel David

This chapter gives a summary of the flow through a wind tunnel. It explains each element using the example of the previous wind tunnel 'David' from the Department of Applied Physics.

In 2011 the Department of Applied Physics of the TU/e left the building “N-laag” to be moved at the end of 2014 to the new physics building. The new setting has space only for one of their two previous wind tunnels. The smallest one, David, was moved to vertigo 1.10 for the unit BPS. Wind tunnel David was built in the eighties to produce a flow with a low turbulence degree. David is a closed circuit wind tunnel with test section dimension of 40 cm $\times$ 50 cm (W $\times$ H) with a flow velocity of 15 m/s.

3.2.1 How a closed circuit wind tunnel works

Imagine walking through the tunnel David which is presented in figure 3.2. In the fan (1) the air velocity is generated, here in anti-clockwise direction through the wind tunnel. The fan, as the only part of the wind tunnel, has a circular cross section. The transition section (2a) changes the wind tunnel shape from circular to rectangular. From there the cross section is increased in a diffuser (3a) to have a lower velocity before the flow reaches the first corner (4a). Two turns of 90 degrees follow. To avoid rotation of the flow, the corner contains turning vanes. The quarter circle metal sheets guide the flow “layer by layer” around the corner. After the 180 degree turn the flow needs to get ready for the experiments. For the

Department of Applied Physics this was a flow with a rather high velocity and low turbulence degree ($< 0.2\%$).

To create this straight flow, straightening elements with a large resistance are used. Flow resistance is proportional to the square of the velocity, therefore they are located in a low velocity area. This large area is reached by a wide angle diffuser (3b). In this largest cross section of the wind tunnel the settling chamber (5) is located with the straightening elements. After this the flow is speeded up by the contraction (6) before the test section is reached. The test section (7) is a narrow area where the flow rapidly passes over and around the tested object. The air is to return to the fan by a diffuser (3c) and two corners (4c and 4d). The corner elements are connected by a straight part (8). From the last corner the flow passes through a transition section back to the circular fan. In the fan the flow regains the energy it has lost while passing through the tunnel, and it starts all over again, and again.

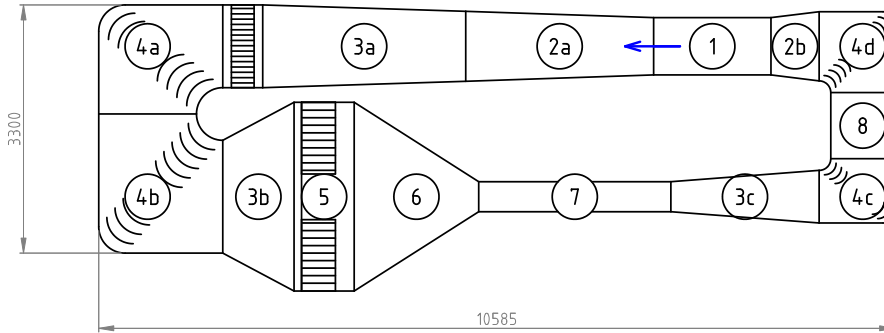


Figure 3.2: Previous wind tunnel David from the Department of Applied Physics (dimensions in mm). The numbers indicate the various parts: (1) fan, (2) transition section, (3) diffuser, (4) corner, (5) settling chamber, (6) contraction, (7) test section, and straight part (8).

In the next paragraphs the working of the various elements is explained in more detail.

3.2.2 Fan

The fan produces the air flow in the wind tunnel. It compensates for the pressure that is lost through the wind tunnel parts. For steady flow wind tunnel conditions, the entire losses experienced by the flow need to be compensated for by the fan. It is the challenge to calculate the fan power that regenerates exactly the pressure which is lost along the elements of the wind tunnel.

For open wind tunnels either an axial or a centrifugal fan is used. A centrifugal fan spins the air outward perpendicular to the inlet (see figure 3.3(b)). It produces

a continuous flow over a large range of velocities and can therefore also be used for experiments with low velocities (Stathopoulos, 1984). The centrifugal fan is mostly used in blow down wind tunnels (Gousseau, 2009a). An axial fan accelerates the flow parallel to the duct (see figure 3.3(a)), and is most often used in closed circuit wind tunnels. The recirculated air is less disturbed by this fan.

The flow exiting an axial fan swirls due to the rotation of the fan. Often flow straightening is included in the fan directly downstream of the rotating blades. Flow straightening is usually done by 4 to 6 blades in flow direction guiding the flow in parallel direction to the centre line.

Fan output is most uniform when operating near its maximum power (Mehta, 1977). Therefore it is important to calculate the fan power as accurate as possible (see chapter 4.1.8). The power scales with the cube of the velocity. Consequently, a twice as high velocity requires a fan (2^3) 8 times more powerful. Overestimation of the fan can lead to a less uniform flow, also a more powerful fan is significantly more expensive.

A fan can have the motor inside or outside the duct. A motor outside the duct does not add the generated heat to the heat gains in the wind tunnel, which is preferable.

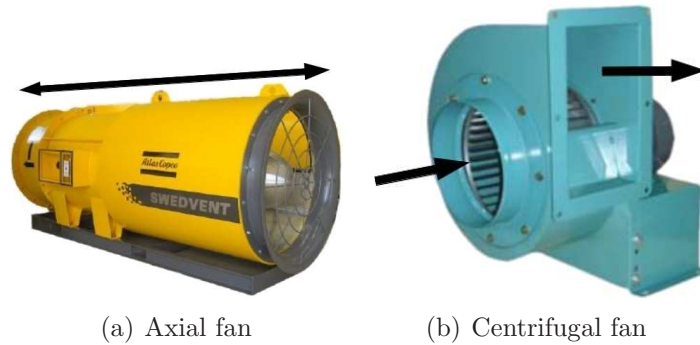


Figure 3.3: The axial fan (a), is most commonly used in wind tunnels and exits air parallel to the fan duct. The centrifugal fan (b) is most commonly used in open circuit blower wind tunnels and exits air perpendicular to the air inlet (Copco, 2013).

A suitable fan for a wind tunnel has to match the system curve, see figure 3.4.

The system curve of a wind tunnel shows the pressure loss through the wind tunnel for increasing (test section) velocity / flow rate. The larger the flow rate through the wind tunnel the larger the pressure loss. The fan curve shows the flow rate it can generate with the pressure loss it has to overcome. A fan without wind tunnel parts, does not have to overcome any extra pressure, and generates the highest flow rate. The larger the pressure loss in the wind tunnel the lower

the flow rate the fan can generate.

A fan with a changeable frequency has more performance curves. When the fan rotates at higher rounds per minute, it generates a higher flow rate. With a lower rotation the fan can overcome less pressure and generates a lower lower rate, see figure 3.4. A fan implemented in a wind tunnel operates at the point where the pressure loss in the wind tunnel equals the pressure rise by the fan. The generates flow rate is the velocity that will be operated in the test section.

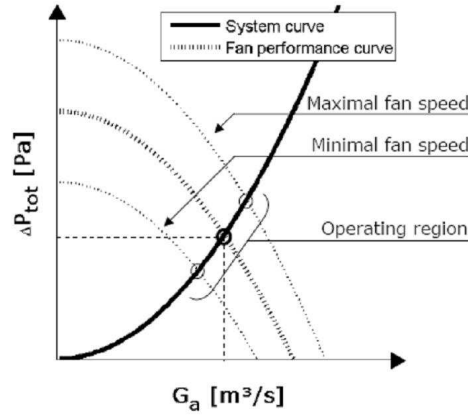


Figure 3.4: The fan performance curve (dotted) and the wind tunnel system curve (black). A frequency changeable fan has a different curve for a different rotation speed. The system curve shows a possible wind tunnel curve. (Moonen et al., 2006b)

3.2.3 Transition section

The transition section is the element placed between two parts with a different cross sectional shape. Often the fan housing has a circular cross section, the majority of the other parts have a rectangular cross section. From aerodynamic point of view a circular cross section is preferred over rectangular. In a circular cross section the streamlines narrow or spread homogeneous over the cross section, with decreasing or increasing the cross section. In a rectangular cross section separation is more likely to occur due to secondary flows. The merging of streamlines is not homogeneously over the cross section (Johl et al., 2004). However, rectangular cross sections are easier to build. Also the risk of separation can be reduced by using corner fillets of 45 degrees, resulting in an (irregular) octagon (Johl et al., 2004, Mehta, 1978, Su, 1991).

Another way of reducing the risk of separation is the efficient use of the transition section. A circular cross section downstream the fan can take better advantage

of the swirls generated by the fan. The transition section can combine the advantage of a circular cross section with the advantage of a low angle diffuser. This combination is used in David's transition part 2.



Figure 3.5: Transition section

3.2.4 Diffuser

A diffuser is used to decrease the velocity by increasing the cross section. The diffuser converts dynamic pressure (kinetic energy) into static pressure (potential energy). This process is reversible.

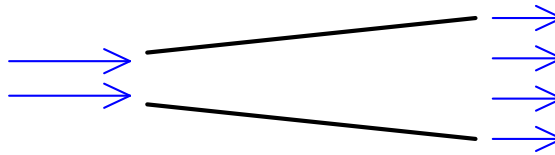


Figure 3.6: Diffuser, increases the cross section and decreases the velocity.

The performance of the diffuser is largely dependent on the flow quality near the wall (Johl et al., 2004). When a diffuser reaches a certain angle the flow near the wall separates (see chapter 2.5). The angle at which the flow separates depends largely on the cross section area (A), the incoming velocity profile (mostly unknown), and the presence of boundary layer control (BLC) devices. Separation increases the pressure loss. The exact angle at which the flow separates is not predictable. The flow character in diffusers is not yet fully understood (Diana et al., 1998, Mehta, 1977, Roberts and Webster, 2002).

A diffuser with a longer length has a smaller angle between the opposite walls. A diffuser angle around 3 degrees gives minimum pressure loss (Daly, 1978), see figure 3.7. Due to the resulting length it is difficult to add a diffuser with an angle of 3 degrees (or less) in wind tunnels (Gousseau, 2009b, Johl et al., 2004).

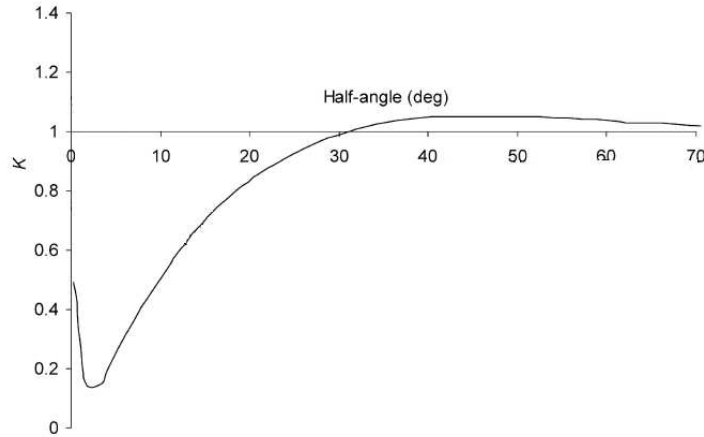


Figure 3.7: Example of the pressure loss coefficient (K), see chapter 4, versus the diffuser angle. A diffuser angle around 3 degrees compensates optimally for the friction along the walls (Singhal et al., 2004).

Wide angle diffuser

A wide angle diffuser (see figure 3.8) is a diffuser in which the cross sectional area increases so rapidly that separation can only be avoided by using boundary layer control (Mehta, 1977, Snyder et al., 1979). Wide angle diffusers are used when cross section increase is more important than pressure loss. In wide angle diffusers boundary layer control (BLC) is used to avoid separation.

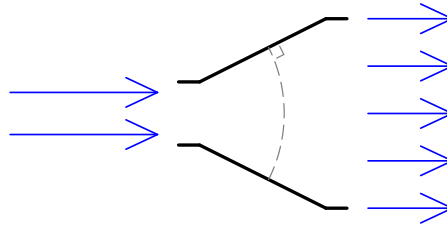


Figure 3.8: Wide angle diffuser with straight walls and a screen for boundary layer control

Boundary layer control can be done in various ways; from splitter plates and screens to suction or blowing through the diffuser walls to. Non-screen boundary layer control has the advantage to be able to cope with solid particles (e.g. for flow visualization) though this is an expensive solution (Mehta, 1977). Screens as boundary layer control are most common and economical. A screen generates turbulence and thereby mix the higher velocities in the free-stream with the lower velocity in the boundary layer. The screen preferable makes an angle of 90 degrees

with the wall (see figure 3.8) to avoid secondary flows (Mehta, 1977). The larger the diffuser angle the more screens are needed to avoid flow separation, see figure 3.9.

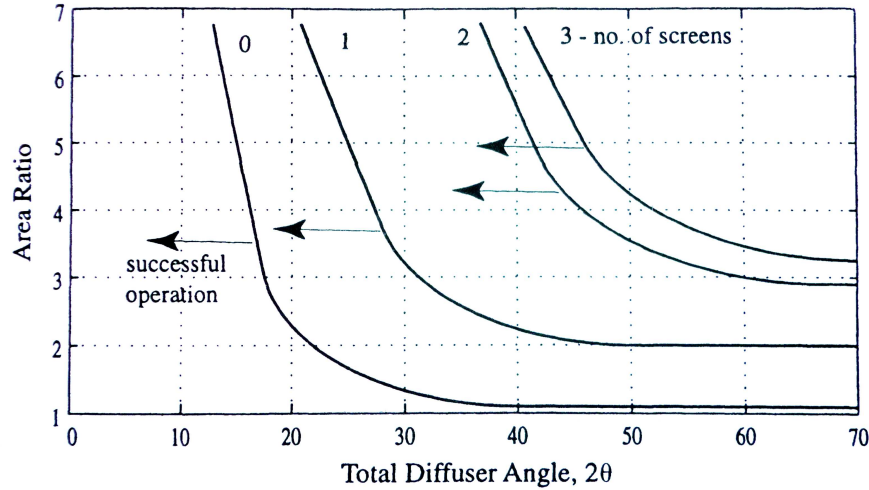


Figure 3.9: The number of screens required to avoid separation in a wide angle diffusers depend on the area ratio and the diffuser angle (Barlow et al., 1999).

Exit diffuser

In an open circuit wind tunnel an exit diffuser is placed at the exit of the wind tunnel. At the exit of an open circuit wind tunnel, the flow exits in vortices and all the kinetic energy is discharged (Vlajinac, 1970). By increasing the exit cross section, it reduces the velocity and thereby the kinetic energy at the exit, see chapter 4.1.1. Flow separation in a diffuser as well as an exit diffuser should be avoided. Separation influences the continuous flow and the pressure loss by separation will be larger than the (static) pressure gain by increasing the cross section.

3.2.5 Straight part

A straight part is a duct in the wind tunnel without extra features. They are used to fit the wind tunnel parts together or to settle the flow before encountering a new element. The latter is used after screens (see section 3.2.7) and often after corners (section 3.2.6). A special straight part is the test section (section 3.2.10) which is by far the longest straight part in an ABL wind tunnel.

3.2.6 Corner

Corners are used to deflect the air flow. A corner without additions results in a rotation of the flow and a non uniform flow at the outlet, see figure 3.10(a). The most common method of deflecting the air flow are guiding vanes, see figure 3.10(b). Other options are splitter plates, which divide the corner in multiple lanes or thick (hollow) vanes which are more aerodynamic see figure 3.11. The latter and does not vibrate with the flow, giving them an advantage over the usual flat turning vanes (Diana et al., 1998). The most practical and economical shape is a quarter of a circle (see figure 3.12) and can be extended by a short straight part at the end, to reduce vortices at the edges (Mehta, 1977).

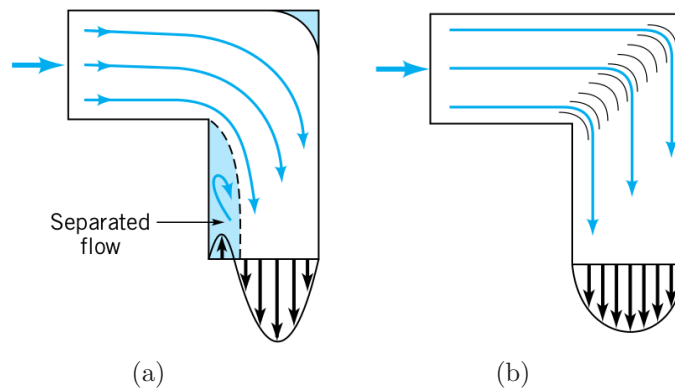


Figure 3.10: Corner without guiding vanes (a), and with guiding vanes (b) (Munson et al., 1998). A corner without guiding vanes induces separation. A corner with guiding vanes turns the flow layer by layer.

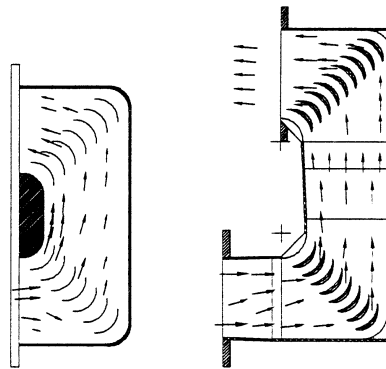


Figure 3.11: Flat quarter circular guiding vanes (left) and hollow vanes (right) (Diana et al., 1998).

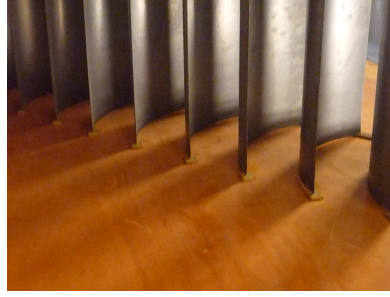


Figure 3.12: Guiding vanes in the corners of wind tunnel David.

3.2.7 Settling Chamber

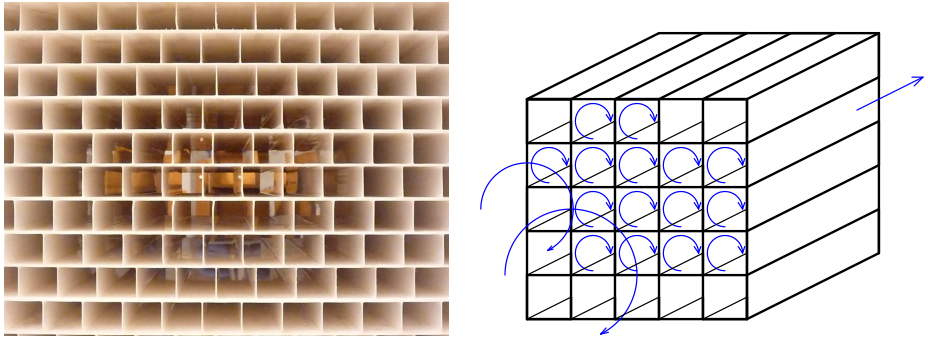
Uniforming the flow is the goal of the settling chamber. The air in the wind tunnel is subjected to rotation of the fan and flow separation along the wind tunnel, while in the test section a mimic of the atmospheric boundary layer is required. Therefore, knowledge is needed of the turbulence structure the models are subjected to. This is obtained in the settling chamber by introducing a honeycomb and one or more screens.

Honeycomb

A honeycomb turns the flow parallel to the center line and reduces fluctuations in the plane perpendicular to the flow, see figure 3.13(b). The honeycomb looks, exactly as the name suggests, like a grid made of hollow tubes (see figure 3.13(a)). The cell sizes of the honeycomb should be smaller than the smallest lateral wavelength of the flow in order to reduce these variations to the same wavelength (Stathopoulos, 1984), see figure 3.13(b). When the air flow reaching the honeycomb is too diverse, with flow angle variations of more than 10 degrees (e.g. after a wide angle diffuser) the flow under an angle “blocks” the tube, reducing the efficiency (Lindgren and Johansson, 2002). If a severe swirl is expected a screen can be placed upstream of the honeycomb to reduce the flow angles and to allow the honeycomb to work properly (Johl et al., 2004, Mehta, 1977).

Screen

A screen homogenizes the velocity profile. The screen uses many small obstacles (wires) to induce a large amount of small vortices with the same size, see figure 3.14. The flow resistance of a wire, scales to the square of the velocity. Therefore the resistance is greater at high velocities than at lower velocities. A screen brings the velocity over the cross section closer to the average, and so homogenizes the velocity profile see figure 3.15. The screen with the largest open area is used



(a) Honeycomb in wind tunnel David. (b) Principle of damping lateral fluctuation in a honeycomb.

Figure 3.13: The honeycomb for flow straightening.

upstream, the finer screen mesh downstream. A fine screen mesh can have an open area as low as 30% (Mehta, 1977, Moonen, 2012). For an ABL wind tunnel, two screens are in general sufficient to straighten the flow (Mehta, 1977).

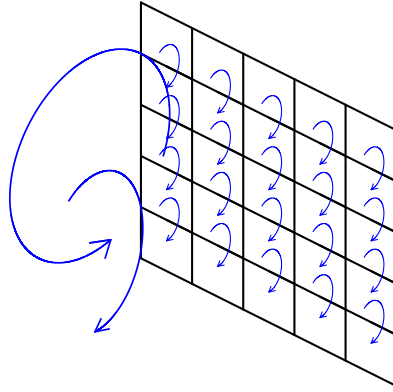


Figure 3.14: Homogenizing the flow through a screens.

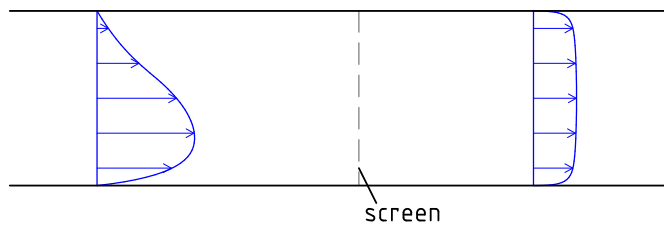


Figure 3.15: Principle of reducing velocity variations through a screen.

The influence of screens on the flow depends on the configuration of the combined parts in a wind tunnel and are not known in advance. After screens have

been placed in a wind tunnel, velocity and turbulence profiles have to be measured, and adjusted accordingly.

3.2.8 Contraction

A contraction is used to increase the velocity by decreasing the cross section. A contraction stretches the flow, thereby reducing the velocity variations in flow direction (Mehta, 1977). A contraction is located upstream the test section to create the high velocity.

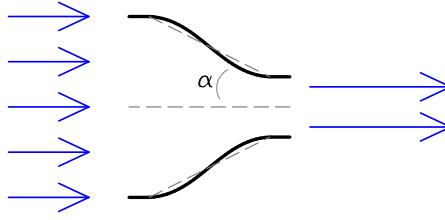


Figure 3.16: Curved contraction, with gradual angle changes. α is the contraction angle.

A contraction can consist of straight or curved walls. Curved walls gradually change the angle of the wall. An increasing angle in the first half of the contraction guides the flow to a smaller cross section. The second half turns the flow back to a parallel flow. The curve smoothly changes parallel to the tube, giving it an S-shape, see figure 3.16. This shape has the largest angle in middle of the contraction. The “S-shape” is less sensitive to flow separation (Johl et al., 2004, Mehta, 1977, Su, 1991). Curved contractions result in a fully attached flow up to 30 degrees, and up till 25 degrees for an inlet contraction (Bellmouth) (Johl et al., 2004, Shope, 2006).

3.2.9 Fetch

The fetch is the area upstream of the test section where the atmospheric boundary layer is created. The quality of environmental wind tunnels depends primarily on the quality of the fetch (Stathopoulos, 1984). The flow enters the fetch as a low-turbulence flow. It then needs to be converted to a scaled version of the full scale atmospheric boundary layer with a logarithmic velocity profile (van Lier, 1997), see figure 3.17.

The creation of the atmospheric boundary layer is done using various obstacles: the spires, fence, and the roughness elements, see figure 3.17. The spires, situated directly up or downstream of the fence introduce large scale turbulence structures with predominantly vorticity in horizontal plane. The fence has the flow



Figure 3.17: Cross section of the contraction and the fetch. The various elements in black represent (from left to right) the spires, the fence, and the roughness elements. The contraction has a parallel flow, the end of the fetch has a logarithmic velocity profile.

rolling over it, creating span wise vortices (see figure 3.18). Finally the roughness elements, situated between the fence, spire and the test section, gradually create small scale turbulence in the lower layer of the flow (Moonen, 2012).

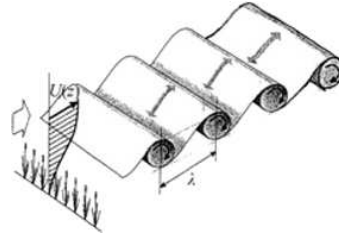


Figure 3.18: Creation of span wise vortices by a laterally stretched object like a fence(McCabe, 2013)

3.2.10 Test section

The actual tests and measurements take place in the test section. The wind tunnel is designed to mimic the atmospheric boundary layer in the test section. The test section with the fetch is the longest straight part in the wind tunnel. In this long part the boundary layer growth along the wall is significant.

The increase in the boundary layer thickness along a straight empty duct reduces the area available for the free-stream, see figure 3.19(a). This smaller cross section downstream results in a higher velocity and a lower static pressure in the free stream, see figure 3.19(b). A lifted ceiling toward the end can compensate for this effect. The increase in cross section ensures that there is no pressure difference over the test section, which is also called a "zero longitudinal pressure gradient".

The influence of a lifted ceiling depends on the configuration, roughness elements and test object, and is not known in advance. After the ceiling has been installed the velocity difference over the length of the fetch with test section has to be measured and the ceiling adapted accordingly.

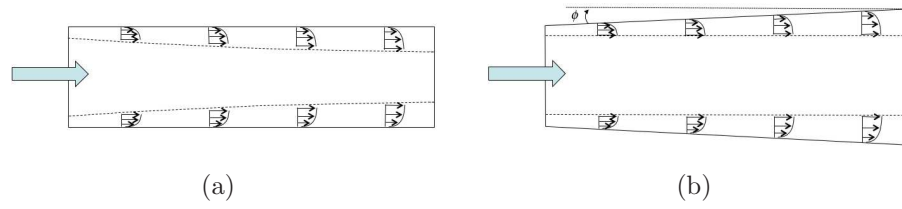


Figure 3.19: Growing boundary layer in a straight parts (Dimitriadis, 2008). (a) Reduced free stream area due to growing boundary layer, (b) constant free stream area due to the lifted ceiling, compensating for the growing boundary layer.

Chapter 4

Development of a Wind tunnel Design Instrument (WinDI)

For the design of a wind tunnel the flow quality, flow rate and fan power are the most important factors. Flow quality depends on the avoidance of flow separation and the accurate mimic of the atmospheric boundary layer (see chapter 2.5). The flow rate is a property of the fan and the pressure loss over the wind tunnel (see section 3.2.2). The fan power compensates for the entire pressure loss experienced by the flow. The pressure loss calculation is a combination of equations, making it difficult to see the influence of a changing parameter on the total pressure loss. The aim of WinDI is to minimize the required input parameters and optimize a wind tunnel for minimum pressure loss.

The first section of this chapter describes the pressure loss that occurs in the different parts of a wind tunnel and the parameters that influence the pressure loss. Furthermore, the choice for the used equation is discussed (see also appendix B). The second section shows the combination of equations to reduce the minimum number of input parameters and the implementation of these equations in WinDI. Section three deals with the validation of the tool. WinDI is validated against seven wind tunnels of which the installed and calculated power are compared.

4.1 Pressure loss in wind tunnels

When the air flows through the various parts of a wind tunnel, as described in the previous chapter, it encounters pressure loss. The properties that influence the pressure loss of each part are accounted for in the pressure loss equation of the part or must fulfill the specific conditions for the equation.

The pressure loss through the wind tunnel must be overcome by the fan. For a stationary flow the pressure loss through the wind tunnel parts (Δp_{wt}) must be

equal to the pressure rise by the fan (Δp_{fan}).

$$\Delta p_{fan} = \Delta p_{wt} \quad (4.1)$$

The total pressure loss in the wind tunnel is equal to the sum of the pressure loss over the wind tunnel parts (see equation 4.2).

$$\Delta p_{wt} = \sum (\Delta p_1 + \Delta p_2 + \Delta p_3 + \dots + \Delta p_{n-1} + \Delta p_n) \quad (4.2)$$

The pressure loss can be calculated by multiplying the pressure loss coefficient (K) with the dynamic pressure (q) (Barlow et al., 1999, Bertin and Smith, 1998, Eckert et al., 1976, Gousseau, 2009a, Johl et al., 2004, Mehta, 1977, Munson et al., 1998, Pope and Harper, 1966, Post, 2011):

$$\Delta p_i = K_i q_i \quad (4.3)$$

$$\begin{aligned} \text{where } K_i &= \text{Pressure loss coefficient of part i} & (-) \\ q_i &= \frac{1}{2} \rho v_i^2, \text{ inlet dynamic pressure of part i} & (Pa) \end{aligned}$$

The pressure loss coefficient is not relevant for the total pressure loss. Therefore, the flow rate needs to be taken into account. To compare the pressure loss among the different parts, the pressure loss coefficient relative to the test section K_{ts} is introduced.

$$K_{ts,i} = K_i \frac{q_i}{q_{ts}} \quad (4.4)$$

For calculation of the pressure loss coefficient the most cited references are Barlow et al. (1999), Eckert et al. (1976), Pope and Harper (1966). They provide the same equations for the straight part, the contraction and the calculation of total fan power. However, the equations for honeycomb, screens, corners, and

	K	q	Δp	K_{ts}
part 1	K_1	q_1	$K_1 q_1$	$K_1 \frac{q_1}{q_{ts}}$
part 2	K_2	q_2	$K_2 q_2$	$K_2 \frac{q_2}{q_{ts}}$
part 3	K_3	q_3	$K_3 q_3$	$K_3 \frac{q_3}{q_{ts}}$
part i	K_i	q_i	$K_i q_i$	$K_i \frac{q_i}{q_{ts}}$
				$\sum (K_{ts}) = \frac{\sum (K_i q_i)}{q_{ts}}$
				$\Delta p = \sum (K_i q_i) \quad \Delta p = \sum (K_{ts}) q_{ts}$

Table 4.1: Overview total pressure loss calculation for a combination of parts.

diffusers are based on empirical relations and therefore differ. Consequently exact values of the pressure loss can not be expected. For more details about the different equations see appendix B.

The cross section shape in a wind tunnel can differ. The equations in this chapter can be used for circular or square cross sections. A rectangular cross section has a higher pressure loss. In a straight part a height-width ratio of 2 has 10% more pressure loss, a height-width ratio of 3 has 70% more pressure loss (Daly, 1978). A height-width ratio of <1.5 is taken as acceptable for relative square cross section.

The diameter used for the equations is the hydraulic diameter. The definition of the hydraulic diameter for rectangular cross sections varies between the references. In Cengel et al. (2012), Daly (1978), and Eckert et al. (1976) the hydraulic diameter for rectangular cross section is $D_h = 2 \frac{HW}{H+B}$. The hydraulic diameter for circle cross section is their general diameter.

4.1.1 Diffuser

The pressure loss that occurs in diffusers contributes to around 30% of the total pressure loss in wind tunnels. An optimum in pressure loss in the diffuser is therefore a large gain in the total pressure loss through a wind tunnel. However, the flow characteristics in diffusers are not yet fully understood, resulting in a variety of equations (Barlow et al., 1999, Mehta, 1977, Munson et al., 1998). The mostly used equation from Barlow et al. (1999) for square diffusers has an error in the coefficients. For the revised equation see equation 4.10.

The pressure loss in a diffuser is due to wall friction and expansion loss; see equation 4.5 (Barlow et al., 1999, Eckert et al., 1976). Expansion loss is the additional pressure loss that occurs in diffusers apart from skin friction loss, it can originate from flow separation. A diffuser angle of more than four degrees results in larger expansion loss than friction loss. At smaller angles than around 3 degrees the skin friction at the wall is relatively larger and contributes predominately to the pressure loss. However, this friction is less than the friction in a straight part, see figure 4.3. Between two to four degrees the diffuser compensates optimal for the friction loss, see figure 4.1, 4.2, and 4.3.

The pressure loss coefficient of a diffuser is calculated according to:

$$K = K_f + K_{ex} \quad (4.5)$$

where:

$$K_f = \left(1 - \frac{1}{Ar^2}\right) \frac{f}{8\sin(\alpha)} \quad (4.6a)$$

$$K_{ex} = K_e \left(\frac{Ar - 1}{Ar}\right)^2 \quad (4.6b)$$

$$(4.6c)$$

The Area ratio is the ratio outlet cross section to the inlet cross section (see equation 4.7). For the calculation of the pressure loss in diffusers the angle is taken to be the equivalent conical angle. This is the angle a circular cross section would have in order to allow the same mass flow rate as the existing rectangular cross section. The equivalent conical angle can be calculated via equation 4.8 (Barlow et al., 1999, Eckert et al., 1976):

$$Ar = \frac{A_{out}}{A_{in}} \quad (4.7)$$

$$\alpha = \tan^{-1} \left(\frac{\sqrt{A_{out}} - \sqrt{A_{in}}}{\frac{L}{\pi}} \right) \quad (4.8)$$

$$f = \left[2 \log_{10} \left(Re \sqrt{f} \right) - 0.8 \right]^{-2} \quad (4.9)$$

For the angles for which the expansion loss (K_{ex}) dominates, the pressure loss is due to the area ratio (Ar) and the diffuser angle (via K_e), see equation 4.5. The larger the angle or the area ratio, the larger the pressure loss. With a fixed area ratio a larger diffuser angle gives a larger pressure loss, up to angles of about 30 degrees. A larger angle results in a shorter diffuser length, therefore the wall at which separation occurs is also shorter. For angle larger than 30 degrees the pressure loss no longer increases if the angle increases. The pressure loss levels out, or even decreases (Rennels and Hudson, 2012), see figure 3.7.

The cross section shape of the diffuser influences the expansion loss, see equation 4.10, 4.11 and appendix B figure B.2. In square diffusers separation starts at a lower angle due to secondary flows in the corners (Rennels and Hudson, 2012). The expansion loss coefficients are different for circular or square cross sections. Rectangular diffusers have even larger expansion loss. Diffusers with a height-width ratio of <1.5 can be counted as square (Daly, 1978)). The equation for rectangle diffusers can be found in (Eckert et al., 1976).

For the calculation of square diffusers the equation from Barlow et al. (1999) contains an error in the coefficients. The correct coefficients have not been pub-

lished, but have been verified by the author of Barlow et al. (1999)¹. The difference in calculated pressure loss between the printed equation in Barlow et al. (1999) and the revised equation can be over 100%. The correct coefficients for a square diffuser can be found in equation 4.10.

Ke for a square cross section:

$$\begin{aligned}
K_e &= 0.09623 - 0.004152 \alpha && \text{for } \alpha < 1.5 \\
K_e &= 0.122156 - 0.0458960 \alpha + 0.02202816 \alpha^2 \\
&\quad - 0.003269152 \alpha^3 - 0.0006144896 \alpha^4 \\
&\quad + 0.00027999008 \alpha^5 - 0.000023373888 \alpha^6 && \text{for } 1.5 \leq \alpha \leq 5 \\
K_e &= -0.1322 + 0.05866 \alpha && \text{for } \alpha > 5
\end{aligned} \tag{4.10}$$

Ke for a circular cross section:

$$\begin{aligned}
K_e &= 0.1033 - 0.02389 \alpha && \text{for } \alpha < 1.5 \\
K_e &= 0.1709 - 0.1170 (\alpha) + 0.03260 \alpha^2 \\
&\quad + 0.001078 \alpha^3 - 0.0009076 \alpha^4 \\
&\quad - 0.00001331 \alpha^5 + 0.00001345 \alpha^6 && \text{for } 1.5 \leq \alpha \leq 5 \\
K_e &= -0.09661 + 0.04672 \alpha && \text{for } \alpha > 5
\end{aligned} \tag{4.11}$$

The equations of diffusers are based on empirical relations, accordingly references give different equations (see appendix B). In graph 4.1 the equations from Barlow et al. (1999) and Eckert et al. (1976) and two measurements (by D.J. Cockrell and A.H. Gibson) are compared. The equation from Barlow differs with the measurements and the equation from Eckert with a peak between 1.5 and 5 degrees. This equation is incorrect. The author has revised this equation (named Barlow update in figure 4.1). The updated coefficients corresponds with the equation from Eckert et al. (1976). Both equations have a pressure loss that keeps increasing with increasing angle, which does not correspond with the measurements. However in practice these equations will not be used for large angles as boundary layer control will be used in wide angle diffusers, which requires a different equation. For calculations of wide angle diffusers without boundary layer control a maximum pressure loss coefficient of $K = 1$ has to be used to comply with the experimental results in diffusers. An example of this equation can be found in plot figure 4.2.

¹In correspondence with the author of Barlow et al. (1999), he confirmed the incorrect coefficients and sent the revised version. The revised equation is printed in this report.

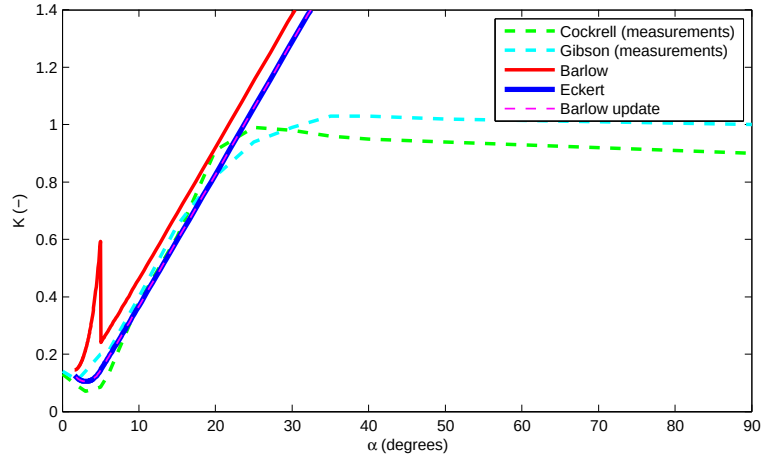


Figure 4.1: Comparison of pressure loss coefficient in rectangular diffusers from references Barlow et al. (1999) and Eckert et al. (1976) and two measurements (Cockrell and Gibson), for an area ratio of 9. The revised equation of Barlow (Barlow update) is also plotted in purple lines and is equal to the equation from Eckert (in blue) for large area ratio's.

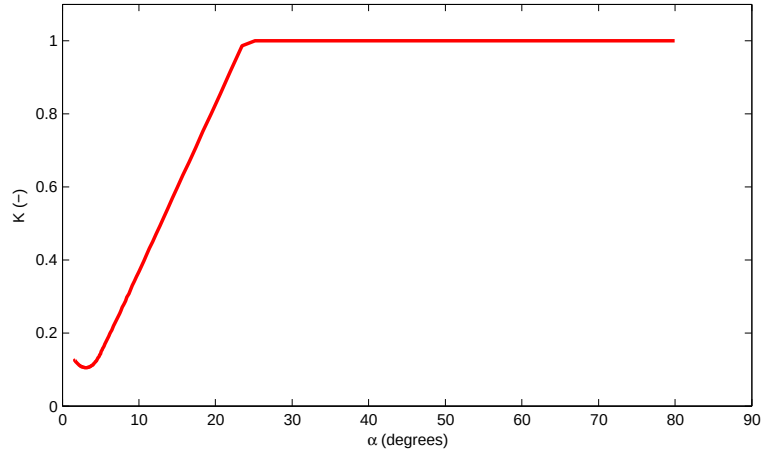


Figure 4.2: Example plot of the pressure loss coefficient versus the diffuser angle according to the equation used in WinDI

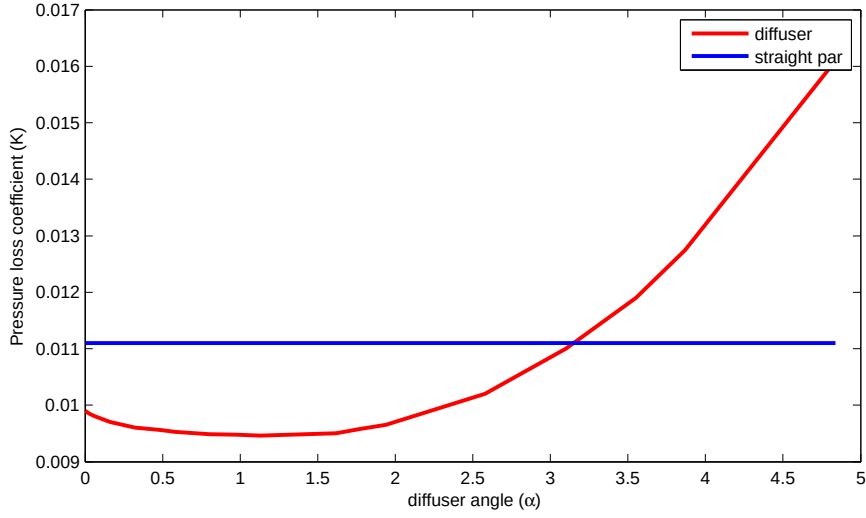


Figure 4.3: pressure loss coefficient of a straight part versus a diffuser

Wide angle diffuser

To avoid separation, boundary layer control (BLC) is often applied at wide angle diffusers. By avoiding separation also the pressure loss is decreased. The equation for a wide angle diffuser with BLC assumes that no separations occurs. Therefore sufficient screens or other BLC devices need to be added. The first separation occurs at the end of the diffuser in the widest part. Therefore the first screen, when screens are used as BLC), needs to be added in the middle of the diffuser to re-energize the slowing boundary layer before it separates (Rennels and Hudson, 2012).

The pressure loss coefficient of a wide angle diffuser with BLC is the sum of the pressure loss coefficient of the BLC diffuser and the BLC devices, e.g. the screens (Rennels and Hudson, 2012).

$$K = K_{screens} + K_{BLC,diffuser} \quad (4.12)$$

For K_{screen} see chapter 4.1.4.

$$K_{BLC,diffuser} = 0.29 + 0.012 (\alpha - 10) \quad (4.13)$$

where $\alpha > 10$ degrees

The number of screens that need to be added depends on the area ratio and the diffuser angle, see figure 3.9.

Sudden expansion

To have minimum pressure loss over a diffuser a combination of a sudden expansion with a small angle diffuser (< 10 degrees) can have a lower pressure loss than a wider angle diffuser. However, separation always occurs in a sudden expansion. Making it only an option in areas where the flow is already disturbed e.g. directly down or upstream the fan. The pressure loss of a sudden expansion is calculated via equation 4.14 (Cengel et al., 2012, Munson et al., 2009).

$$K = 1 - \left(\frac{A_{in}}{A_{out}} \right)^2 \quad (4.14)$$

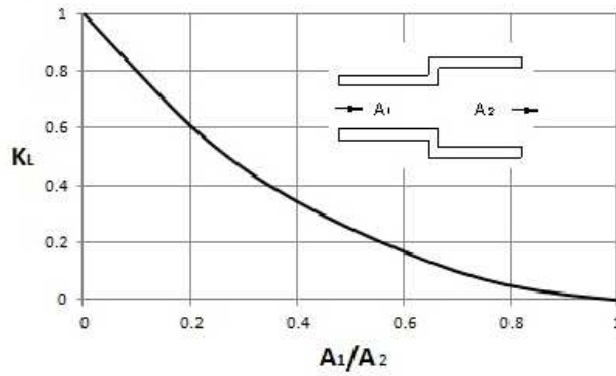


Figure 4.4: Pressure loss coefficient of a sudden expansion (Munson et al., 2009).

Exit diffuser

In an open wind tunnel there is an additional pressure loss when the air exits the wind tunnel. All the kinetic energy of the flow is then discharged. The pressure loss at the exit is equal to the dynamic pressure at the exit. An exit is a special sudden expansion with a $K > 0.99$ (when the exit covers $< 10\%$ of the cross section of the room). In general the exit pressure loss coefficient is taken to be (Munson et al., 2009):

$$K_{exit} = 1 \quad (4.15)$$

The pressure loss of an exit diffuser is the sum of the pressure loss in the diffuser and the pressure loss by exiting the wind tunnel.

4.1.2 Straight part

The pressure loss in an empty straight parts is due to friction. The pressure loss coefficient is calculated via (Barlow et al., 1999, Eckert et al., 1976, Pope and Harper, 1966):

$$K = f \frac{L}{D} \quad (4.16)$$

where f = friction factor $(-)$
 L = length (m)
 D = hydraulic diameter (m)
Surface roughness is assumed to be “smooth”

The equation for the straight part cannot be used for the fetch and the test section as these are not empty, see section 4.1.6.

For straight parts, the longer and/or smaller the hydraulic diameter of a part the larger the friction loss. The friction factor is calculated using the Reynolds number, for which the local velocity and diameter are required as input (see chapter 2.4). Both a larger diameter and a larger velocity result in a larger friction factor. However, in total, the pressure loss coefficients for larger diameters decreases. The pressure loss coefficient for straight parts is less than $K = 0.1$ and therefore only significant at velocities of about the maximum velocity of a wind tunnel. Equation 4.16 is used for the pressure loss calculation.

4.1.3 Corner

Corners, together with diffusers, are responsible for most of the total pressure losses in closed circuit wind tunnels. Corners with guiding vanes contribute to about $\frac{1}{3}$ to the total pressure loss in wind tunnels. A corner without turning vanes has almost all pressure loss due to rotation. The addition of guiding vanes reduces the total pressure loss by about 80%. Even though guiding vanes increase the friction, the reduction of separation reduces the total pressure loss. The pressure loss coefficient in corners with guiding vanes is around $K = 0.2$ (Bloomer, 2000) and can be calculated using (Barlow et al., 1999, Eckert et al., 1976, Pope and Harper, 1966):

$$K = 0.10 + \left(\frac{4.55}{(\log_{10} (Re))^{2.58}} \right) \quad (4.17)$$

where Re = Reynold number, based on the chord length $(-)$
 c = chord length (m)
 h = gap size between chords (m)

Equation 4.17 is valid under the conditions:

The spacing between flat vanes is assumed to be: $0.2 < \frac{h}{c} < 0.3$.

The spacing between hollow vanes is assumed to be: $0.1 < \frac{h}{c} < 0.25$.

The degree of flow turning is assumed to be 90 degrees.

In case of no guiding vanes assume $K = 1$.

For corners directly downstream of the test section a safety margin of 10% is added to compensate for the extra vortices in the flow.

The minimum number of guiding vanes is accounted for in the gap to chord ratio ($\frac{h}{c}$). In the special case of diffusing corners, longer guiding vanes help to reduce the pressure loss. The guiding vanes then divide the diffusing corner into many parts with smaller angles (Eckert et al., 1976).

4.1.4 Settling Chamber

The pressure loss in settling chambers is the sum of the pressure losses of a honeycomb and of one or more screens. To obtain the pressure loss of the settling chamber, several influencing parameters need to be taken into account. The pressure loss in the settling chamber is highly dependent on arrangements and assumptions. However, when placed in a low velocity area, the influence on the total pressure loss of the wind tunnel is minimal ($<10\%$) (Mehta, 1977, Pope and Harper, 1966). However a larger cross section of the settling chamber, requires a larger diffuser and contraction angle and increases their pressure loss, see chapter 5.

Honeycomb

For the calculation of the pressure loss coefficient of the honeycomb the empirical relations by Prandtl (1933) are still used (Barlow et al., 1999, Eckert et al., 1976):

$$K = \lambda \left(\frac{L_{cell}}{D_{cell}} + 3 \right) \left(\frac{1}{\beta} \right)^2 + \left(\frac{1}{\beta} - 1 \right)^2 \quad (4.18)$$

$$\begin{aligned}
\text{where } \lambda &= 0.375 \left(\frac{\Delta}{D_{cell}} \right)^{0.4} Re_{\Delta}^{-0.1} && \text{for } Re_{\Delta} \leq 275 \quad (-) \\
\lambda &= 0.214 \left(\frac{\Delta}{D_{cell}} \right)^{0.4} && \text{for } Re_{\Delta} > 275 \quad (-) \\
L_{cell} &= \text{cell length} && (m) \\
D_{cell} &= \text{cell hydraulic diameter} && (m) \\
\beta &= \text{honeycomb porosity} && (-) \\
Re_{\Delta} &= \frac{v\Delta}{\nu} && (-) \\
\Delta &= \text{roughness} && (-)
\end{aligned}$$

The pressure loss coefficient increases with the length of the honeycomb cells. λ depends on the roughness and cell diameter. A larger roughness gives a higher λ which increases the pressure loss coefficient. A smaller cell diameter also increases the pressure loss coefficient. However, a larger cell length and a smaller cell diameter create a more uniform flow. A good balance between minimum pressure loss and flow straightening is $D_{cell} = \frac{D_{\text{settling chamber}}}{150}$ and a cell length of 8-12 times the cell diameter (Barlow et al., 1999, Snyder et al., 1979).

Screen

The pressure loss in screens is in general larger than in the honeycomb. Screens have in general a smaller open area for the flow (larger mesh density). Equations for screens are also empirical and can be calculated as follows (Barlow et al., 1999, Eckert et al., 1976, Mehta, 1977):

$$K = K_{mesh} K_{re} (1 - \beta) + \left(\frac{1 - \beta}{\beta} \right)^2 \quad (4.19)$$

$$\begin{aligned}
\text{where } K_{re} &= 0.785 \left(\frac{Re_{wire}}{241} + 1.0 \right)^{-4} + 1.01 && (-) \\
\beta &= \text{screen porosity, open area} && (-) \\
Re_{wire} &= \text{Reynolds number based on wire diameter} && (-) \\
K_{mesh} &= \text{wire smoothness coefficient (between 1.0 - 2.1)} && (-)
\end{aligned}$$

A smooth wire decreases the pressure loss. The porosity (open area for the flow) is the most important factor. A screen with a large pressure loss coefficient induces a large resistance upon the flow, resulting in suppressing nearly all variations in velocity and direction of the flow.

4.1.5 Contraction

The pressure loss in contractions depends on their area ratio and their length which together determine the contraction angle. Furthermore, the shape of the

contraction largely influences the pressure loss. In straight walled diffusers separation occurs at smaller angles. Contractions in wind tunnels contribute in general to less than 5% of the total pressure loss.

Pressure losses in curved contractions are due to friction only as long as no separation occurs. The generally used equation for the pressure loss coefficient of a contraction is i.a. stated by Barlow et al. (1999) and Eckert et al. (1976):

$$K = 0.32 f_{av} \frac{L}{D_{out}} \quad (4.20)$$

where f_{av} = friction at the average diameter $(-)$
 L = length of centre line (m)
 D_{out} = outlet diameter (m)
 Wall shape is assumed to be curved

The equation is taken to be valid for curved wall contractions only, see appendix B. The pressure loss in straight wall contractions are higher than in curved contractions, the following pressure loss coefficients are mentioned (Abdullah, 2005, Bloomer, 2000, Daly, 1978, Munson et al., 2009):

$K = 0.07$ for 60 degrees
 $K = 0.045$ for 45 degrees
 $K = 0.02$ for 30 degrees

Based on these values equation 4.21 can be constructed which can be used for straight contractions.

$$K = 0.045 + 0.001667(\alpha - 45) \quad (4.21)$$

4.1.6 Fetch and test section

The pressure loss in the fetch is due to the friction of the roughness elements and the study objects themselves. The pressure loss of a single object due to vortex generation and blockage effect can be calculated. However, a group of objects influencing each others' flow, makes it impossible to calculate the resulting pressure loss (Stathopoulos, 1984). The test section together with the fetch needs to possess a constant static pressure and free-stream velocity for reliable measurements (see chapter 3.2.10). The total pressure loss of the fetch is based on a zero pressure gradient in the free stream in combination with a cross section increase to achieve this.

$$p_{t,in} - p_{t,out} = \Delta p \quad (4.22a)$$

$$p_t = p_{stat} + \frac{1}{2}\rho v^2 \quad (4.22b)$$

A lifted ceiling is used to realize a zero pressure gradient (Gousseau, 2010).

$$p_{stat,in} = p_{stat,out} \quad (4.23a)$$

$$\Delta p_t = \frac{1}{2}\rho (v_{in}^2 - v_{out}^2) \quad (4.23b)$$

$$K = \frac{\Delta p_t}{q_{ts}} = \frac{v_{in}^2}{v_{ts}^2} - \frac{v_{out}^2}{v_{ts}^2} \quad (4.23c)$$

Via continuity of mass (see equation 2.3)

$$K = \left(\frac{H_{ts}}{H_{in}} \right)^2 - \left(\frac{H_{ts}}{H_{out}} \right)^2 \quad (4.24)$$

where H_{ts} = test section design height (m)
 H_{in} = height inlet fetch (m)
 H_{out} = height outlet test section (m)

4.1.7 Additional parts

Wind tunnels can comprise more parts than described before. Most of these parts are mentioned in this section.

The pressure loss coefficients for the following parts are fixed values (Bloomer, 2000, Cengel et al., 2012, Munson et al., 1998, Nakayama and Boucher, 1998, Post, 2011):

Rounded inlet	$K = 0.1$
Straight inlet with $D_{wall} \geq D_{inlet}$, as in a centrifugal inlet	$K = 0.5$
Duct inlet with $D_{wall} < D_{inlet}$, closely followed by a wall	$K = 0.8$
Duct inlet $D_{wall} < D_{inlet}$, as in an axial inlet	$K = 1$
Safety net	$K = 0.2$

Transition parts are calculated as a contraction or diffuser depending on their increase or decrease in cross section.

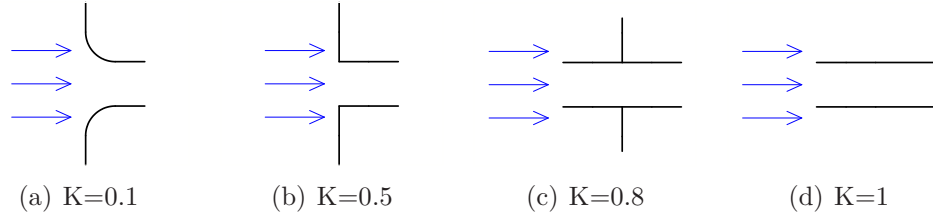


Figure 4.5: Pressure loss coefficient (K) of various inlet types. (a) Rounded inlet, (b) Straight inlet, (c) Duct with wall inlet, (d) Duct inlet.

4.1.8 Fan

The power needed to operate a wind tunnel is calculated by multiplying the total pressure loss with the flow area (Eckert et al., 1976, Johl et al., 2004, Mehta, 1977):

$$P = A_{ts} v_{ts} \sum \Delta p_{wt} \quad (4.25)$$

where	P	= power of the wind tunnel flow	(W)
	Δp_{wt}	= total pressure loss of all wind tunnel parts	(Pa)
	A_{ts}	= cross section of the test section	(m ²)
	v_{ts}	= velocity in the test section	(m/s)

Little knowledge exists on the flow interaction between connecting parts. This interaction is not directly taken into account, but a reserve factor of 10% is taken to include unknown interactions such as flow interaction. Also fan and motor efficiency are taken into account here (Gousseau, 2009a, Johl et al., 2004):

$$P_{fan} = P \frac{RF}{\mu_{tot}} \quad (4.26)$$

where	P_{fan}	= fan power	(W)
	RF	= reserve factor (taken 1.1)	(-)
	μ_{tot}	= $\mu_{fan} \times \mu_{motor}$	(-)
	μ_{fan}	= fan efficiency	(-)
	μ_{motor}	= motor efficiency	(-)

4.2 Development of a Wind tunnel Design Instrument (WinDI)

One of the aims for wind tunnel design is to find the minimum required fan power for given test section dimensions and target velocity. The Wind tunnel Design

Instrument (WinDI) has been developed to find this optimum. WinDI combines the pressure loss equations from chapter 4.1 to calculate the total pressure loss and fan power of a wind tunnel and optimize parameters on minimum pressure loss. The equations in chapter 4.1 are also reduces to fewer basic parameters. WinDI makes it possible to vary many wind tunnel parameters in a time efficient way and reduce the risk of (manual) human errors. Appendix A goes more into detail about the working of WinDI.

WinDI consists of 6 components (see figure 4.6), and works as follows. The *settings* deal with properties that are independent of the wind tunnel (e.g. air density and viscosity). In the *wind tunnel specifications* general wind tunnel information is added. These are the test section velocity and cross section dimensions.

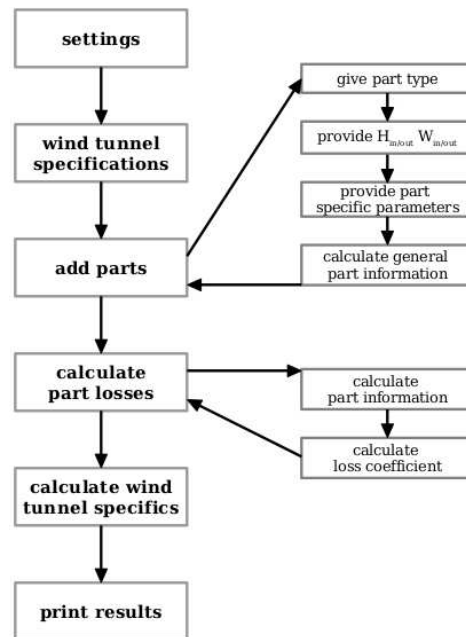


Figure 4.6: The schematic operation of WinDI.

In *add parts* the following type of parts can be combined to represent an open or closed circuit wind tunnel:

- Contraction
- Corner
- Diffuser (with and without boundary layer control)
- Fetch

- Honeycomb
- Inlet
- Outlet
- Safetynet
- Straight part
- Screen
- Sudden expansion
- Test section

All parts require as input the inlet and outlet height and width and, dependent on the part, extra information e.g. length or wire diameter. In *add parts* the general part parameters are also calculated, which are the "cross section in out and average", "hydraulic diameter in out and average" and "velocity in out and average".

After the parts and their parameters are calculated, WinDI calculates the pressure losses per part in *calculate part losses*. The pressure loss calculations of these parts can be found in chapter 4.1. Here the parameters that are needed for the pressure loss coefficient (K) are calculated. These can be the friction factor (f), Reynolds number (Re), angle (α), density (β), etc. For a list of required input and given output per part, see appendix A.2.3. All parameters that have been calculated can be retained for validation (figure 4.8).

In *calculate wind tunnel specifics* the dynamic pressure (q), the pressure loss (Δp), the pressure loss coefficient relative to the test section (K_{ts}), and the percentage it contributes to the total pressure loss (%) are calculated per part. This results in a total pressure loss over the wind tunnel and the required fan power. Finally *print results* combines all information in a table, see figure 4.7. Instructions on how to use WinDI can be found in appendix A.

For the optimization over a combination of elements the input parameters have to be set as a range instead of a fixed value. The results of the optimization are plotted in a graph (in the case of one or two variables), or in an index plot (with more than two variables). Design optimization plots are found in chapter 5.2.

4.3 Validation of WinDI

Before use, WinDI has been validated against existing wind tunnels. The validation of WinDI is based on seven references. Four are calculations of the total

nr	Part name	K	q	delta p	Kts	%
1	Diffuser transition	0.0116	532.3	6.18	0.0161	0.9
2	Diffuser	0.0486	328.2	15.95	0.0417	2.4
3	corner A large	0.1549	87.6	13.57	0.0355	2.0
4	Straight part	0.0038	87.6	0.33	0.0009	0.0
5	corner B large	0.1549	87.6	13.57	0.0355	2.0
6	Straight part	0.0034	87.6	0.29	0.0008	0.0
7	Screen BLC 1	0.7902	87.6	69.25	0.1809	10.5
8	Diffuser_BLC	0.3656	87.6	32.04	0.0837	4.8
9	Screen BLC 2	0.7933	34.6	27.47	0.0718	4.1
10	Screen 3	0.7997	16.5	13.21	0.0345	2.0
11	Honeycomb	2.5456	16.5	42.06	0.1099	6.4
12	Screen 4	0.7997	16.5	13.21	0.0345	2.0
13	Screen 5	0.7997	16.5	13.21	0.0345	2.0
14	Screen 6	0.7997	16.5	13.21	0.0345	2.0
15	Screen 7	0.7997	16.5	13.21	0.0345	2.0
16	Screen 8	0.7997	16.5	13.21	0.0345	2.0
17	Contraction	0.0100	564.7	5.65	0.0148	0.9
18	Test section	0.5148	382.8	197.07	0.5148	29.8
19	2d contraction	0.0100	453.9	4.54	0.0119	0.7
20	corner C small	0.1614	453.9	73.26	0.1914	11.1
21	Straight part	0.0071	453.9	3.23	0.0084	0.5
22	corner D small	0.1614	453.9	73.26	0.1914	11.1
23	contr transition	0.0100	532.3	5.32	0.0139	0.8
Total				662.34	1.7302	100.0
Wind tunnel information:						

efficiency_fan =		0.8				
efficiency_motor =		0.8				
reserve_factor =		1.1				
power_input =		54643				
power_fan =		85380				

Figure 4.7: Example of overall WinDI output *print results*.

wind tunnel power by literature examples, and three are the operational power of existing wind tunnels. The calculations are based on the examples: (1) Barlow et al. (1999), (2) the NASA wind tunnel at Ames Research Center (Eckert et al., 1976), (3) the Indian Institute of Science open wind tunnel at Bangalore Eckert et al. (1976), and (4) a project Atmospheric Boundary Layer Wind Tunnel at the TU/e Gousseau (2009a). WinDI is also compared with the operational power of existing wind tunnels. The existing wind tunnels WinDI is compared with are the Atmospheric Boundary Layer Wind Tunnel EMPA in Zurich Empa (2011), and two wind tunnels currently present at the TU/e (“David” and “Leuven”).

Table 4.2 shows the validation of pressure loss coefficient relative to the test section. The difference of 12 % between the calculations of Gousseau (2009a) and WinDI are purely due to the different used diffuser equation (see appendixD),

name:	'Diffuser'
type:	'diffuser'
crosssection_test_section:	3
velocity_test_section:	25
height_in:	1.8000
width_in:	1.8000
height_out:	2.2000
width_out:	2.8500
diameter_in:	1.8000
diameter_out:	2.4832
diameter_average:	2.1423
crosssection_in:	3.2400
crosssection_out:	6.2700
crosssection_average:	4.5895
velocity_in:	23.1481
velocity_out:	11.9617
velocity_average:	16.3417
cross_section_shape:	'rectangle'
length_center_line:	11.7933
length_wall:	0
radius_in:	1.0155
radius_out:	1.4127
angle_radian:	0.0336
angle_degrees:	1.9278
area_ratio:	1.9352
reynolds_number:	2.3119e+06
friction_factor:	0.0101
loss_friction:	0.0276
loss_e:	0.0899
loss_expansion:	0.0210
loss_coefficient:	0.0486
pressure_loss_dynamic:	328.2000
pressure_drop:	15.9480
loss_coefficient_ratio:	0.0417
loss_percentage:	2.4078

Figure 4.8: Example of the calculated part specific parameters for a diffuser. Highlighted parameters are given as input.

	Values from reference (K_{ts})	Values from WinDI (K_{ts})	deviation (%)
Barlow 1999	0.1834	0.1764	-3.8
Eckert - Nasa Ames	0.1239	0.1272	+2.7
Eckert - Bangalore	0.1464	0.1485	+1.4
Gousseau 2009	1.701	1.505	-12

Table 4.2: Comparison of pressure loss coefficient relative to the test section (K_{ts}) (Barlow et al., 1999, Eckert et al., 1976, Gousseau, 2009a) and the calculated values by WinDI and their deviation.

	Installed power (kW)	Calculated power (kw)	deviation (%)
David	0.605	0.579	-4.1
Leuven	11.5	11.4	-0.9
EMPA	87.0	85.4	-0.6

Table 4.3: Comparison of installed and calculated fan power of two existing wind tunnels at the TU/e (David and Leuven) and the ABL wind tunnel EMPA in Zurich.

WinDI used the revised diffuser equation by Barlow (see 4.1.1 and figure 4.1, while Gousseau (2009a) uses the original diffuser equation by Barlow et al. (1999). Table 4.3 compares the calculated power by WinDI with the installed power in three wind tunnels. For the underlying assumptions and the pressure loss comparison per part see appendix D.

The difference between the calculations of WinDI and the installed power is less than 5%. An overall accuracy of $\pm 10\%$ is seen as acceptable (Daly, 1978). WinDI is therefore assumed to be validated and will be used for the design of an atmospheric boundary layer wind tunnel in the next part of this thesis.

Chapter 5

Design of a new ABL wind tunnel

5.1 Requirements

The following requirements are used for the design of an ABL wind tunnel for the group *Urban Physics and Environmental Fluid Mechanics* at the TU/e.

Location

For an ABL wind tunnel at the TU/e, room VRT1.10 on the first floor of the Vertigo building is available. The room, which has an L-shape (see figure 5.12, 5.19, and 5.25), has a length of 22.6 m and a width of 9.2 m on the smallest part and 14.0 m at the widest part and a height of 5.4 m. The free height is 4.5 m. For maintenance and safety reasons a free space of 2 m on all horizontal sides of the wind tunnel has to be taken into account. This leaves a space of 18.6 m $\times$ 5.2 m $\times$ 4.5 m for the wind tunnel.

Diffuser

For the diffuser, low angles (<10 degrees) are preferred. For angles larger than 10 degrees, boundary layer control (BLC) has to be added to avoid separation. One to three screens are added as BLC (see chapter 3.2.4). A diffuser with a single screen requires the screen to be installed in the middle of the diffuser. For screens that serve as boundary layer control, a lower mesh density than for homogenizing screens suffices, and is set at:

- Screen:

$$D_{wire} = 1 \text{ mm}$$

$$D_{mesh} = 10 \text{ mm}$$

Smoothness coefficient $K_{mesh} = 1.3$

The maximum height width ratio of the cross section for all parts is 1.5 (see chapter 4.1).

Corner

The maximum diameter of the corner is limited to $\frac{1}{2} \times 5.2$ m minus the $\frac{chord}{\sqrt{2}}$. This leaves a maximum width of the corner of around 2.5, and a maximum corner cross section of 9 m². Corner with flat guiding vanes are used for the design calculations.

Settling chamber

The diameter of the settling chamber is limited to 3.5 m width and 4.5 m height (cross section of 15.75 m²). Straightening elements for ABL wind tunnels can be limited to a honeycomb and two screens. The honeycomb and screen dimensions are taken to be (Barlow et al., 1999, Johl et al., 2004, Mehta, 1977, Moonen, 2012, Snyder et al., 1979):

- Honeycomb:

$$D_{cell} = D_{duct}/150$$

$$L_{cell} = D_{cell} \times 8$$

$$\text{Porosity } \beta = 0.80$$

$$\text{Roughness } \Delta = 8 \times 10^{-5}$$

- Screen 1:

$$D_{wire} = 0.5 \text{ mm}$$

$$D_{mesh} = 1.5 \text{ mm}$$

$$\text{Smoothness coefficient } K_{mesh} = 1.3$$

- Screen 2:

$$D_{screen} = 0.5 \text{ mm}$$

$$D_{mesh} = 2 \text{ mm}$$

$$\text{Smoothness coefficient } K_{mesh} = 1.3$$

Spacing between the end of the diffuser and the first screen has to be 0.1 to 0.2 times the settling chamber hydraulic diameter. The same spacing needs to be present between the screens and the contraction. The total length of the settling

chamber is therefore taken to be 0.7 m. To check and adjust the velocity and turbulence profiles, as well as for maintenance purposes the screens have to be placed in frames that can be removed.

Contraction

The contraction reduces velocity differences and the turbulence intensity. In an ABL wind tunnel, extra turbulence is created in the fetch, making it less important to reduce the turbulence in the contraction (Snyder et al., 1979). A contraction ratio of 3 is sufficient. The maximum angle of the contraction is 30 degrees.

Test section

The test section has to be large enough to accommodate an urban model, or a part of it. Here the space for the model is taken between 1 m and 1.5 m. In the test section the walls next to the tested object influence the flow which are not present in reality. The boundary layer growth along the walls grows (see chapter 2.5), this will be around 15 cm along for a 12 m fetch (Engineering Science Data Unit, 2013). To eliminate these side effect of the walls the section has to be at least 30 cm wider than the model placed in it.

Buildings are located in the boundary layer of the atmosphere (see chapter 2.5). Therefore the flow in the wind tunnel has to simulate the lower part of the atmospheric boundary layer. The atmospheric boundary layer has to be scaled with the scale of the models Wang et al. (1996). The use of lego bricks is a good scale for the ABL around 1:200 to 1:600. Larger scales require smaller roughness elements such as a carpet. Smaller scales require larger roughness elements such as egg cubes. The top of the boundary layer needs to be well above the largest building, a height of 1 m is usual for a scaling factor of 1:500 (Kozmar, 2009, Kumar, 2005, Wang et al., 1996). Test section dimensions of ABL wind tunnels are presented in appendix C.

For reliable tests the Reynolds number has to be above the critical value of 10.000 for objects with sharp edges in the test section, see chapter 2.4. Smaller model diameters require higher velocities to have a similar wake (see chapter 2.4).

To keep a constant pressure in the test section, ideally an adjustable or lifted ceiling has to be made. To avoid a lower pressure downstream of the test objects, the blockage ratio of the models should not exceed 5% at most, but is preferable less than 3% (Mehta, 1977, Moonen et al., 2006a). The test section has to be provided with a turntable for measurements at various flow angles. Accessibility of the test section is important, preferable without the use of screwdrivers. To support visual observation, transparent parts and/or windows have to be part of the wall.

The test section is taken to at least 1 m in height, 1.3 m in width, and the fetch length 8 times the height of the test section. For the lifted ceiling 1 cm lift per 1m fetch is taken. The velocity range is preferred between 10 to 20 m/s (wind force 6 to 8).

Open circuit wind tunnel

In case of an open circuit wind tunnel a straight part of 0.5 times the hydraulic inlet diameter is required upstream of the entrance contractor to stabilize the flow. The spacing between the exit of the diffuser and the wall has to be at least 1.5 times the diffuser hydraulic inlet diameter.

5.2 Design variations

Design optimizations are calculated in order to fulfill the requirements. Five options have been designed. The first design is the transformation of the former wind tunnel David from the Applied Physics Department into an ABL wind tunnel (the ‘ABL-David’, see section 5.2.1). The second design uses the existing centrifugal blower fan, from the KU Leuven, in combination with the existing parts from wind tunnel David, see section 5.2.2. The third uses the centrifugal fan in a newly designed wind tunnel, see section 5.2.3. The fourth and fifth options provide the largest test section cross section that fits in the room for respectively a closed and an open circuit wind tunnel, see section 5.2.4 and 5.2.5.

Special attention will be paid to the optimization of combined elements. A larger settling chamber reduces the pressure loss, though consequently the diffusing angle increases resulting in a larger pressure loss in the diffuser. For all wind tunnels the cross sections of the corners and settling chamber as well as the length of the diffusers have been optimized using WinDI.

5.2.1 Transformation of the former wind tunnel David into ABL-David

The former wind tunnel David from the Applied Physics Department (see figure 3.2, and figure E.1) can be transformed into an ABL wind tunnel with the smallest number of new elements. To create the ABL at least a fetch needs to be added to the wind tunnel. While some of the straightening elements, that were present for low turbulence physical experiments, can be removed. The smallest corners in David have a cross section of $0.60 \text{ m} \times 0.70 \text{ m}$. This cross section is also used for the cross section of the test section in the design. The main limit for this design is the fan power, which is 605 W.

Following the requirements, the length of the fetch is taken to be 5 m. With a contraction ratio of 3, the cross section of the settling chamber needs to be at least 1.26 m^2 . Within these boundaries the following have been varied to optimize for minimum fan power and consequently maximum flow rate, see also figure 5.1:

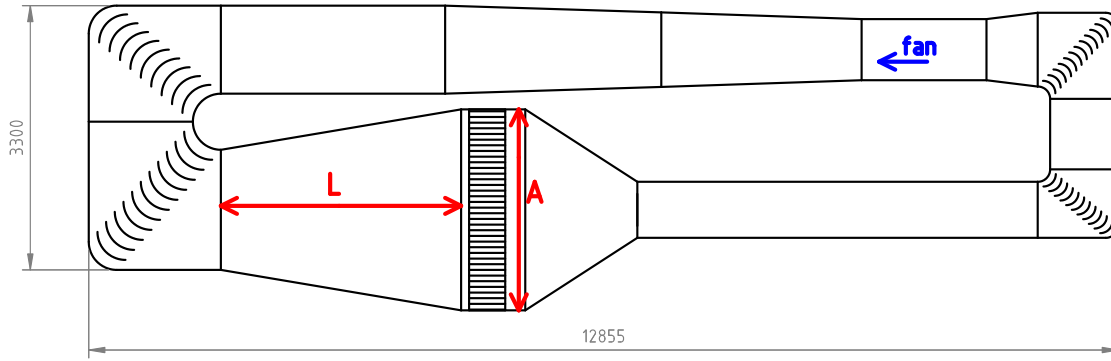


Figure 5.1: Optimization parameters for ABL-David wind tunnel (dimensions in mm): length of the diffuser (L) and cross section of the settling chamber (A).

- Cross section settling chamber (A)
The gain in required fan power with a larger cross section than the existing settling chamber from David is less than 3%.
The used settling chamber cross section is 5 m^2 (reused).
- Diffuser length (L)
For a diffuser with an angle lower than 10 degrees, and resulting length 3 m or larger, the pressure loss decreases by 7%. For lengths larger than 3 m the pressure loss reduces maximum by another 1.5%, see figure 5.2.
The used diffuser length is 3 m.

For design of ABL David with fewest new parts, only a new test section with dimensions of $5 \text{ m} \times 0.6 \text{ m} \times 0.7 \text{ m}$, a new diffuser (with a length of 3 m) and a

straight connecting part need to be constructed. The total pressure loss and the pressure loss per part can be found in table 5.1. The drawing of the ABL-David can be found in figure 5.3, in green are the new parts.

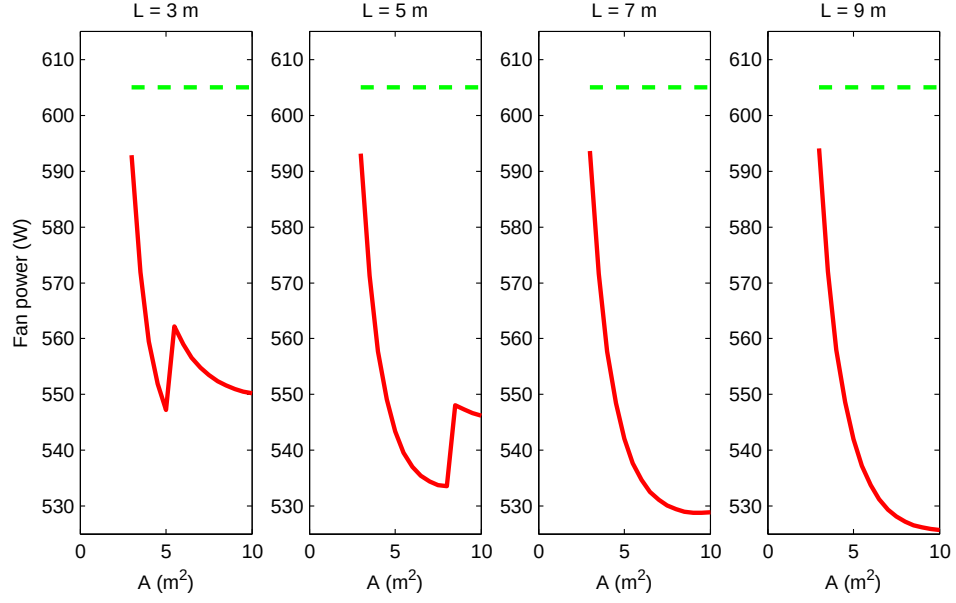


Figure 5.2: In red the required fan power for ABL David wind tunnel for the diffuser length (L), and settling chamber cross section (A). In green the maximum fan power of the fan in wind tunnel David. The jump in the graph is the diffuser angle that exceed 10 degrees, resulting in a larger pressure loss.

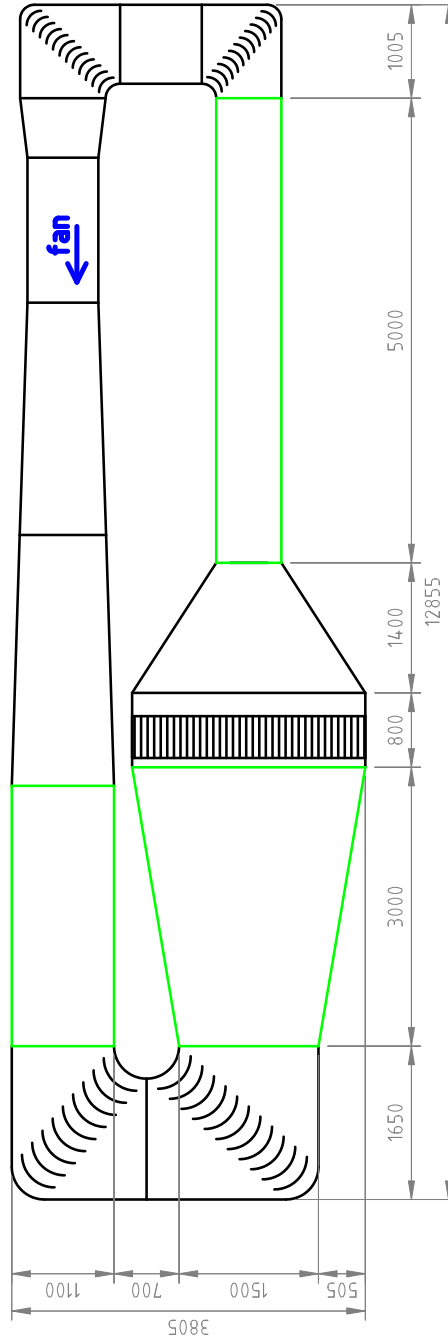


Figure 5.3: Dimensions of the ABL-David wind tunnel (dimensions in mm). Test section dimensions of $5\text{ m} \times 0.6\text{ m} \times 0.7\text{ m}$ and velocity of 12 m/s . The green parts are new.

nr	Part name	K	q	Δp	Kts	%
1	Test section	0.0918	95.7	8.78	0.0918	14.2
2	Corner small 1	0.1704	95.7	16.31	0.1704	26.4
3	Diffuser corner 1	0.0170	95.7	1.62	0.0170	2.6
4	Straight part	0.0166	64.9	1.08	0.0113	1.7
5	Corner small 2	0.1729	64.9	11.22	0.1172	18.1
6	Diffuser corner 2	0.0155	64.9	1.00	0.0105	1.6
7	Safety net	0.2000	55.4	11.08	0.1158	17.9
8	diffuser before fan	0.0098	55.4	0.54	0.0057	0.9
9	Diffuser small angle	0.0356	50.6	1.80	0.0188	2.9
10	Diffuser small angle2	0.0316	20.3	0.64	0.0067	1.0
11	Straight part	0.0378	9.7	0.37	0.0038	0.6
12	Corner large 3	0.1700	9.7	1.65	0.0172	2.7
13	Diffuser corner 3	0.0159	9.7	0.15	0.0016	0.2
14	Corner large 4	0.1727	6.9	1.20	0.0125	1.9
15	Diffuser corner 4	0.0148	6.9	0.10	0.0011	0.2
16	Diffuser	0.1856	5.2	0.97	0.0101	1.6
17	Straight part honey	0.0029	0.7	0.00	0.0000	0.0
18	Screen 1	2.6231	0.7	1.77	0.0185	2.9
19	Screen 2	1.2350	0.7	0.83	0.0087	1.3
20	Honeycomb	0.8756	0.7	0.59	0.0062	1.0
21	Contraction	0.0208	4.9	0.10	0.0011	0.2
Total				61.82	0.6459	100.0

Wind tunnel information

efficiency_fan	=	0.8
efficiency_motor	=	0.8
buffer_margin	=	1.1
power_input	=	357.
power_fan	=	558.

Table 5.1: Pressure loss overview per part for wind tunnel ABL David.
Test section velocity: 12.5 m/s
Test section cross section: 0.42 m²

5.2.2 Closed circuit wind tunnel 1 with a centrifugal fan (CF1)

A centrifugal blower fan of 11.5 kW is available at the TU/e. This fan comes from the former open circuit ABL wind tunnel from the Katholieke Universiteit Leuven. To create a larger test section and a larger velocity, this fan can be incorporated in a closed circuit wind tunnel by simultaneously using the parts from wind tunnel David. Centrifugal fans have been used in closed circuit wind tunnels before. The Von Karman Institute for Fluid Dynamics in Rhode-St-Genèse, Belgium, has a centrifugal blower fan in their closed circuit Wind Tunnel CWT-1. Also the Karlsruhe Institute for Technology uses a centrifugal fan in their atmospheric boundary layer wind tunnel. The centrifugal fan has a small straight part with a length of 15 cm attached to the fan. This straight part has to be removed to take better advantage of the optimization for pressure loss. This straight part is not a part of

this wind tunnel design.

To optimize this design the settling chamber cross section and diffuser length have been varied as they have the largest pressure loss (see figure 5.4).

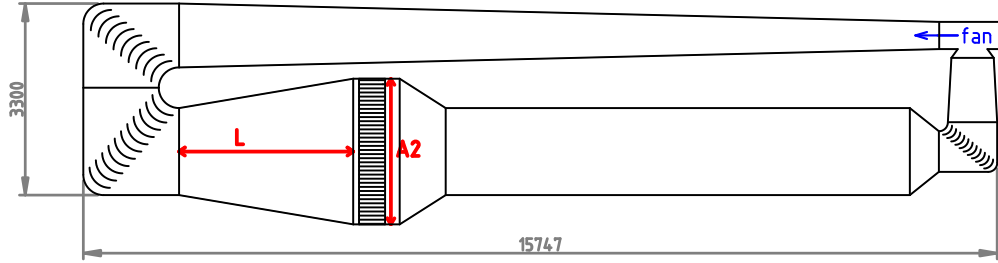


Figure 5.4: Optimization parameters for the closed circuit wind tunnel 1 with centrifugal fan (dimensions in mm), with the used corners from wind tunnel David: diffuser length (L), settling chamber cross section (A2).

- Settling chamber (A2)
The larger the cross section of the settling chamber the lower the pressure loss until 11 m^2 , though this gain is less than 4% compared to the existing settling chamber of 5 m^2 , see figure 5.5.
The used settling chamber cross section is 5 m^2 (reused).
- Diffuser length (L)
The diffuser length has an optimum at 3 m, this length is used in this design.

For this design only a new test section with dimensions of $8 \text{ m} \times 1.5 \text{ m} \times 1 \text{ m}$, a new diffuser and connecting parts need to be constructed. The other parts comprise of the parts from David, the centrifugal fan from Leuven, and connecting parts (see figure 5.6). The total pressure loss and the pressure loss per part can be found in table 5.2, the pressure loss in flow direction can be found in graph 5.7. The drawing of the closed circuit wind tunnel 1 with centrifugal fan can be found in figure 5.6, in green are the new parts.

The maximum flow rate for this design is around $15 \text{ m}^3/\text{s}$. A test section of 1 m^3 can operate a velocity of 15 m/s a test section of 1.5 m^2 a velocity of 10 m/s . The fan performance curve (in de efficient range) can be found in appendix F figure F.1. The wind tunnel system curve with the approximate fan performance curve can be found in figure 5.14. The total fan curve from the centrifugal fan is unknown. Before deciding on this design the maximum flow rate of the fan without wind tunnel parts has to be measured. The maximum flow rate, for no pressure loss, has to be added to figure 5.14 to find a more accurate operation point.

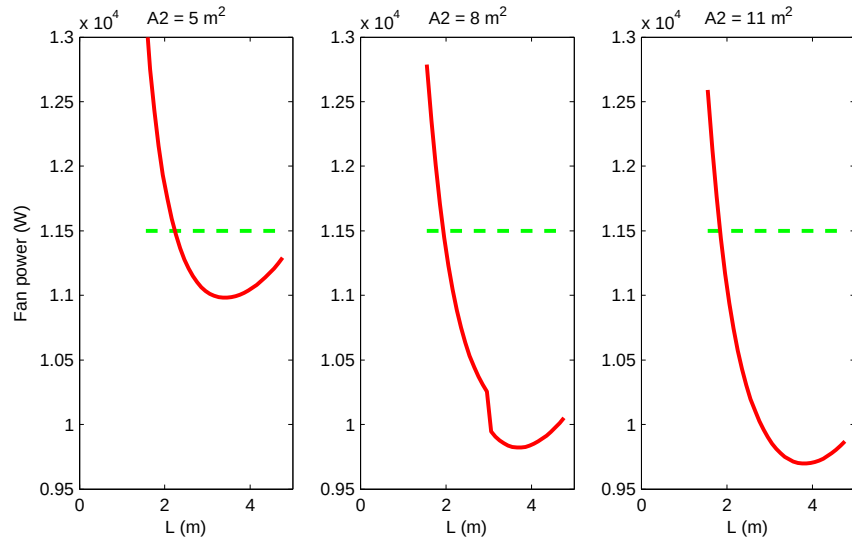


Figure 5.5: In red the required fan power for the closed circuit wind tunnel 1 with centrifugal fan. With variable diffuser length (L) and settling chamber cross section (A_2). In green the available fan power. The existing settling chamber from former wind tunnel David is 5 m^2 . The jump in graph $A_2 = 8 \text{ m}^2$ is due to the change from diffuser ($\alpha < 10$ degrees) to diffuser with BLC.

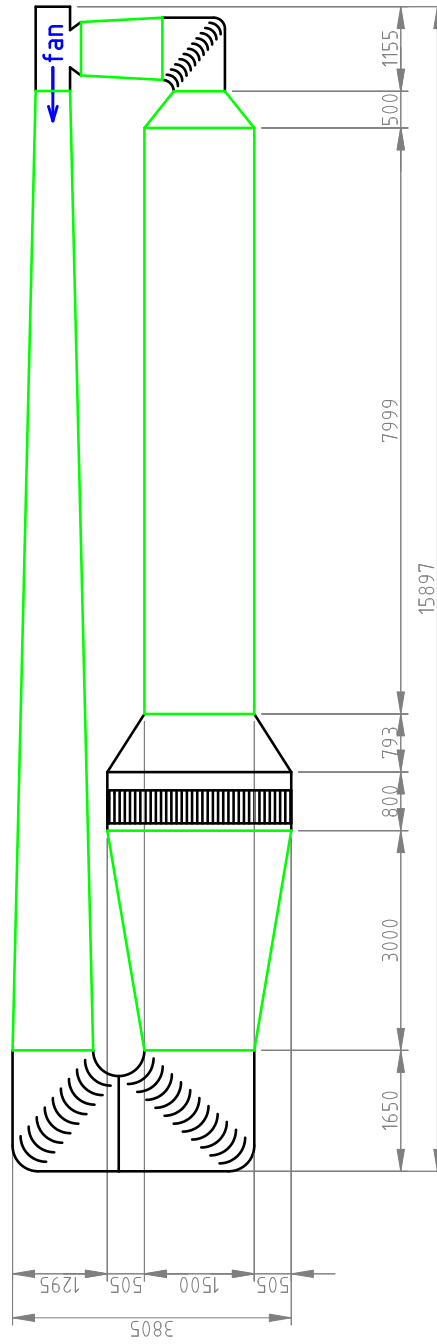


Figure 5.6: Dimensions of the closed circuit wind tunnel 1 with centrifugal fan (dimension in mm). The green parts are new.

nr	Part name	K	q	Δp	Kts	%
1	Rectangular diffuser	0.1005	1109.1	111.42	1.2632	30.3
2	Corner David A	0.1545	113.9	17.60	0.1995	4.8
3	Corner David B	0.1564	81.5	12.75	0.1446	3.5
4	Diffuser corner A	0.0242	113.9	2.75	0.0312	0.7
5	Diffuser corner B	0.0227	81.5	1.85	0.0210	0.5
6	Diffuser	0.1835	61.3	11.24	0.1274	3.1
7	Honeycomb David	0.7870	7.9	6.25	0.0708	1.7
8	Screen	1.0846	7.9	8.61	0.0976	2.3
9	Screen	2.4060	7.9	19.10	0.2165	5.2
10	Contraction David	0.0292	88.2	2.57	0.0292	0.7
11	Fetch	0.0913	88.2	8.05	0.0913	2.2
12	Straight contraction	0.0224	695.9	15.60	0.1769	4.2
13	Corner David D	0.1598	695.9	111.21	1.2609	30.2
14	Diffuser corner D	0.0126	763.0	9.62	0.1090	2.6
15	Straight contraction	0.0100	2933.4	29.33	0.3326	8.0
Total				367.96	4.1719	100.0

Wind tunnel information

efficiency_fan	=	0.6
efficiency_motor	=	0.9
buffer_margin	=	1.1
power_input	=	7286.
power_fan	=	11384.

Table 5.2: Pressure loss overview closed circuit wind tunnel 1 with centrifugal fan.
Test section velocity: 10 m/s
Test section cross section: 1.5 m²

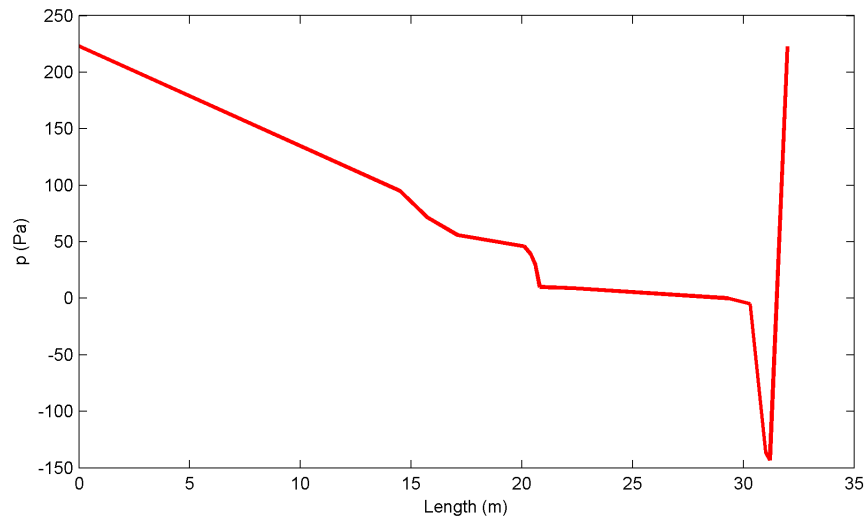


Figure 5.7: Pressure loss in flow direction. Closed circuit wind tunnel 1 with centrifugal fan, see table 5.2 for corresponding parts.

5.2.3 Closed circuit wind tunnel 2 with a centrifugal fan (CF2)

The centrifugal fan can also be used in a wind tunnel optimizing for most parts (see figure 5.8). Parts between fixed elements (usually the fan and the test section) are optimized separately. In this design the fan and the test section size are fixed. The corner cross section (A1), diffuser length (L), and settling chamber cross section (A2) are optimized simultaneously. Also the possibility of a sudden expansion directly downstream the fan is optimized, as is the corner cross section (A3).

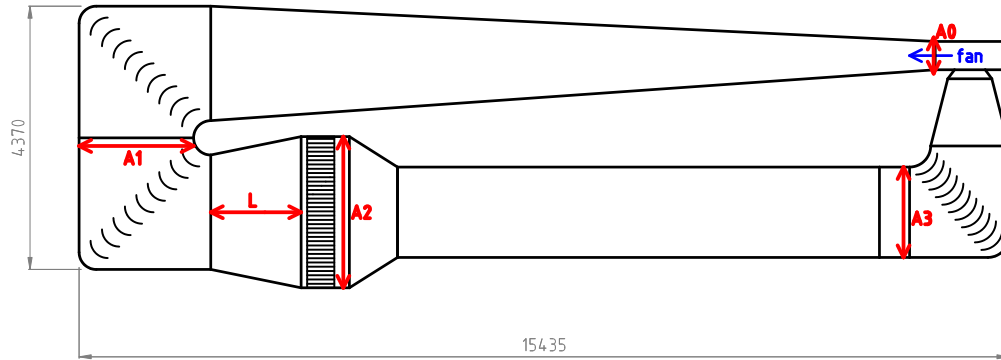


Figure 5.8: Optimization parameters for the closed circuit wind tunnel with centrifugal fan 2 (dimensions in mm): sudden expansion (A0), corner cross section (A1), diffuser length (L), settling chamber cross section (A2), and corner cross section (A3)

- Sudden expansion (A0)
Adding a sudden expansion directly downstream of the fan only increases the pressure loss, see figure 5.11.
No sudden expansion is used.
- Corner cross section (A1)
These corners have an optimum cross section at 3.5 m^2 , see figure 5.9. It reduces the pressure loss by almost 40 % in this setting compared to the use of the corners from David with an cross section average of 1.56 m^2 .
The used corner cross section is 3.5 m^2 .
- Diffuser length (L)
The length of the diffuser does not influence the pressure loss largely as long as the angle is below 10 degrees, which is in this case at diffuser angle of 1.5 m and larger, see figure 5.9
The used a diffuser length is 1.5 m

- Settling chamber cross section (A2)
The use of a larger settling chamber than the existing from David reduce the fan power by 10%.
The used settling chamber cross section is 5 m^2 (reused).
- Corner cross section (A3)
The most efficient cross section for the corner downstream of the test section is a cross section of 2.3 m^2 , see figure 5.10. However, the gain in fan power is less than 1% compared to the use of one largest corner from David with a cross section of 1.86 m^2 .
The used corner cross section is 1.86 m^2 (reused).

This design reuses the centrifugal fan, the settling chamber, the contraction and the largest corner. The dimension of the test section is $8 \text{ m} \times 1.5 \text{ m} \times 1 \text{ m}$. The total pressure loss and the pressure loss per part can be found in table 5.3, the pressure loss in flow direction can be found in graph 5.13. The drawing of the ABL-David can be found in figure 5.12, in green are the new parts.

The maximum flow rate for this design is around $17 \text{ m}^3/\text{s}$. A test section of 1 m^2 can operate a velocity of 17 m/s a test section of 1.5 m^2 a velocity of 11 m/s . The fan performance curve (in the efficient range) can be found in appendix F figure F.1. The wind tunnel system curve with the approximate fan performance curve can be found in figure 5.14. The total fan curve from the centrifugal fan is unknown. Before deciding on this design the maximum flow rate of the fan without wind tunnel parts has to be measured. The maximum flow rate, for no pressure loss, has to be added to figure 5.14 to find a more accurate operation point.

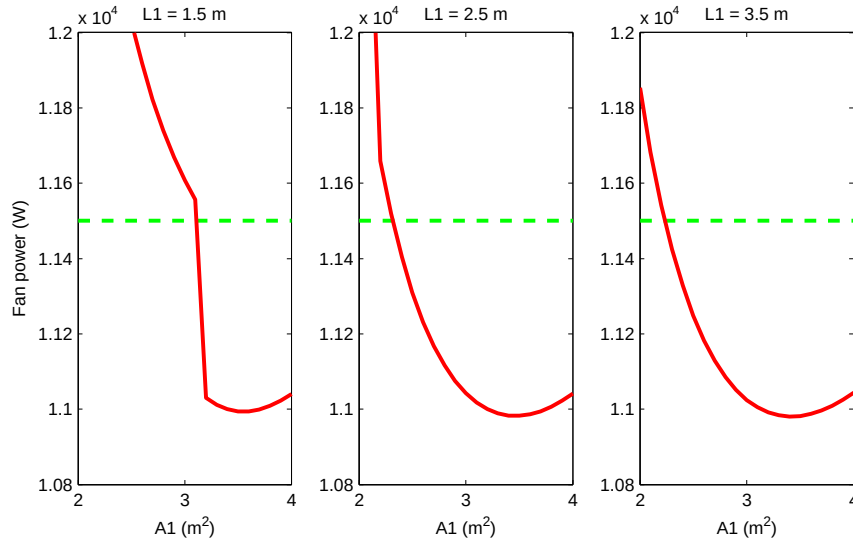


Figure 5.9: In red the required fan power for the closed circuit wind tunnel 2 with centrifugal fan (dimensions in mm). With variable diffuser length ($L1$) and corner cross section ($A1$). In green the available fan power. The jump in the graph is the change to low angle diffuser (< 10 degrees), resulting in a less pressure loss. The existing average corner cross section from David is $1.56 m^2$.

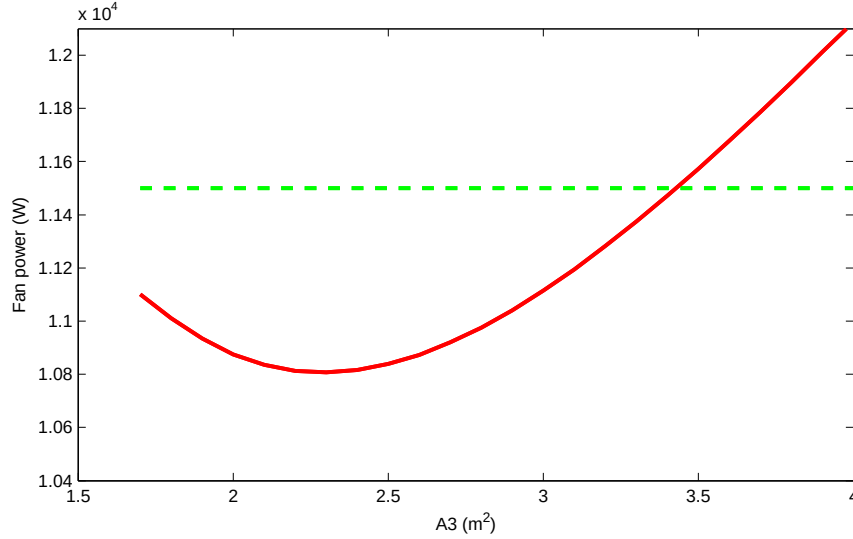


Figure 5.10: In red the required fan power for the closed circuit wind tunnel 2 with centrifugal fan. With variable corner cross section ($A3$). In green the available fan power. The largest existing corner cross section from David is $1.86 m^2$.

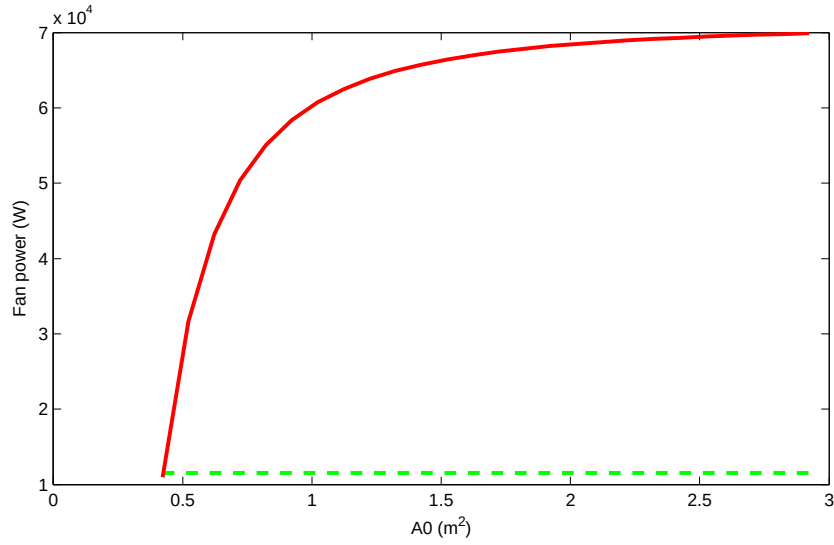


Figure 5.11: In red the required fan power for the closed circuit wind tunnel 2 with centrifugal fan. With variable sudden expansion (A_0), at a cross section of 0.47 m^2 there is no sudden expansion. In green the available fan power.

nr	Part name	K	q	Δp	Kts	%
1	Rectangular diffuser	0.1015	1641.8	166.62	1.2762	55.2
2	Corner	0.1645	22.7	3.73	0.0286	1.2
3	Corner	0.1645	22.7	3.73	0.0286	1.2
4	Diffuser	0.0289	22.7	0.66	0.0050	0.2
5	Honeycomb David	0.7739	11.8	9.09	0.0697	3.0
6	Screen	1.0698	11.8	12.57	0.0963	4.2
7	Screen	2.3784	11.8	27.95	0.2141	9.3
8	Contraction David	0.0292	130.6	3.81	0.0292	1.3
9	Fetch	0.0913	130.6	11.92	0.0913	3.9
10	Rectangular diffuser	0.0077	111.9	0.86	0.0066	0.3
11	Corner	0.1705	104.1	17.75	0.1359	5.9
12	Straight contraction	0.0100	4342.2	43.42	0.3326	14.4
Total				302.10	2.3139	100.0

Wind tunnel information

efficiency_fan	=	0.5
efficiency_motor	=	0.9
buffer_margin	=	1.1
power_input	=	7278.
power_fan	=	11371.

Table 5.3: Pressure loss overview closed circuit wind tunnel 2 with centrifugal fan.
Test section velocity: 11 m/s
Test section cross section: 1.5 m^2

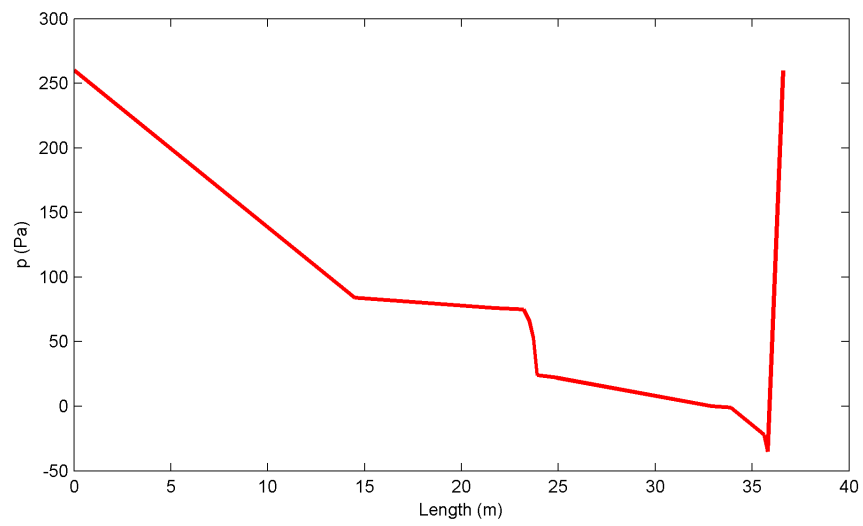


Figure 5.13: Pressure loss in flow direction. Closed circuit wind tunnel 2 with centrifugal fan, see table 5.3 for corresponding parts.

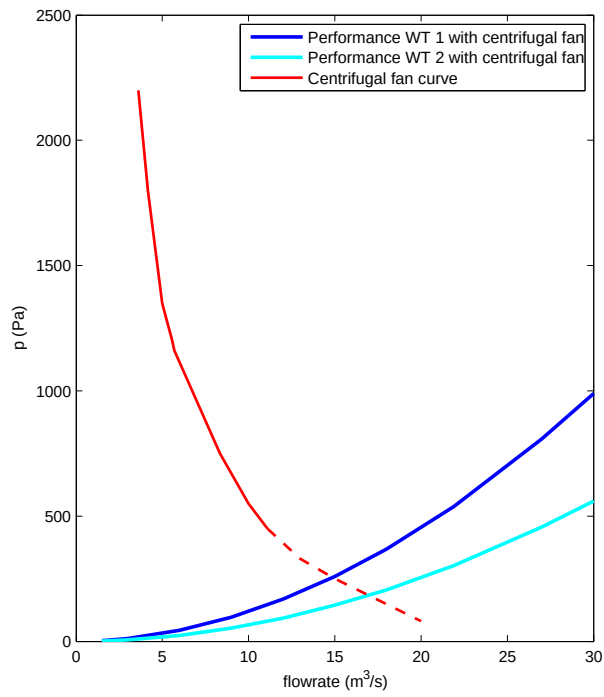


Figure 5.14: The performance curve of the closed circuit wind tunnel 1 and 2 with a centrifugal fan. In red the fan performance curve of the centrifugal fan, in dashed red a possible contour of the total curve.

5.2.4 Closed circuit wind tunnel with axial fan (AF1)

A larger test section cross section has more possibilities for experiments. In this design the test section dimensions are taken maximum for the available space. The minimum required length for the corners, diffuser, the settling chamber and the contraction is around 7,5 m. This leaves 11 m for the fetch and the test section. The design cross section for the test section is $2 \text{ m} \times 1.5 \text{ m}$ and a velocity of 20 m/s.

Optimizations have been made by changing the cross section of the corners and the settling chamber, and the length of the diffuser and the contraction (see figure 5.15). Corner cross section (A1) and settling chamber cross section (A2) have been optimized simultaneously, see figure 5.16. Diffuser (L1) and contraction (L2) have also been optimized for their combined maximum length.

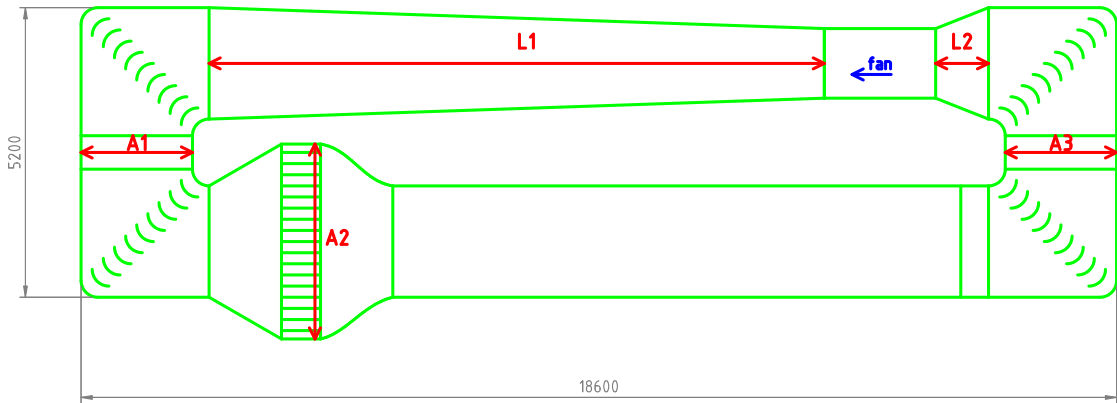


Figure 5.15: Optimization parameters for closed circuit wind tunnel with axial fan (dimensions in mm): cross section of corner (A1), settling chamber (A2), and corner (A3). Diffuser length L1, and contraction length L2. All (green) parts are new.

- Diffuser length (L1)
The diffuser length (L1) and contraction length (L5) have been optimized within their total length of 12 m. These have an optimum with a straight contraction at a diffuser length of 11.05 m.
The used diffuser length is 11.05 m
- Corner cross section (A1)
The corners have a optimum pressure loss at 7.3 m^2 , see figure 5.16.
The used corner cross section is 7.3 m^2 .
- Settling chamber cross section (A2)
The largest possible settling chamber cross sections (A2) possible in the

available room gives the minimum required fan power. However, at cross sections larger than 13 m^2 the power gain is less than 4%, see figure 5.16. The used settling chamber cross section is 13 m^2 .

- Corner cross section (A3)
The pressure loss through the corners downstream of the test section (A3) is has an optimum at 4.1 m^2 , see figure 5.17.
The used corner cross section is 4.1 m^2 .
- Contraction length (L2)
Changing the diffuser (L1) and a curved contraction as (L2) does not have an optimum, see figure 5.18. With use of a straight contraction it has an optimum at 0.95 m^2 .
The used straight contraction length is 0.95 m .

Other dimensions are taken to be maximum for the available room. Within the existing room this is:

- Wide angle diffuser length is 1.3 m
- Contraction length is 1.3 m
- Test section is $11 \text{ m} \times 2 \text{ m} \times 1.5 \text{ m}$
- Connecting part is 0.5 m

All dimensions can be found in figure 5.19. A fan of 27 kW can operate a velocity of 20 m/s in a test section. The total pressure loss and the pressure loss per part can be found in table 5.4, the pressure loss in flow direction can be found in graph 5.21. The fan diameter in this design is taken to be 1.5 m . A larger diameter results in less total pressure loss through the wind tunnel, and visa versa.

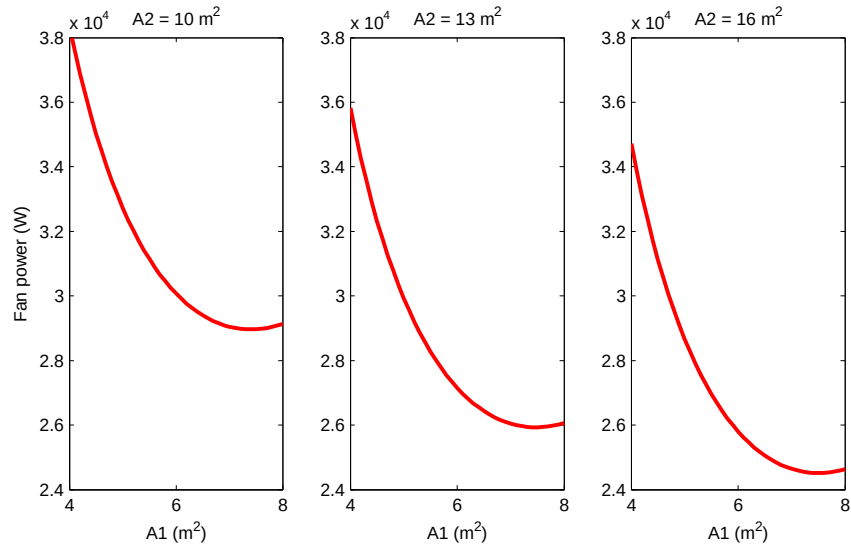


Figure 5.16: Required fan power for closed circuit wind tunnel with axial fan. With variable corner cross section (A_1), and settling chamber cross section (A_2).

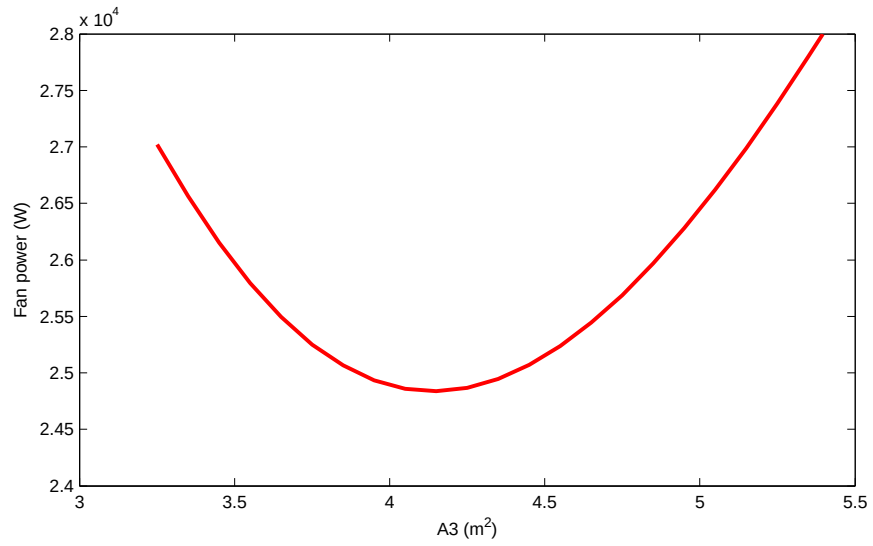


Figure 5.17: Required fan power for closed circuit wind tunnel with axial fan. With variable corners cross section (A_3)

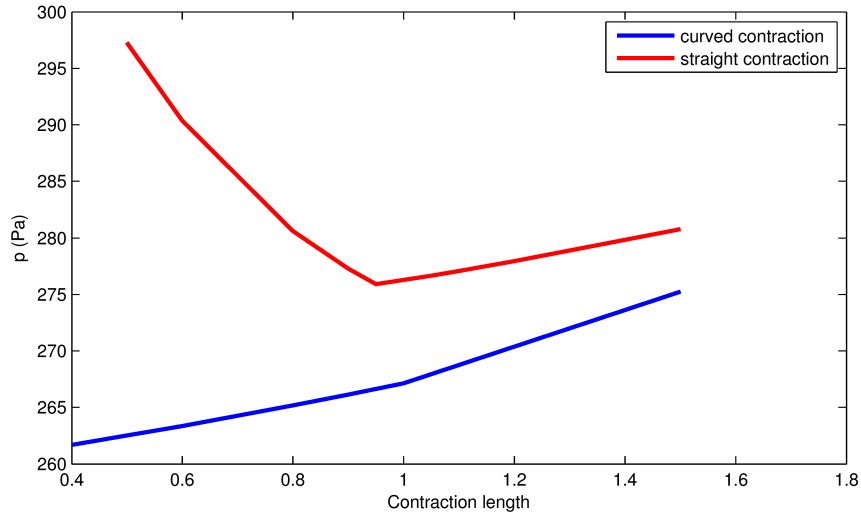


Figure 5.18: Required fan power for length diffuser plus length contraction = 12 m.

nr	Part name	K	q	Δp	Kts	%
1	Rectangular diffuser	0.0930	903.2	84.02	0.3430	34.8
2	Corner	0.1559	39.2	6.11	0.0249	2.5
3	Corner	0.1559	39.2	6.11	0.0249	2.5
4	Diffuser_BLC	0.7322	39.2	28.70	0.1171	11.9
5	BLC Screen	0.3098	17.4	5.40	0.0220	2.2
6	Honeycomb	0.4470	9.8	4.38	0.0179	1.8
7	Screen	1.0762	9.8	10.55	0.0430	4.4
8	Screen	2.3907	9.8	23.43	0.0956	9.7
9	Curved contraction L=1m	0.0020	245.0	0.50	0.0020	0.2
10	Fetch	0.0587	245.0	14.37	0.0587	6.0
11	Rectangular diffuser	0.0261	212.7	5.55	0.0227	2.3
12	Corner downstream ts	0.1688	137.8	23.26	0.0950	9.6
13	Corner	0.1535	137.8	21.15	0.0863	8.1
14	Safety net	0.2000	137.8	27.56	0.1125	10.5
15	Straight contraction	0.0219	903.2	19.80	0.0808	0.5
Total				282.17	1.0715	100.0

Wind tunnel information

efficiency_fan	=	0.8
efficiency_motor	=	0.8
buffer_margin	=	1.1
power_input	=	17326.
power_fan	=	27072.

Table 5.4: Pressure loss overview closed circuit wind tunnel with axial fan.

Test section velocity: 20 m/s

Test section cross section: 3 m²

Case straight contraction, L = 2.5 m, total pressure loss + 1%

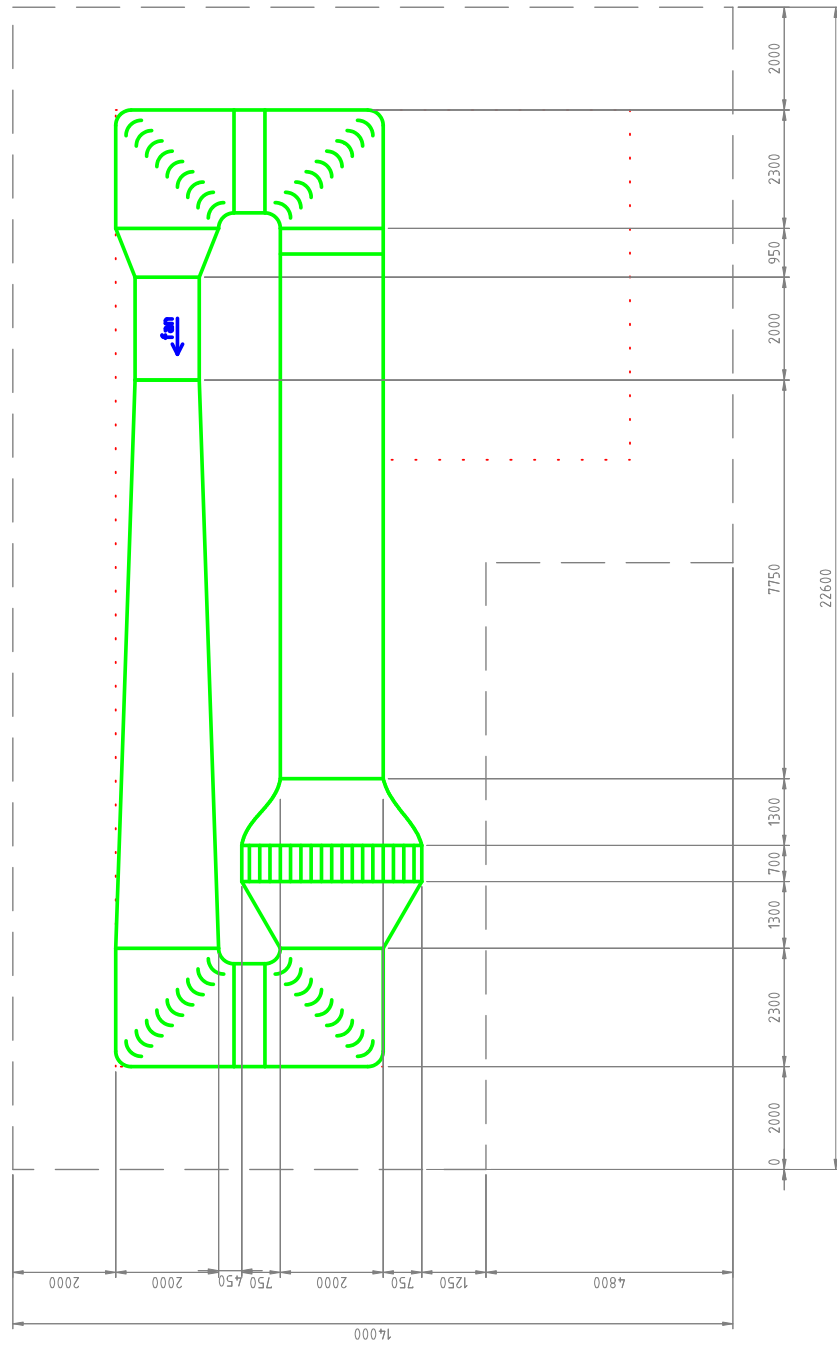


Figure 5.19: Closed circuit wind tunnel with an axial fan of 27 kW (dimensions in mm). Test section dimension are 11 m × 2 m × 1.5 m and velocity of 20 m/s. All (green) parts are new.

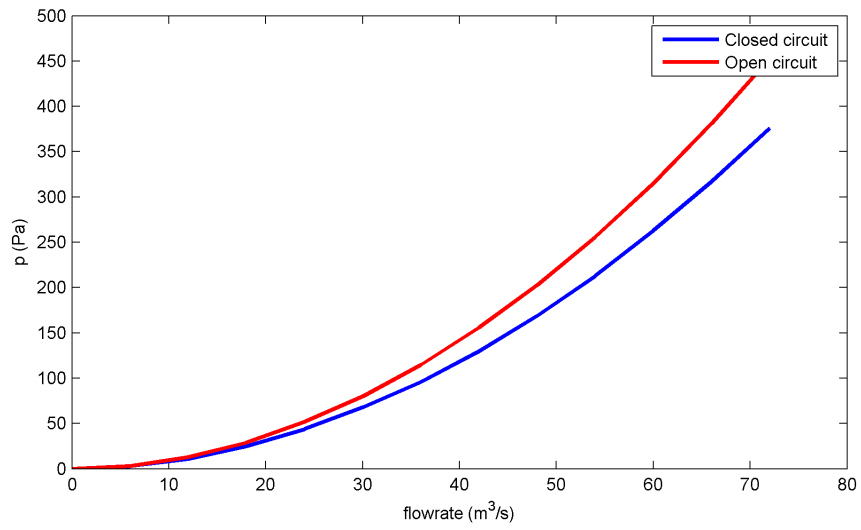


Figure 5.20: Wind tunnel performance curve for the closed circuit wind tunnel with axial fan and the open circuit wind tunnel.

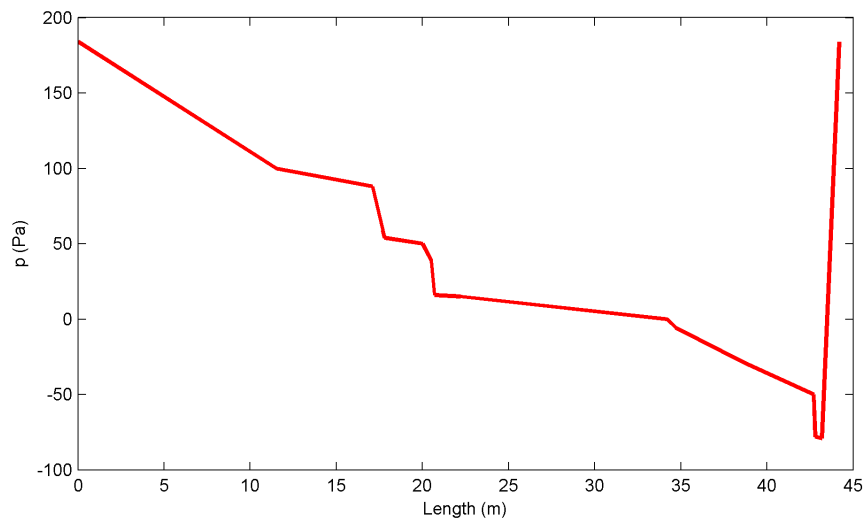


Figure 5.21: Pressure loss in flow direction. Closed circuit wind tunnel with axial fan, see table 5.4 for corresponding parts.

5.2.5 Open circuit wind tunnel with axial fan (AF2)

A larger test section cross section has more possibilities for experiments. In this design the test section dimensions are taken maximum for the available space. For an open circuit wind tunnel the maximum fetch length is determined by the length of the inlet, settling chamber, contraction, diffuser and corner. These parts take at least 7.5 m, leaving 11 m for the fetch and the test section. Optimizations have been made changing the inlet cross section (A1), the corner cross section (A2), the outlet cross section (A3) and contraction length (L2) and diffuser length (L3), see 5.25.

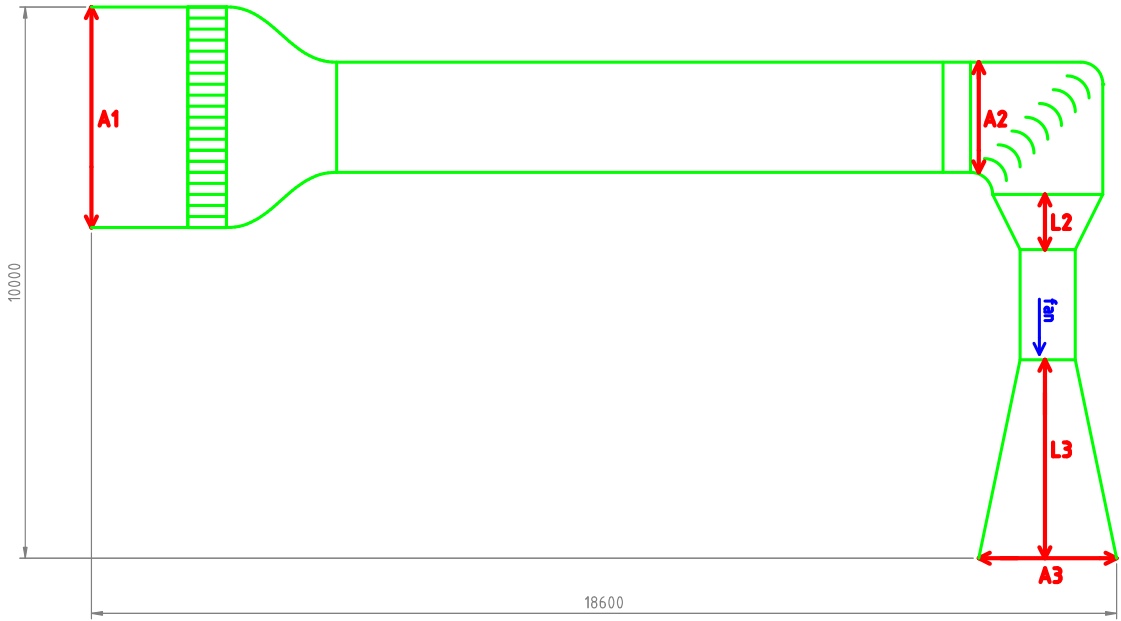


Figure 5.22: Optimization parameters for open circuit wind tunnel (dimensions in mm): cross section of the inlet and settling chamber (A1), cross section of the corner (A2), length of the contraction (L2), length of the exit diffuser (L3) and cross section of the exit (A3). All (green) parts are new.

- Inlet cross section (A1)

For the curved inlet contraction (bellmouth) the larger the inlet cross section the lower the pressure loss. In a bellmouth the gain in pressure loss only has to compensate for the increased contraction loss. The pressure loss approaches a limit, see figure 5.23. However, for cross sections larger than 12 m^2 the gain is less than 5%. For flow quality reasons this angle should not exceed 25 degrees.

The used inlet cross section is 12 m^2

- Corner cross section (A2)
The corner has an optimum pressure loss through at 3.8 m^2 , see figure 5.24.
A fan of almost 20% more power is needed when an existing corner from wind tunnel David (with cross section 1.86 m^2) would be used.
The used corner cross section is 3.8 m^2 .
- Contraction length (L2)
The maximum length for the contraction and the diffuser (L3) is 4.5 m. With a straight contraction they have an optimum at
The used contraction length is 0.9 m.
- Diffuser length (L3)
With a maximum length of 4.5 m for contraction length (L2) and the diffuser the diffuser has an optimum at
The used diffuser length is 3.6 m.
- Exit cross section (A3)
For a diffuser length of 3.6 m the optimum exit cross section is at 4 m^2 (diffuser angle of 4 degrees) see figure 5.24
The used exit cross section is 4 m^2 .

Other dimensions are taken to be maximum for the available room. Within the existing room this is:

- Inlet straight part is 1.75 m
- Contraction length is 2 m
- Test section is $11 \text{ m} \times 2 \text{ m} \times 1.5 \text{ m}$
- Connecting part is 0.5 m
- Fan diameter is 1.5 m

All dimensions can be found in figure 5.25. A fan of 35 kW can operate a velocity of 20 m/s in a test section. The total pressure loss and the pressure loss per part can be found in table 5.5.

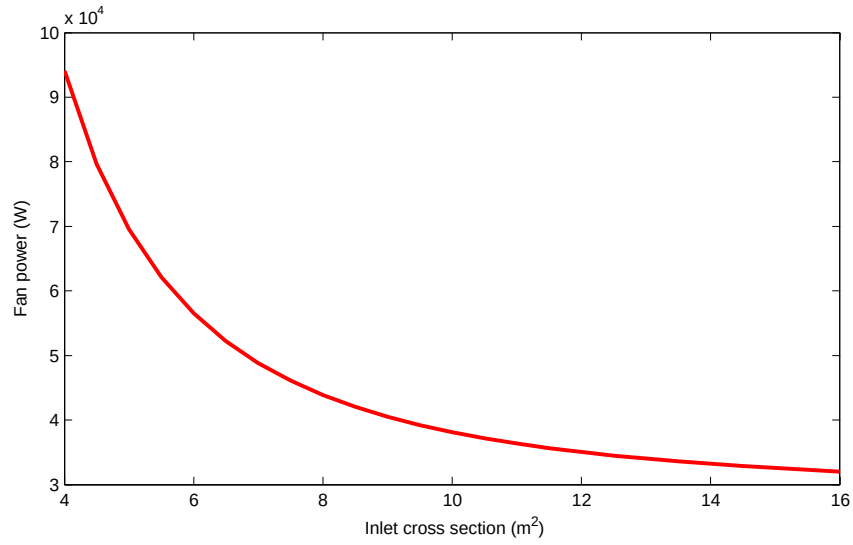


Figure 5.23: Required fan power for the open wind tunnel. With variable inlet and settling chamber cross section(A_1).

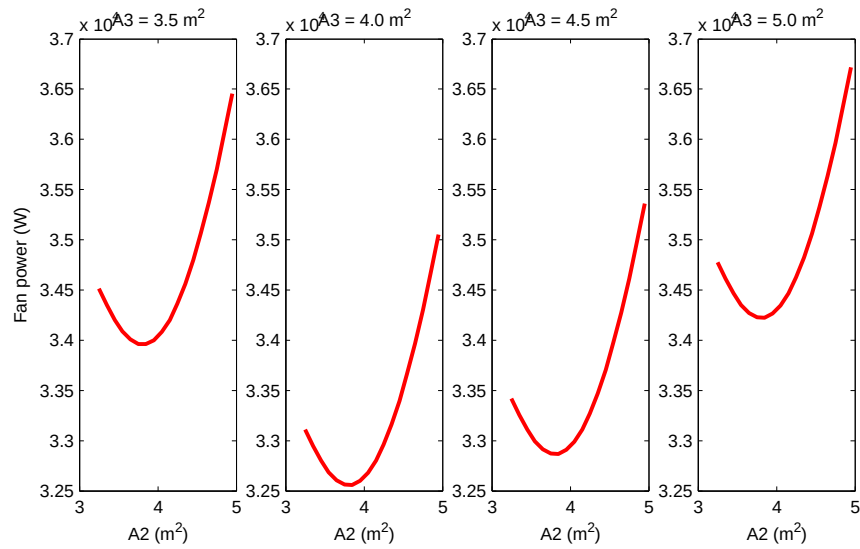


Figure 5.24: required fan power for the open wind tunnel. With variable corner cross section (A_2) and the exit cross section (A_3).

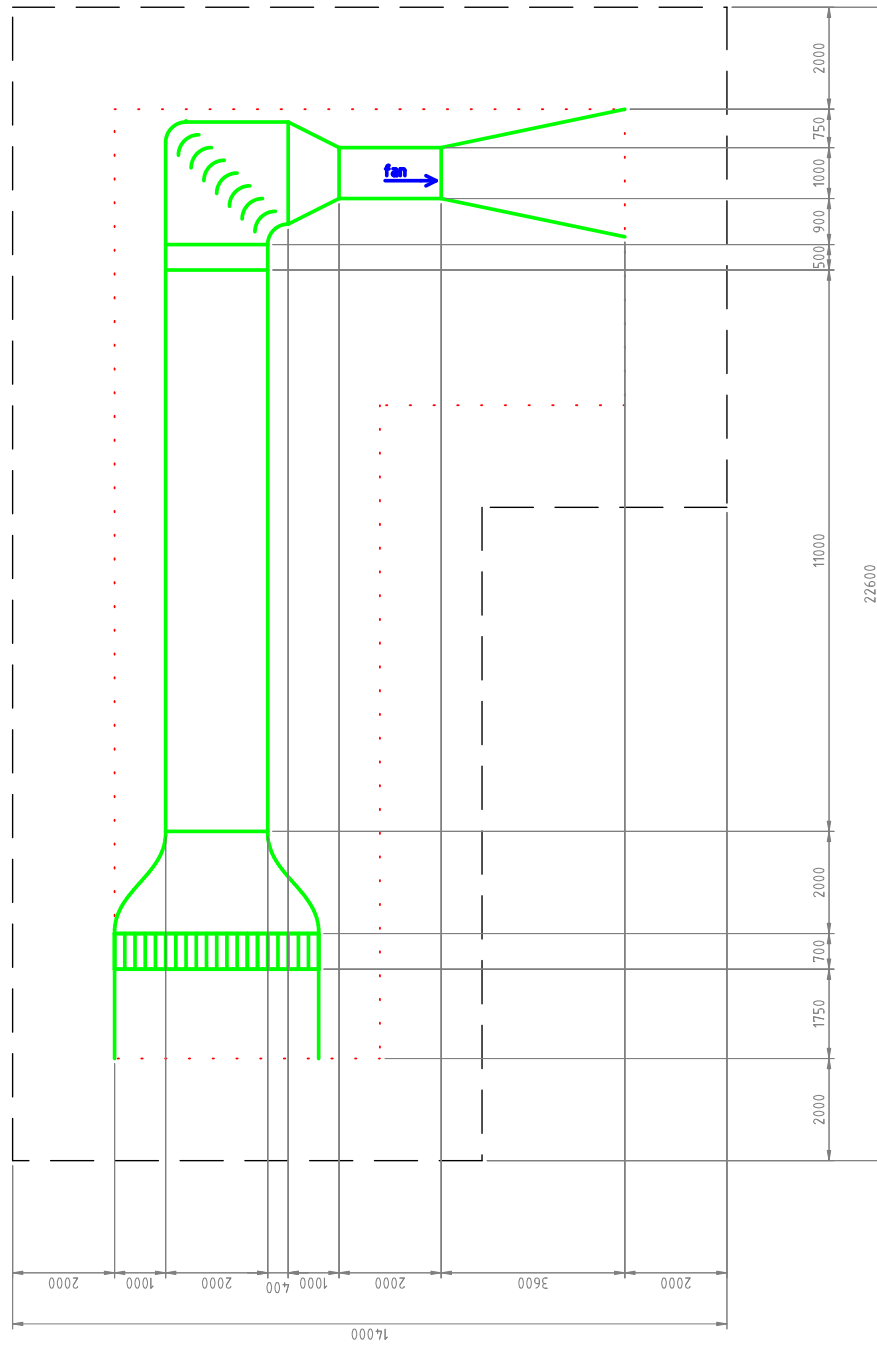


Figure 5.25: Open circuit wind tunnel with an axial fan of 35 kW (dimensions in mm).
 Test section dimensions of 12 m × 2 m × 1.5 m and velocity of 20 m/s.
 All (green) parts are new.

nr	Part name	K	q	Δp	Kts	%
1	Inlet	0.5000	9.8	4.90	0.0200	1.6
2	Straight part	0.0058	9.8	0.06	0.0002	0.0
3	Honeycomb	0.4470	9.8	4.38	0.0179	1.4
4	Screen	1.0762	9.8	10.55	0.0430	3.3
5	Screen	2.3907	9.8	23.43	0.0956	7.4
6	Curved contraction	0.0031	245.0	0.75	0.0031	0.2
7	Fetch	0.0587	245.0	14.37	0.0587	4.6
8	Rectangular diffuser	0.0126	212.7	2.67	0.0109	0.8
9	Corner	0.1529	152.7	23.35	0.0953	7.4
10	contraction	0.0283	347.9	9.84	0.0402	3.1
11	Rectangular diffuser	0.0925	903.2	83.59	0.3412	26.5
12	Outlet	1.0000	137.8	137.81	0.5625	43.7
Total				315.70	1.2886	100.0
Wind tunnel information						
efficiency_fan			=	0.8		
efficiency_motor			=	0.8		
buffer_margin			=	1.1		
power_input			=	20836.		
power_fan			=	32556.		

Table 5.5: Pressure loss overview. Open circuit wind tunnel.

Test section velocity: 20 m/s

Test section cross section: 3 m²

5.3 Cost indication

The life cycle costs of a wind tunnel include: material, man hours for construction, maintenance (materials and man hours), power consumption, and demolition costs. The two main factors are the costs for construction and the running costs. The major cost in construction is the fan price and will be discussed in the next section.

5.3.1 Finding a suitable fan

For the first three design options the existing fan from David or open circuit wind tunnel from KU Leuven is reused. In the case of the largest possible test section in TU/e room VRT1.10 a fan has to be found for a flow rate of 60 m³/s (20 m/s $\times$ 2 m $\times$ 1.5 m) and a fan power of minimum 27 kW and 263 Pa (for the closed circuit) or 35 kW and 315 Pa (for the open circuit). The operation curve for both wind tunnels can be found in figure 5.20. Fan information has been obtained from Geveke, Rucon, Helios, Savio, and Witt&Sons, see table 5.6. A fan with an external motor which is preferable (and discussed in chapter 3.2.2), has higher costs than a fan with internal motor. Fan performance curves of Helios, Rucon, Witt & Sons and the performance curve from the existing centrifugal fan from Leuven can be found in appendix F.

When selecting a fan, the diameter of the fan will most likely differ with the

Company	Location	Type	P_{fan} kW	Flow rate m^3/s	p_{stat} Pa	q Pa	p_{tot} Pa	D_{fan} mm	Price €
Geveke	Netherlands	Axipal BZI	45	30	800	366	1166	1250	10352
Rucon	Netherlands	EFC2004/H(G)	33.5	30	230	224	454	2000	22835
Rucon	Netherlands	EF1806/H(G)	40.3	30	238	353	591	1800	17000
Helios	Germany	ADV 1250/4	45	33.3	600	443	1043	1250	15000
Savios	Italy		55					1120	3500 - 4500
Witt & Sohn	Germany	A-M8L5/V1.4/	41	56	413	194	600	2000	25579

Table 5.6: Fan information, all fans are frequency changeable. the fan from Rucon EFC2004/H(G) has an external motor

diameter of the fan used in the calculations. The fan diameter influences the pressure loss. Therefore the WinDI has to be run with the chosen fan diameter to calculate the accurate pressure loss.

5.3.2 Construction costs

The construction costs depend on material, man hours and the fan price. In the calculations the man hours are not included. The material costs are mainly based on the materials for the ducts. Wood is easy to handle and less likely to drum than steel (Mehta, 1977). Wooden plates can be bought at do-it-yourself markets. Normal prices at these stores are (effective March 2013) Multiplex (€20 / m^2), underlayment (€12 / m^2), plywood concrete €10¹. Also wooden poles to support the construction should be accounted for. For the test section transparent material, such as Plexiglas (approximately €1000), is required and a turntable (approximately €1000) for the measurements (Gousseau, 2009a) and taken into account the inflation (Stathopoulos, 1984). Measurement equipment is not taken into account in the cost indication.

The running costs depend on the amount of time the wind tunnel is used. The calculations are based on a 5 hours usage per week, with 40 running weeks and 200 running hours per year. The price per kWh in the Netherlands (effective March 2013) is €0.22.

An overview of the cost indication can be found in table 5.7.

5.3.3 Cost efficient design

An important design decision for the wind tunnel is the choice between using existing parts or optimizing a wind tunnel without taking existing parts into account. The transformation of the former wind tunnel David from the Applied Physics

¹For example: <http://www.praxis.nl>

		ABL David	CF 1	CF 2	Closed AF1	Open AF2
test section velocity	(m/s)	12	12	14.8	20	20
Test section cross section	(m ²)	0.42	1.5	1.5	3	3
Fan power	(kW)	0.605	11.5	11.5	27	35
Plywood concrete (21 mm)	10 €/m ²	400	600	1500	2500	1500
Wooden poles 0.07×0.07×3 m	(€)	100	150	750	1000	400
Flow straighteners	(€)	0	0	0	1000	1000
Corners + guiding vanes	(€)	0	0	500	2000	2000
Test section plexiglas	(€)	500	1000	1000	1000	1000
Turntable	(€)	0	1000	1000	1000	1000
Frequency controller	(€)	0	0	0	5000	5000
Transport	(€)	0	0	0	1000	1000
Fan	(€)	0	0	0	5000 - 25000	5000 - 25000
Unforeseen (€500 or 10%)	(€)	500	500	600	2000	2000
Total construction cost	(€)	1500	3250	5500	21500 - 41500	20000 - 40000
200 running hours / year	(€)	30	500	500	1200	1500

Table 5.7: Cost indication.

Department is with €1500 the most cost efficient option. However the test section cross section is only 0.42 m², which does not fulfill the requirements. The transformation of David with implementation of the centrifugal fan from the KU Leuven enables a test section of 1 m × 1.5 m with a velocity of around 10 m/s. When this wind tunnel is combined with newly built corners, the same fan can operate a slightly velocity of 11 m/s.

When choices are for a larger test section with a higher velocity no parts can be reused and all parts need to be build and bought. Both open and closed configuration are designed for the same test section cross section. The total construction costs for the open en closed configuration are comparable. The operation costs only exceed the construction costs after 15 to 20 years.

An overview of the five design options can be found in table 5.8. Generally, the larger the test section size and the test velocity, the more expensive the wind tunnel.

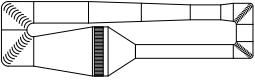
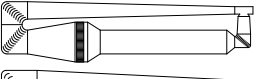
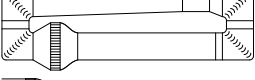
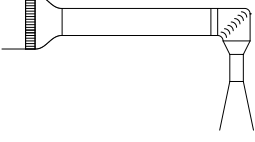
	Type	A_{ts} (m ²)	v_{ts} (m/s)	P_{fan} (kW)	Construction costs (€)
	closed ABL-David	0.4	12	0.605	1500
	closed CF1	1.5	10	11.5	3300
	closed CF2	1.5	11	11.5	5500
	closed AF1	3.0	20	27.	22000 - 42000
	open AF2	3.0	20	35.	20000 - 40000

Table 5.8: Overview design options

Chapter 6

Summary and Conclusions

The Wind tunnel Design Instrument (WinDI) has been developed to calculate the pressure loss through a wind tunnel. It can be used to minimize pressure loss or, with a given fan power, optimize for maximum test section velocity and dimensions. The WinDI is usable for pressure loss calculations and optimizations for different types of wind tunnels and other duct systems such as ventilation systems.

In this research the following was found using WinDI:

- The largest gain in pressure loss is to find in simultaneously optimizing for the settling chamber cross section, the corners cross section, and the diffuser length between these elements.
- Generally assumed for wind tunnel design is that pressure loss sensitive parts, such as corners and settling chambers, have to be placed in an as large as possible cross section / low as possible velocity. Using WinDI it was found that increasing the cross section does not necessarily decrease the total pressure loss for pressure loss sensitive elements. The gain in pressure loss and increase in pressure loss in the diffuser and contraction have an optimum.
- For diffusers it's advised to have an as low as possible angle, preferable around 3 degrees. This is true, however the gain is largest if the angle is under the critical value of (for ABL wind tunnels) 10 degrees. Larger angles require boundary layer control which significantly increases the pressure loss.
- For a curved inlet contraction it was found that the larger the inlet cross section the lower the pressure loss. The pressure loss approaches a limit. For flow quality reasons the contraction angle should not exceed 25 degrees.

- The equation for square diffusers from Barlow et al. (1999) contains an error in the coefficients. The difference in calculated pressure loss can be over 100%. The correct coefficients have been verified by the author and can be found in equation 4.10.

6.1 Answers to the research questions

For the design of an ABL wind tunnel the following research questions have been formulated in chapter 1.3:

1. What are the various parts in wind tunnels?

1. What are their properties?
2. What are their functions?
3. What causes the pressure loss of each part?
4. How can the pressure loss of each part be calculated?

Diffuser, the opposite walls diverge, this increases the cross section and decreases the flow velocity. The pressure loss are due to expansion and friction loss. Calculation of the pressure loss according to equation 4.5

Straight part, is a parallel duct, settles the flow and connects parts. Pressure loss due to friction. Calculation of the pressure loss according to equation 4.16

Corner, usually with turning vanes, turns the flow. Pressure loss due to rotation and friction. Calculation of the pressure loss according to equation 4.17

Settling chamber, consist of honeycombs and screens, uniforms the flow. Pressure loss due to friction. Calculation of the pressure loss according to equation 4.18 and 4.19

Contraction, the opposite walls converge, this decreases the cross section and increases the velocity. Pressure loss due to friction. Calculation of the pressure loss according to equation 4.20

Fetch, roughness elements in a lengthy straight part, generation of the atmospheric boundary layer. Pressure loss due to friction and the drag of the roughness elements. Calculation of the pressure loss according to equation 4.24

Transition section, from circular to rectangular cross section and visa versa, connects parts of different shapes. Pressure loss due to friction and expansion. Calculation of the pressure loss according to chapter 4.1.7.

Inlet, opening at the inlet of the wind tunnel, allows air to enter the wind tunnel. Pressure loss due to separation. Calculation of the pressure loss according to chapter 4.1.7.

Outlet, opening at the exit of the wind tunnel, outlet for the air flow. Pressure loss due to expansion loss. Calculation of the pressure loss according to chapter 4.1.7.

See chapter 3.2 for more information on properties and function, and chapter 4.1 for information on pressure losses and pressure loss calculation.

2. What are the needed parameters to calculate the fan power of a wind tunnel and how can this be calculated from minimum input parameters?

To calculate the pressure loss of a wind tunnel, various parameters are required. Most wind tunnel parts use at least one of the following parameters: the friction factor (f), Reynolds number (Re), Area ratio (Ar), porosity (β), length, and the angle (α). These parameters can be derived from the basic exterior parameters, which are the inlet and outlet dimensions, in combination with additional part specific parameter(s). Part specific parameters are for example: length, cross section shape, and wire diameter. For the list of all part specific parameters and the calculated parameters per part see appendix A.2.3.

3. How can a wind tunnel design be optimized for minimum pressure loss?

To find an optimum design for a wind tunnel, ideally all parts have to be optimized for minimum pressure loss. Parts between fixed objects can be optimized separately. In a closed circuit wind tunnel the parts between the fan and the test section, and the test section and the fan can be optimized separately. The largest gain in pressure loss can be found in simultaneously optimizing for the settling chamber cross section, corners cross section, and diffuser length between these elements.

To avoid the need of boundary layer control in a diffuser (angles < 10 degrees) is a large pressure gain. The corner and settling chamber cross section have to be optimized according to this maximum preferred diffuser angle.

4. What are the design requirements for the ABL wind tunnel?

1. What is the ideal velocity?
2. What are the required dimensions?
3. What are the cost limitations?

The most important aspects of the design requirements are:

- The air velocity of at range of 10 m/s to 20 m/s (wind force 6 to 8). To have a comparable flow turbulence in the test section as in reality the flow around the test objects has to be above the critic Reynolds number (see chapter 2.4). To be above this critic Reynolds number an object in an air velocity of 10 m/s has to be larger than 1.5 cm height and width to have reliable experiments. Lower velocity require even larger object diameters.
- The wind tunnel outer dimensions are limited to $18.6 \text{ m} \times 5.2 \times 4.5 \text{ m}$. An as large as possible test section is preferable, but with a minimum of 1 m height and 1.3 m width.
- Minimum costs are preferable.

5. How can parts of the previous wind tunnels be transformed into an ABL wind tunnel?

1. What are the various options and which parts are being reused?
2. What are the compromises to be made?

Wind tunnel David can be transformed into an ABL wind tunnel with only a new fetch and test section. The centrifugal fan from KU Leuven can also be used in a closed circuit wind tunnel in combination with parts from wind tunnel David. This implementation has a larger cross section. Options with a new fan and new parts have a larger test section and test velocity.

Transformation of David with just a new test section results in a test section cross section and velocity too small for urban experiments. The reuse of the centrifugal fan from Leuven produces an appropriate test section and velocity. The option with no parts reused from former wind tunnel has the most favorable cross section and velocity, though needs a also new fan which is consequently more expensive. An overview of the design options can be found in table 5.8.

6. What are the life cycle cost indications for the ABL wind tunnel, and which costs are most demanding?

The most demanding costs for the designs of an ABL wind tunnel are the construction costs. With 200 operation hours per year will the operation costs only exceed the construction costs after 15 to 20 years. The elaborated results of cost indications can be found in table 5.7.

7. What is the most efficient design for the ABL wind tunnel?

The decision for a certain wind tunnel is a consideration between a wider range of test possibilities and the limitation of a budget. From the variant proposed in chapter 5.2, the most cost efficient design that fulfills the requirements is the combination of the parts from former wind tunnel David with the reuse of the centrifugal fan.

6.2 Conclusions for the ABL wind tunnel

For design of an ABL wind tunnel for the unit BPS at the TU/e, five options have been designed. The basic requirements are:

- velocity: 10 - 20 m/s
- minimum test section dimension: 1 m $\times$ 1.3 m
- minimum fetch length: 8 times test section height
- minimum contraction ratio: 3

Three options use existing parts from former wind tunnels, two use only new parts. The available wind tunnels that can be reused are a closed circuit wind tunnel with an axial fan of 605 W from the Department of Applied Physics, one is an open circuit ABL wind tunnel with an centrifugal fan of 11.5 kW from the KU Leuven.

From the former wind tunnels is the fan power from David too low to fulfill the requirements. The parts however can be reused with the centrifugal fan. This can generate a flow rate around 10 m/s for a test section of 1.5 m $\times$ 1 m. Before deciding on this design the maximum flow rate of the fan without wind tunnel parts has to be measured and added to the performance curve of the fan to an accurate operation point.

Furthermore, two wind tunnel with all newly built parts have been designed for maximum test section cross section. The available room has an L shape, within the boundaries the maximum wind tunnel dimensions are $18.6\text{ m} \times 5.2\text{ m} \times 4.5\text{ m}$. In this room a test section of $11\text{ m} \times 1.5\text{ m} \times 2\text{ m}$ is the largest possible, regardless of an open or closed configuration. A fan power of 27 kW can generate a velocity of 20 m/s in the closed circuit wind tunnel. An open circuit wind tunnel requires a fan of 35 kW to generate a velocity of 20 m/s.

Considering the budget and the fulfillment of the requirements the reuse of parts from wind tunnel David in combination with reuse of the centrifugal is the best option. This wind tunnel can be built for €3300. The construction costs for a completely new wind tunnel, either open or closed, are in the range €20000 to €40000. The running costs, based on 200 running hours per year, are respectively €1200 and €1500 for the closed circuit and the open circuit wind tunnel.

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Appendix A

Wind tunnel Design Instrument (WinDI)

This appendix provides details about the Wind tunnel Design Instrument (WinDI). WinDI has been developed to calculate and optimize an ABL wind tunnel for minimum pressure loss. WinDI can also be used for other types of wind tunnel or duct systems.

WinDI is written in Matlab (and currently not compatible with GNU Octave). The first section explains the usage of the Instrument. The second section provides details on the design and the code.

A.1 How to use WinDI

WinDI is used to support the designing process of a wind tunnel. The Instrument allows to design a wind tunnel by providing the following information:

- fluid properties
- general wind tunnel characteristics,
- the (ordered) list of wind tunnel parts, and
- the relevant part information for every part.

Based on the provided information WinDI calculates:

- the hydraulic diameter at the inlet, outlet and average for each part
- the velocity at the inlet, outlet and average diameter for each part

- the parameters needed for the pressure loss calculation, such as the friction factor, Reynolds number, angle, area ratio, ect.
- the pressure loss coefficient per part,
- the dynamic pressure per part,
- the pressure loss per part,
- the total pressure loss,
- the required power,

The file `example_script.m` is kept up-to-date and can be seen as reference file. When constructing a wind tunnel with WinDI, the example file can be used as starting point.

A.2 Design of WinDI

WinDI uses a structured approach to design a wind tunnel. A graphical representation of the structure of the tool is provided in figure A.1. The main functions are described in the following sections.

Using WinDI

For WinDI to be able to work, Matlab needs to expand it's environment to include all required folders. The script `startup.m` takes care of this, it has to be run before starting to work with the tool.

A.2.1 Settings

The settings are grouped in the function `make_settings`, which can be found at `./process/make_settings.m`. In the settings a number of parameters are set:

1. air density
2. kinematic viscosity
3. fixed values for the pressure loss coefficients (K) of the parts: safetynet, sudden expansion, outlet, inlet shapes, and corner without vanes,
4. possibility to use the average or incoming velocity and cross section for the calculations.

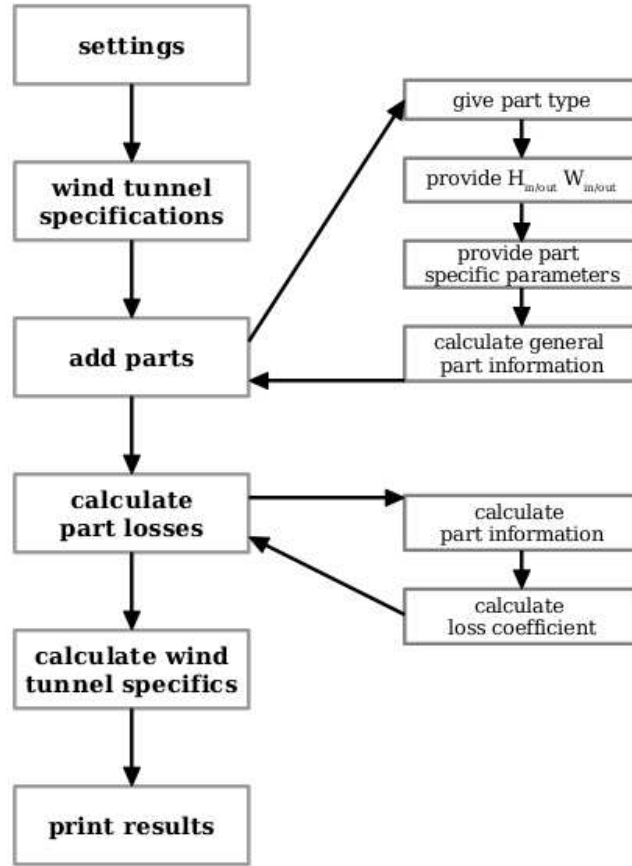


Figure A.1: The schematic structure of WinDI.

5. steps for the calculation of the friction factor. The friction factor is used in the calculations of the diffuser, contraction and straight part,

The friction factor is calculated with brute-force-search (via loops). The speed and accuracy of depends on:

1. the start and stop values of the search (a bigger difference results in longer calculation/search times),
2. the interval value (a smaller value results in a higher accuracy, but also in longer calculation/search times).

A.2.2 Wind tunnel specifications

The wind tunnel specifications are added at the top of the file for each wind tunnel. The following general wind tunnel information has to be provided here:

1. Test section cross section (m^2)
2. Test section velocity (m/s)
3. Fan efficiency (if unknown chose 0.9)
4. Motor efficiency (if unknown chose 0.8)

A.2.3 Adding wind tunnel parts

WinDI supports thirteen part types. Each of these parts require the standard parameters and, depending on the part, also part specific parameters. The standard parameters, that have to be given for every part, are `height_in` `width_in` and `height_out` and `width_out`. The part specific parameters are:

- Straight part
 - `length` (`length`)
- Fetch
 - `length` (`length`)
 - test section height (`height_testsection`) - this height has to correspond with the test section cross section and velocity added earlier in WinDI.
- Honeycomb
 - cell length (`cell_length`)
 - cell diameter (`cell_diameter`)
 - cell wall thickness (`cell_wall_thickness`)
 - material roughness (`material_roughness`)
- Screen
 - cell diameter (`cell_diameter`)
 - wire diameter (`wire_diameter`)
 - wire roughness coefficient (`wire_roughness`) - (usually between 1.0 for smooth new wire and 2.1 for rough wire)
- Corner
 - There are two supported corner types:
 - corners upstream of the test section (`corner`) and a

corner directly downstream the test section
(`corner_downstream_testsection`).

Both have the same parameter:

vane chord (`chord`) - when no vanes are present enter 0, WinDI then uses the pressure loss coefficient for corner without vanes.

- Contraction

shape (`shape`) - enter either `curved` or `straight`.

center length (`center_length`)

- diffuser:

cross section shape (`cross_section_shape`) - enter either `circle` or `rectangle`. - Note: rectangle cross section with a height-width ratio over 1.5 should not use this equation!

center length (`length_center_line`) - only enter centre length or wall length

length of the wall (`length_wall`) - only enter centre length or wall length

- Diffuser with boundary layer control:

center length (`length_center_line`) - only enter centre length or wall length

length of the wall (`length_wall`) - only enter centre length or wall length

This diffuser has to be used when angle > 10 degrees, and has to be combined with boundary layer control (see chapter 4.1.1).

- Inlet

inlet shape (`inlet_shape`) - enter

(`rounded`) or

(`straight`) or

(`duct_with_wall`) or

(`duct`) see chapter 4.1.7

The following parts do not require additional parameters:

- Safetynet

- Sudden expansion
- Outlet

The list of parameters of a part looks as follows:

```
parts = add_part(wind_tunnel, parts, 'partname', 'part type', height_in, ...
                width_in, height_out, width_out, 'parametername-1', parameter 1, ...
                'parametername-2', parameter2);
```

A list comprising of all wind tunnel parts looks as follows:

```
% Wind tunnel information , Part name , Part type , Part sizes , Parameters
add_part(wind_tunnel, parts, 'Straight part' , 'straight_part' , 1.0, 1.0, 1.0, 1.0 , 'length', 1.0);
add_part(wind_tunnel, parts, 'Fetch' , 'fetch' , 1.0, 1.0, 1.1, 1.0 , 'length', 10.0, 'h'
add_part(wind_tunnel, parts, 'Honeycomb' , 'honeycomb' , 1.0, 1.0, 1.0, 1.0 , 'cell_length', 0.3
add_part(wind_tunnel, parts, 'Screen' , 'screen' , 1.0, 1.0, 1.0, 1.0 , 'wire_diameter', 0
add_part(wind_tunnel, parts, 'Corner' , 'corner' , 1.0, 1.0, 1.0, 1.0 , 'chord', 0.5);
add_part(wind_tunnel, parts, 'Corner downstream ts' , 'corner_downstream_ts' , 1.0, 1.0, 1.0, 1.0 , 'chord', 0.5);
add_part(wind_tunnel, parts, 'Straight contractor' , 'contraction' , 2.0, 2.0, 1.0, 1.0 , 'shape', 'straight
add_part(wind_tunnel, parts, 'Rectangular diffuser' , 'diffuser' , 1.0, 1.0, 2.0, 2.0 , 'cross_section_shap
add_part(wind_tunnel, parts, 'Diffuser_BLC' , 'diffuser_blc' , 1.0, 1.0, 2.0, 2.0 , 'length_center_line
add_part(wind_tunnel, parts, 'Inlet' , 'inlet' , 1.0, 1.0, 1.0, 1.0 , 'shape', 'curved');
add_part(wind_tunnel, parts, 'Safety net' , 'safetynet' , 1.0, 1.0, 1.0, 1.0);
add_part(wind_tunnel, parts, 'Sudden expansion' , 'sudden_expansion' , 1.0, 1.0, 1.0, 1.0);
add_part(wind_tunnel, parts, 'Outlet' , 'outlet' , 1.0, 1.0, 1.0, 1.0);
```

All the parts and their parameters are presented in the example script. The parts can be copy-pasted in the order to represent a wind tunnel. The parts have to be provided in the order they appear in the wind tunnel. An ordered list of parts enables WinDI to check if the parts connect correctly.

Geometric check

The wind tunnel validation function checks if the parts outlet sizes match the next parts' inlet size. In the case of a closed circuit wind tunnel, the last parts' outlet size should match the first parts' inlet size. If the sizes do not match (and the parts do not fit), then a warning is printed. WinDI continues the calculations, even if a warning occurs.

A.2.4 Part pressure loss

Here WinDI first calculated the parameters which are required for the calculation for the pressure loss coefficient (K). These parameters can be the friction factor (f), Reynolds number (Re), angle (α), density (β), etc, see chapter 4.1. All parameters that have been calculated can be retained for validation. When the parameters are calculated, WinDI calculates the pressure loss coefficient K per part. With the pressure loss coefficient and the dynamic pressure, WinDI calculated the pressure loss per part.

A.2.5 Calculation of the wind tunnel specifications

The remaining part of the script contains the calculations for the following part specific and the wind tunnel characteristics. The overall wind tunnel output is:

1. the total pressure loss
2. the total pressure loss coefficient ratio
3. the power input
4. the fan power, takes into account the efficiency of the motor and the fan

A.2.6 Summary of the calculated information

After running the script, a table with the part specific and overall output is printed, see figure A.2. To review part specific information the function `parts` (`parts.m`) can be used, see figure A.3.

nr	Part name	K	q	delta p	Kts	%
1	Diffuser transition	0.0116	532.3	6.18	0.0161	0.9
2	Diffuser	0.0486	328.2	15.95	0.0417	2.4
3	corner A large	0.1549	87.6	13.57	0.0355	2.0
4	Straight part	0.0038	87.6	0.33	0.0009	0.0
5	corner B large	0.1549	87.6	13.57	0.0355	2.0
6	Straight part	0.0034	87.6	0.29	0.0008	0.0
7	Screen BLC 1	0.7902	87.6	69.25	0.1809	10.5
8	Diffuser_BLC	0.3656	87.6	32.04	0.0837	4.8
9	Screen BLC 2	0.7933	34.6	27.47	0.0718	4.1
10	Screen 3	0.7997	16.5	13.21	0.0345	2.0
11	Honeycomb	2.5456	16.5	42.06	0.1099	6.4
12	Screen 4	0.7997	16.5	13.21	0.0345	2.0
13	Screen 5	0.7997	16.5	13.21	0.0345	2.0
14	Screen 6	0.7997	16.5	13.21	0.0345	2.0
15	Screen 7	0.7997	16.5	13.21	0.0345	2.0
16	Screen 8	0.7997	16.5	13.21	0.0345	2.0
17	Contraction	0.0100	564.7	5.65	0.0148	0.9
18	Test section	0.5148	382.8	197.07	0.5148	29.8
19	2d contraction	0.0100	453.9	4.54	0.0119	0.7
20	corner C small	0.1614	453.9	73.26	0.1914	11.1
21	Straight part	0.0071	453.9	3.23	0.0084	0.5
22	corner D small	0.1614	453.9	73.26	0.1914	11.1
23	contr transition	0.0100	532.3	5.32	0.0139	0.8
Total				662.34	1.7302	100.0
Wind tunnel information:						

efficiency_fan =		0.8				
efficiency_motor =		0.8				
reserve_factor =		1.1				
power_input =		54643				
power_fan =		85380				

Figure A.2: Overall output WinDI.

name:	'Diffuser'
type:	'diffuser'
crosssection_test_section:	3
velocity_test_section:	25
height_in:	1.8000
width_in:	1.8000
height_out:	2.2000
width_out:	2.8500
diameter_in:	1.8000
diameter_out:	2.4832
diameter_average:	2.1423
crosssection_in:	3.2400
crosssection_out:	6.2700
crosssection_average:	4.5895
velocity_in:	23.1481
velocity_out:	11.9617
velocity_average:	16.3417
cross_section_shape:	'rectangle'
length_center_line:	11.7933
length_wall:	0
radius_in:	1.0155
radius_out:	1.4127
angle_radian:	0.0336
angle_degrees:	1.9278
area_ratio:	1.9352
reynolds_number:	2.3119e+06
friction_factor:	0.0101
loss_friction:	0.0276
loss_e:	0.0899
loss_expansion:	0.0210
loss_coefficient:	0.0486
pressure_loss_dynamic:	328.2000
pressure_drop:	15.9480
loss_coefficient_ratio:	0.0417
loss_percentage:	2.4078

Figure A.3: Part specific calculated parameters for a diffuser, highlighted parameters are given as input.

Appendix B

Various equations for different wind tunnel parts

To validate the equations to be used in WinDI, equations per part from various references have been compared and plotted (see also chapter 4). For the calculation of the pressure loss in diffusers, corners and contractions, different equations are used. Especially the diffuser gives a wide range of used equations. An overview of the equations and their interpretation can be found in this appendix.

B.1 Diffuser

From: Henry (1944)

One of the earlier guidelines for pressure loss coefficients uses a graphs combined with an equation to calculate the pressure loss coefficient in a diffuser.

The pressure loss coefficient is

$$K = K_2 \left(1 - \left(\frac{A_{in}}{A_{out}} \right)^2 \right) \quad (\text{B.1})$$

Where K_2 = plotted in graph B.1

From: Low speed wind tunnel testing Pope and Harper (1966)

In divergent parts both wall friction and expansion losses occur. They are both added up in equation B.2.

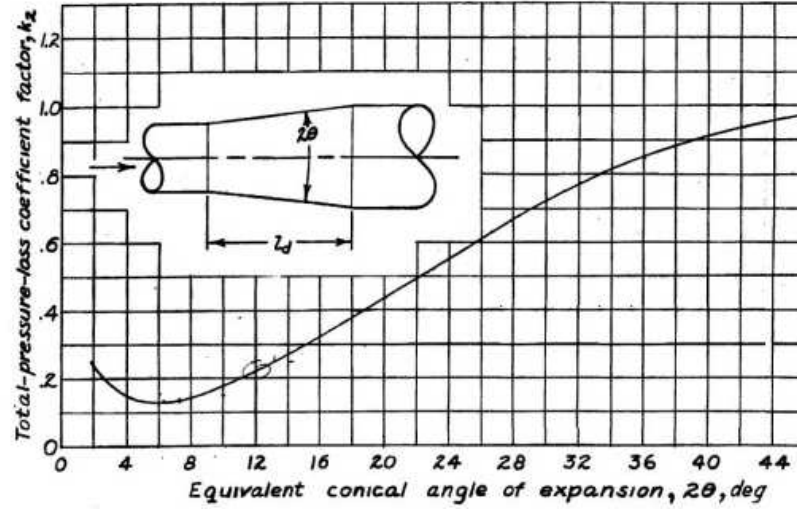


Figure B.1: Plots of the diffuser equation by Henry (1944)

$$K = \left(\left(\frac{f}{8 \tan\left(\frac{\alpha}{2}\right)} \right) + 0.6 \tan\left(\frac{\alpha}{2}\right) \right) \left(1 - \left(\frac{D_1^4}{D_2^4} \right) \right) \quad (\text{B.2})$$

where α = angle between opposite walls ($< 7^\circ$)
 D_1 = smaller diameter
 D_2 = large diameter
 f = friction factor

This equation requires no specifications for the cross section shape.

A small angle gives a smaller loss. In general the expansion losses are larger than the friction losses. However at very small angles the skin friction is very large and increases the pressure loss. This minimum loss factor, or the “valley” of the graph, can be found by equaling the derivative to zero.

From: Design, construction and evaluation of a subsonic wind tunnel Vlainac (1970)

M. Vlainac from Massachusetts Institute of Technology uses an equation based on the inlet and outlet diameters only.

$$K = (3.14f + 0.0262) \left(1 - \frac{D_i^4}{D_o^4} \right) \quad (\text{B.3})$$

From: NASA Aerodynamic design guidelines and computer program for estimation of subsonic wind tunnel performance tunnel testing Eckert et al. (1976)

Eckert is the first to calculate the diffuser pressure loss by adding the losses due to friction and expansion. Friction losses also occur in straight parts, but the expansion losses causes the larger pressure losses in a diffuser.

The equation from W. Eckert is the first which introduces the use of the equivalent cone angle to describe the angle between the walls. Its calculated by taking the area of the cross section (in and out) and changing those angles to virtual circle. The virtual circle should have the same area and therefore the same flow rate that passes the cross section. When using the equivalent conical, the diameter will slightly increase and consequently the angle will be slightly larger. Eckert gives a small allowances for diffuser with a square cross section without rounded corners, their angle should “probably be 0.5° to 1° smaller than the conical angle”.

Assumptions made for this equation are that the fan re-energizes the boundary layer and gives an uniform flow. And up- and downstream parts have no effect on the losses that occur in the diffuser. Both assumptions may not be appropriate.

The following equation, usable for square and circular cross sections, is used by W. Eckert to obtain the pressure loss coefficient:

$$K = K_{exp} + K_f \quad (B.4)$$

with:

$$K_f = \frac{f}{8 \sin(\alpha)} \left(1 - \frac{1}{A_r^2} \right) \quad (B.5a)$$

$$\alpha = \frac{\tan^{-1}(Rh_{out} - Rh_{in})}{L} \quad (B.5b)$$

$$Rh = \sqrt{\frac{A}{\pi}} \quad (B.5c)$$

For a square cross section diffuser the expressions for K_{exp} are:
for $0^\circ < 2\alpha < 3^\circ$:

$$K_{exp} = 9.62274 \cdot 10^{-2} - 4.15164 \cdot 10^{-3} \alpha \quad (B.6)$$

for $3^\circ \leq 2\alpha \leq 10^\circ$:

$$\begin{aligned} K_{exp} = & 1.22156 \cdot 10^{-1} - 2.29480 \cdot 10^{-2} \times 2\alpha \\ & + 5.50704 \cdot 10^{-3} \times (2\alpha)^2 - 4.08644 \cdot 10^{-4} \times (2\alpha)^3 \\ & - 3.84056 \cdot 10^{-5} \times (2\alpha)^4 + 8.74969 \cdot 10^{-6} \times (2\alpha)^5 \\ & - 3.65217 \cdot 10^{-7} \times (2\alpha)^6 \end{aligned} \quad (B.7)$$

for $2\alpha > 10^\circ$:

$$K_{exp} = -0.01321685 + 0.058663\alpha \quad (\text{B.8})$$

The equation for a circle or square cross section are quite similar, the rectangular cross section however gives for most of the usual angle a very much larger loss coefficient. When these are plotted they give the graph as shown in figure B.2.

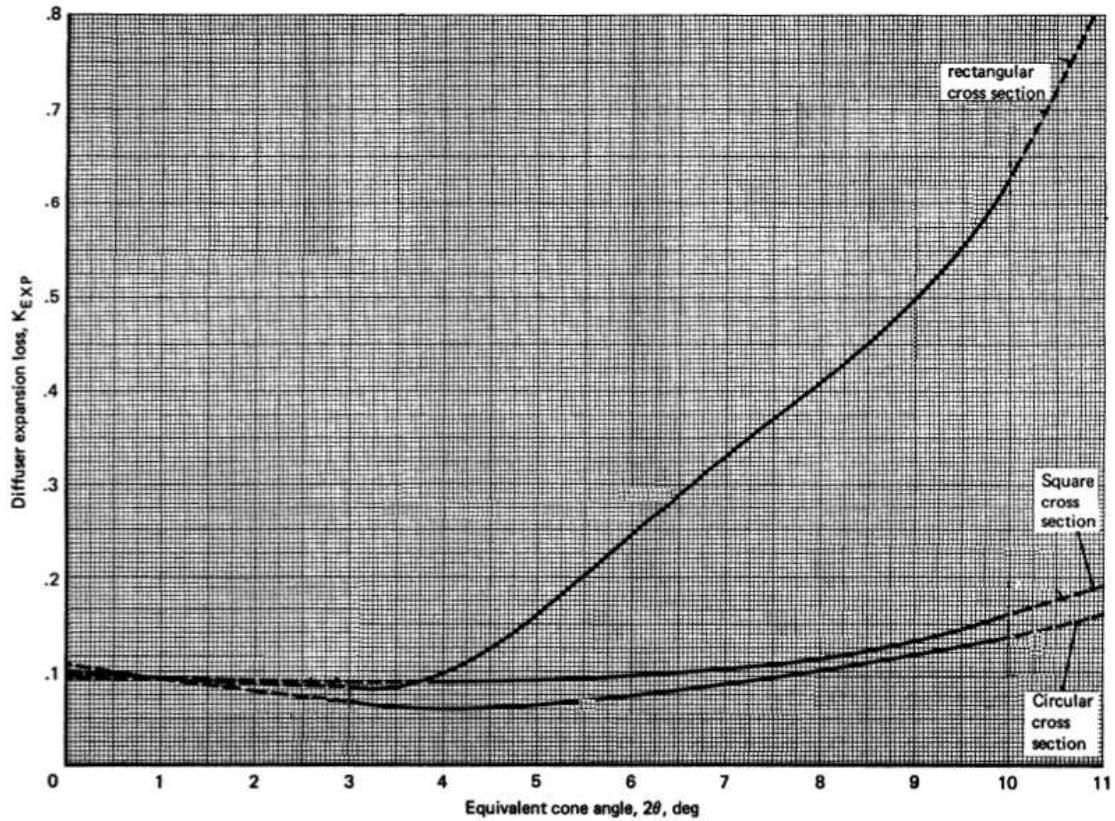


Figure B.2: Plots of the diffuser equation for a circular, square and rectangular cross section by Eckert et al. (1976)

From: Wood practiced guide to fan engineering Daly (1978)

B. Daly determines the pressure loss coefficient of a diffuser via a graph.

The variables in figure B.3(a) are the area ratio in combination with the part length divided by the inlet diameter of the diffuser. This graph is made for conical diffusers, however it can also be used for rectangular cross sections. In the latter case the hydraulic diameter has to be used.

A second graph (figure B.3(b)) from the same guidelines uses different inlet variables to determine the pressure loss coefficient. Here the inverse area ratio

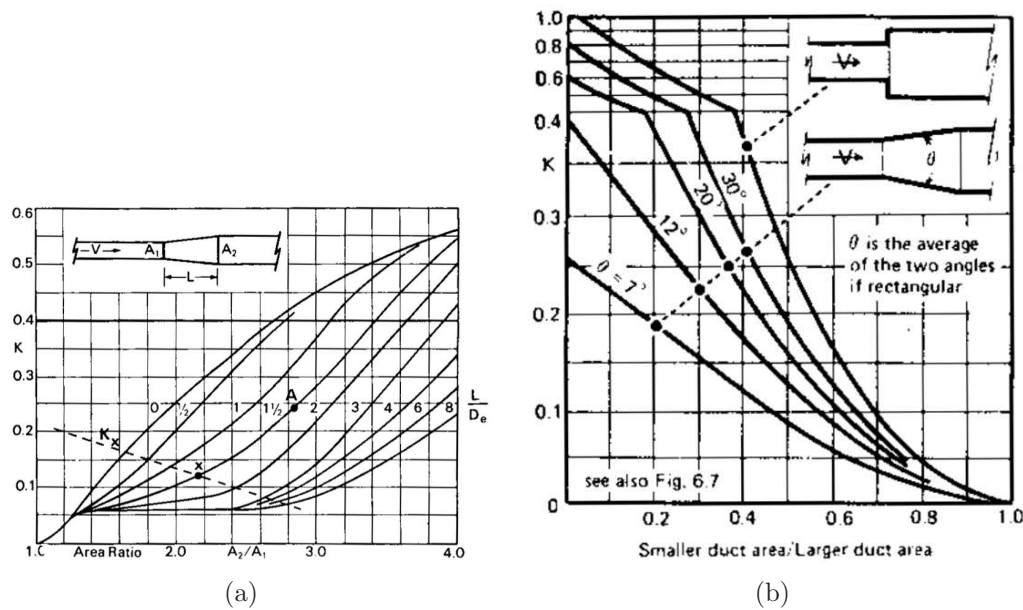


Figure B.3: Graphs to determine the pressure loss coefficient for diffusers from Daly (1978).

is used in combination with the angle between opposite walls. Whichever graph is more convenient can be used, both give similar values for the pressure loss coefficient K .

From: Low speed wind tunnel testing Barlow et al. (1999)

The calculation of the diffuser pressure loss is quite different from the first edition of this book. It uses as input value the friction loss (variable only by Reynolds number), angle between center line and the area ratio (instead of inlet and outlet diameter). The new equation includes the separation of losses due to expansion and friction as done before by Eckert et al. (1976). Furthermore Barlow introduces value K_e , which determines a value based on the angle of the diffuser. The equation assumes skin friction coefficient and density are constant over the length as well as over the cross section. Both are generalizations as the skin fraction varies with the Reynolds number and therefore with the diameter.

To obtain the expansion coefficient there are (still) only experimental data to rely on. Eckert et al. (1976) and Mehta (1977) and Mehta and Bradshaw (1979) are references used for this equation.

The equation for the friction losses and the expansion losses may be used for square and conical cross section diffusers.

$$K = K_f + K_{ex} \quad (\text{B.9})$$

$$K_f = \left(1 - \frac{1}{Ar^2}\right) \frac{f}{8\sin(\alpha)} \quad (\text{B.10})$$

$$K_{ex} = K_e \left(\frac{Ar - 1}{Ar}\right)^2 \quad (\text{B.11})$$

$$(\text{B.12})$$

f is based on entry Re number

$$K_e, \text{ circle}, \alpha < 1.5 = 0.1033 - 0.02389\alpha \quad (\text{B.13a})$$

$$K_e, \text{ circle}, 1.5 < \alpha < 5 = 0.1709 - 0.117\alpha + 0.0326\alpha^2 + 0.001078\alpha^3 - 0.0009076\alpha^4 - 0.0000133\alpha^5 + 0.00001345\alpha^6 \quad (\text{B.13b})$$

$$K_e, \text{ circle}, \alpha > 5 = -0.09661 + 0.04672\alpha \quad (\text{B.13c})$$

$$K_e, \text{ square}, \alpha < 1.5 = 0.09623 - 0.004152\alpha \quad (\text{B.14a})$$

$$K_e, \text{ square}, 1.5 < \alpha < 5 = 0.1222 - 0.04152\alpha + 0.02203\alpha^2 + 0.003269\alpha^3 - 0.0006145\alpha^4 - 0.00002800\alpha^5 + 0.00002337\alpha^6 \quad (\text{B.14b})$$

$$K_e, \text{ square}, \alpha > 5 = -0.01322 + 0.05866\alpha \quad (\text{B.14c})$$

$$\alpha = \tan^{-1} \left(\frac{R_{large} - R_{small}}{L} \right) \quad (\text{B.15})$$

with

$$R = \sqrt{\frac{A}{\pi}} \quad (\text{B.16})$$

B.2 Corner

The pressure loss coefficient calculation of the corner by Pope and Harper (1966) and also the revised version by Barlow et al. (1999) give:

$$K = 0.1 + \frac{4.55}{(\log Re)^{2.58}} \quad (\text{B.17})$$

Pope and Harper (1966) limits the maximum of this equation to $k = 0.15$ for Reynolds Number of 500.000 and higher.

Eckert et al. (1976) gives the slightly different equation of:

$$K = \frac{K_{tv}}{3} 2 + \frac{\log(Re_{ref})^{2.58}}{\log(Re)} \quad (\text{B.18})$$

where K_{tv} at $90^\circ = 0.15$.

Eckert et al. (1976) also specify an equation for diffusing corners:

$$K = (0.3 + 0.006 (2\alpha 21.5) u(2\alpha 21.5)) \left(\frac{(Ar1)^2}{Ar} + \frac{2}{3} K_{tv} \right) \quad (\text{B.19})$$

B.3 Contraction

A generally used equation for the pressure loss coefficient of a contraction is stated by Barlow et al. (1999) and Eckert et al. (1976).

$$K = 0.32 f \frac{L}{D} \quad (\text{B.20})$$

The equation does not specify whether this equation is valid for curved or straight walled contractions. However, it only takes the friction losses into account, and therefore cannot be used when separation occurs. In the example calculation in Barlow et al. (1999) the equation is used for a curved wall contraction. Also straight walled contractions in other references have larger pressure loss coefficients.

Munson et al. (1998) and Cengel et al. (2012) specify pressure loss coefficients for straight wall contractions:

- $K = 0.07$ for 60 degrees straight wall contraction.
- $K = 0.02$ for 30 degrees straight wall contraction.

Daly (1978) specifies:

- $K = 0.07$ for 60 degrees straight wall contraction.
- $K = 0.04$ for 45 degrees straight wall contraction.
- $K = 0.02$ for 30 degrees straight wall contraction.

Bloomer (2000) gives a fixed value for the pressure loss coefficient of $K = 0.05$ for a straight walled contraction.

Rennels and Hudson (2012) calculate the pressure loss coefficient via the friction loss coefficient K_{fr} (see diffuser equation) and the contraction loss coefficient by separation.

$$K = K_{fr} + K_{contraction} \quad (\text{B.21})$$

$$K_{contraction} = 0.0696 \sin(\alpha) (1 - \beta^5) \lambda^2 + (\lambda - 1)^2 \quad (\text{B.22a})$$

$$\lambda = \alpha + 0.622 \left(\frac{\alpha}{90} \right)^{\frac{4}{5}} (1 - 0.215\beta^2 - 0.785\beta^5) \quad (\text{B.22b})$$

$$\beta = \frac{D_{out}}{D_{in}} \quad (\text{B.22c})$$

Appendix C

Wind tunnel fetch dimensions

Table C.1: Wind tunnel fetch dimensions.

Name	Place	Length	Width	Hight in	Hight max	Length / hight max
Concordia	Montreal. Ca	12.2	1.8	1.4	1.8	6.78
Ruhr University	Bochum. De	9.4	1.8	1.6	1.6	5.88
EVVT	Colorado. US	18.3	3.7	2.1	2.1	8.71
Blasius	Hamburg. De	11.5	1.5	1	1	11.5
BRE	Garston. UK	8	2	1	1	8
	Davis. US	14.6	1.2	1.7	1.7	8.59
Von Karman Institute. WG	St genesiu-Rode. Be	7	1.3	1	1	7
Force	Brondby. Dk	20.7	2.6	1.8	2.3	9
Windtech	Singapore. Sg	19	2.8	1.8	2.3	8.26
Riam	Kyushu. Jp	13.5	1.5	1.2	1.2	11.3
	"Porto Alegre, Br	9.3	1.2	0.9	0.9	10.3
Honda	London. UK	9	3	1.5	1.5	6
	Bristol. UK	6.5	2	1	1	6.5
Von Karman Institute. LIB	St genesiu-Rode. Be	20	3	2	2	10
St. Anthony Falls Lab	Minnesota. US	17	1.7	1.8	1.8	9.44
		Average				7.9

Appendix D

Validation of WinDI

The validation of WinDI is based on seven references. Four are calculations of the total wind tunnel power by examples from a textbook and reports, and three are the operational power of existing wind tunnels. The calculations are based on the literature examples: (1) Barlow et al. (1999), (2) the Nasa wind tunnel at Ames Research Center Eckert et al. (1976), (3) the Indian Institute of Science open wind tunnel at Bangalore Eckert et al. (1976), and (4) a TU/e Atmospheric Boundary Layer Wind Tunnel Gousseau (2009a). The actual wind tunnels to compare WinDI's outcomes with are the Atmospheric Boundary Layer Wind Tunnel EMPA in Zurich Empa (2011), and two wind tunnels currently present at the TU/e (“David” and “Leuven”).

The comparing results and underlying assumptions of these seven validation cases can be found in this appendix. The exact dimensions used can be found in the references as well as in the WinDI code.

Example wind tunnel from low speed wind tunnel testing Barlow et al. (1999)

This wind tunnel enables the following test section:

Cross section 2.32 m²
Velocity 44.7 m/s

The made assumption for this calculation are:

Diffuser cross section circular
Chord length small corner 0.1 m
Chord length large corner 0.4 m

nr	Part name	WinDI				Barlow
		K	q	Δp	K_{ts}	K_{ts}
1	Straight part TS	0.0091	1223.9	11.17	0.0091	0.0093
2	Diffuser	0.0304	1223.9	37.22	0.0304	0.0391
3	Corner small C	0.1365	412.6	56.33	0.0460	0.0460
4	Straight part	0.0075	412.6	3.11	0.0025	0.0026
5	Corner small D	0.1365	412.6	56.33	0.0460	0.0460
6	Straight part	0.0058	412.6	2.39	0.0020	0.0020
7	Diffuser	0.0515	412.6	21.26	0.0174	0.0160
8	Corner large A	0.1400	76.5	10.71	0.0087	0.0087
9	Corner large B	0.1400	76.5	10.71	0.0087	0.0087
10	Straight part	0.0031	76.5	0.23	0.0002	0.0002
11	Contraction	0.0052	1223.9	6.42	0.0052	0.0048
Total				215.89	0.1764	0.1834

Table D.1: Validation case Barlow

Nasa wind tunnel at Ames Research Center, by Eckert et al. (1976)

This wind tunnel enables the following test section:

Cross section TS 6.5 m²
 Velocity TS 133.0 m/s

The made assumptions for this calculation are:

contraction shape to be curved
 Roughness coefficient screens 1.3

nr	Part name	WinDI				Eckert
		K	q	Δp	K_{ts}	K_{ts}
1	Straight part	0.0056	54.1	0.30	0.0000	0.00003
2	Contraction	0.0135	10846.2	146.89	0.0136	0.01213
3	Test section	0.0257	10834.5	278.69	0.0257	0.01322
4	Screen downstr ts	0.0024	10567.2	25.69	0.0024	0.00244
5	Diffuser	0.0009	10567.2	9.56	0.0009	0.07838
6	Diffuser	0.0725	10238.5	741.92	0.0685	0.00799
7	Small corner 1	0.1489	591.0	87.99	0.0081	0.00076
8	Diffuser	0.0150	591.0	8.89	0.0008	0.00448
9	Small corner 2	0.1518	325.7	49.45	0.0046	0.00066
10	Diffuser	0.0224	325.7	7.30	0.0007	0.00001
11	Diffuser	0.0211	128.9	2.72	0.0003	0.00071
12	Contractor	0.0017	82.2	0.14	0.0000	0.00001
13	Diffuser	0.0030	82.2	0.25	0.0000	0.00002
14	Diffuser	0.0056	70.9	0.40	0.0000	0.00003
15	Straight part	0.0014	54.1	0.08	0.0000	0.00001
16	Large corner 3	0.1623	54.1	8.79	0.0008	0.00079
17	Straight part	0.0049	54.1	0.27	0.0000	0.00002
18	Large corner 4	0.1725	54.1	9.34	0.0009	0.00084
Total				1378.66	0.1272	0.12387

Table D.2: Validation case Nasa Ames by Eckert

The Indian Institute of Science open wind tunnel at Bangalore by Eckert et al. (1976)

This wind tunnel enables the following test section:

Cross section TS 11.0 m²
 Velocity TS 96.32 m/s

The made assumptions for this calculation are:

Roughness coefficient screens 1.3

nr	Part name	WinDI K	q	Δp	K_{ts}	Bangalore K_{ts}
1	Inlet	0.5000	31.1	15.54	0.0027	-
2	Screen 1	0.2449	31.1	7.61	0.0013	0.0012
3	Screen 2	0.2449	31.1	7.61	0.0013	0.0012
4	Screen 3	0.2449	31.1	7.61	0.0013	0.0012
5	Screen 4	0.2449	31.1	7.61	0.0013	0.0012
6	Screen 5	0.2449	31.1	7.61	0.0013	0.0012
7	Contraction	0.0100	1591.5	15.92	0.0028	0.0031
8	Contraction	0.0038	4590.4	17.66	0.0031	0.0036
9	Test section	0.0371	5682.5	210.79	0.0371	0.0124
10	Diffuser	0.0544	3954.3	215.11	0.0379	0.0695
11	Contraction	0.0015	1212.4	1.87	0.0003	0.0002
12	Diffuser	0.0188	1212.4	22.83	0.0040	0.0129
13	Corner	1.0000	187.6	187.59	0.0330	0.0178
14	Outlet	0.9230	128.4	128.37	0.0209	0.0199
Total				853.75	0.1485	0.1464

Table D.3: Validation case Bangalore by Eckert

The TU/e Atmospheric Boundary Layer Wind Tunnel by Gousseau (2009a)

This wind tunnel enables the following test section:

Cross section TS 3.96 m²
 Velocity TS 20.0 m/s

For this design no assumptions had to be made.

The difference in the output is due to the different used diffuser equation. Gousseau uses the equations as described in Barlow et al. (1999), where the incorrect diffuser equation is printed.

nr	Part name	WinDI				Gousseau
		K	q	Δp	K_{ts}	K_{ts}
1	Test section	0.4556	245.0	111.63	0.4556	0.456
2	Small corner A	0.1609	245.0	39.42	0.1609	0.161
3	Straight part	0.0206	245.0	5.04	0.0206	0.021
4	Safety net	0.2000	245.0	49.00	0.2000	0.200
5	Small corner B	0.1609	245.0	39.42	0.1609	0.161
6	Diffuser	0.0789	199.4	15.74	0.0642	0.256
7	Straight part	0.0140	25.6	0.36	0.0015	0.001
8	Large corner C	0.1538	25.6	3.94	0.0161	0.016
9	Straight part	0.0127	25.6	0.33	0.0013	0.001
10	Large corner D	0.1538	25.6	3.94	0.0161	0.016
11	Honeycomb	0.4868	25.6	12.46	0.0509	0.051
12	Screen 1	1.0826	25.6	27.72	0.1131	0.113
13	Screen 2	1.1091	25.6	28.39	0.1159	0.116
14	Screen 3	1.2073	25.6	30.91	0.1262	0.125
15	Contraction	0.0054	310.1	1.67	0.0068	0.007
Total				369.97	1.5101	1.701

Table D.4: Validation case by Gousseau

The Atmospheric Boundary Layer Wind Tunnel EMPA in Zurich by Empa (2011)

This wind tunnel enables the following test section:

Cross section TS 2.5 m²
 Velocity TS 25.0 m/s

The made assumptions for this calculation are:

Roughness coefficient screens 1.3 (-)
 Honeycomb cell diameter 0.0015 m
 Honeycomb material roughness 0.00025 m

nr	Part name	WinDI				
		K	q	Δp	K_{ts}	%
1	Diffuser transition	0.0116	532.3	6.18	0.0161	0.9
2	Diffuser	0.0486	328.2	15.95	0.0417	2.4
3	Corner A large	0.1549	87.6	13.57	0.0355	2.0
4	Straight part	0.0038	87.6	0.33	0.0009	0.0
5	Corner B large	0.1549	87.6	13.57	0.0355	2.0
6	Straight part	0.0034	87.6	0.29	0.0008	0.0
7	Screen BLC 1	0.7902	87.6	69.25	0.1809	10.5
8	Diffuser_BLC	0.3656	87.6	32.04	0.0837	4.8
9	Screen BLC 2	0.7933	34.6	27.47	0.0718	4.1
10	Screen 3	0.7997	16.5	13.21	0.0345	2.0
11	Honeycomb	2.5456	16.5	42.06	0.1099	6.4
12	Screen 4	0.7997	16.5	13.21	0.0345	2.0
13	Screen 5	0.7997	16.5	13.21	0.0345	2.0
14	Screen 6	0.7997	16.5	13.21	0.0345	2.0
15	Screen 7	0.7997	16.5	13.21	0.0345	2.0
16	Screen 8	0.7997	16.5	13.21	0.0345	2.0
17	Contraction	0.0100	564.7	5.65	0.0148	0.9
18	Test section	0.5148	382.8	197.07	0.5148	29.8
19	2d contraction	0.0100	453.9	4.54	0.0119	0.7
20	Corner C small	0.1614	453.9	73.26	0.1914	11.1
21	Straight part	0.0071	453.9	3.23	0.0084	0.5
22	Corner D small	0.1614	453.9	73.26	0.1914	11.1
23	Contr transition	0.0100	532.3	5.32	0.0139	0.8
Total				662.34	0.1730	100

Calculated pressure loss WinDI 662 Pa
 Calculated pressure loss Empa 675 Pa
 Calculated Power (WinDI) 71.1 kW
 Calculated Power (Empa) 71.5 kW
 Installed power (Empa) 110 kW

Table D.5: Validation case Empa

Wind tunnel David from the Department of Applied Physics of the TU/e

This wind tunnel enables the following test section:

Cross section TS 0.2 m²
Velocity TS 15.0 m/s

The made assumptions for this calculation are:

Large settling chamber contains of 1 honeycomb and 3 screens

Wire in the screens is the wire present in VRT1.10

3 screens are used as BLC in the wide angle diffuser

length large settling chamber 0.455 m

Wire diameter 0.00048 m

Cell diameter 0.0010 m

Roughness coefficient 1.5

Large honeycomb, length cell 0.3 m

Large honeycomb, Cell diameter 0.013 m

Large honeycomb, Wall diameter 0.004 m

Honeycomb roughness 8×10^{-5} m

nr	Part name	WinDI				
		K	q	Δp	K_{ts}	%
1	Diffuser small angle	0.0742	27.4	2.03	0.0147	2.1
2	Straight part	0.0032	3.2	0.01	0.0001	0.0
3	Honeycomb	1.2502	3.2	3.96	0.0287	4.0
4	Corner large C	0.1795	3.2	0.57	0.0041	0.6
5	Diffuser corner C	0.0175	3.2	0.06	0.0004	0.1
6	Corner large D	0.1827	2.3	0.41	0.0030	0.4
7	Diffuser corner D	0.0163	2.3	0.04	0.0003	0.0
8	BLC Screen 1	8.7344	2.3	19.78	0.1436	20.2
9	BLC Screen 2	8.8689	0.8	7.52	0.0546	7.7
10	BLC Screen 3	8.9525	0.4	3.81	0.0276	3.9
11	Diffuser wide angle	0.5057	1.7	0.86	0.0062	0.9
12	Straight part Settling chamber	0.0038	0.2	0.00	0.0000	0.0
13	Honeycomb	9.5613	0.2	2.11	0.0153	2.1
14	Screen 1	9.0198	0.2	1.99	0.0144	2.0
15	Screen 2	9.0198	0.2	1.99	0.0144	2.0
16	Screen 3	9.0198	0.2	1.99	0.0144	2.0
17	Straight contraction	0.0111	137.8	1.53	0.0111	1.5
18	Test section	0.2346	137.8	32.33	0.2346	32.9
19	Diffuser	0.0423	88.2	3.73	0.0271	3.8
20	Corner small A	0.1788	27.2	4.87	0.0353	5.0
21	Diffuser corner A	0.0178	27.2	0.48	0.0035	0.5
22	Straight part	0.0184	21.2	0.39	0.0028	0.4
23	Corner small B	0.1643	21.2	3.48	0.0253	3.5
24	Diffuser corner B	0.0171	21.2	0.36	0.0026	0.4
25	Safety net	0.2000	18.1	3.62	0.0263	3.7
26	Contractor	0.0100	27.4	0.27	0.0020	0.3
Total				98.18	0.7124	100

Calculated Power (WinDI) 579 W
Installed power (David) 605 W

Table D.6: Validation case David

Previous open ABL wind tunnel from the KU Leuven with centrifugal fan

This wind tunnel enables the following test section:

Cross section TS 0.36 m²
 Velocity TS 16 m/s

The made assumptions for this calculation are:

Settling chamber contains of 1 honeycomb and 2 screens

1 screens is used upstream of the honeycomb

adjustable ceiling connects the contraction gradually with the test section

Wire diameter 0.0005 m
 Cell diameter 0.0015 m
 Roughness coefficient 1.5
 Honeycomb, length cell 0.25 m
 Honeycomb, Cell diameter 0.013 m
 Honeycomb, Wall diameter 0.002 m
 Honeycomb roughness 8×10^{-5} m

nr	Part name	WinDI				
		K	q	Δp	K_{ts}	%
1	Inlet	0.8000	69.8	55.82	0.3651	4.8
2	Screen inlet fan	0.0712	69.8	4.97	0.0325	0.4
3	straight part	0.0027	69.8	0.19	0.0012	0.0
4	contraction	0.0552	292.9	16.18	0.1058	1.4
5	flexible part	0.4004	110.8	44.35	0.2900	3.8
6	Straight part	0.0272	110.8	3.02	0.0197	0.3
7	Diffuser BLC	0.3183	110.8	35.25	0.2306	3.0
8	Screen	2.4142	110.8	267.37	1.7486	22.8
9	Honeycomb	9.4743	17.6	166.94	1.0918	14.2
10	Screen	2.4764	17.6	43.63	0.2854	3.7
11	Screen	10.2337	17.6	180.32	1.1793	15.4
12	Curved contraction	0.0111	391.4	4.33	0.0283	0.4
13	Fetch	1.2714	152.9	194.40	1.2714	16.6
14	Straight part	0.0290	152.9	4.43	0.0290	0.4
15	Outlet	1.0000	152.9	152.90	1.0000	13.0
Total				1174.11	7.687	100

Calculated Power (WinDI) 11.4 kW
 Installed power (Leuven) 11.5 kW

Table D.7: Validation case Leuven

Appendix E

Drawings from reusable wind tunnels at the TU/e

E.0.1 Closed circuit wind tunnel David **from the Department of Applied Physics**

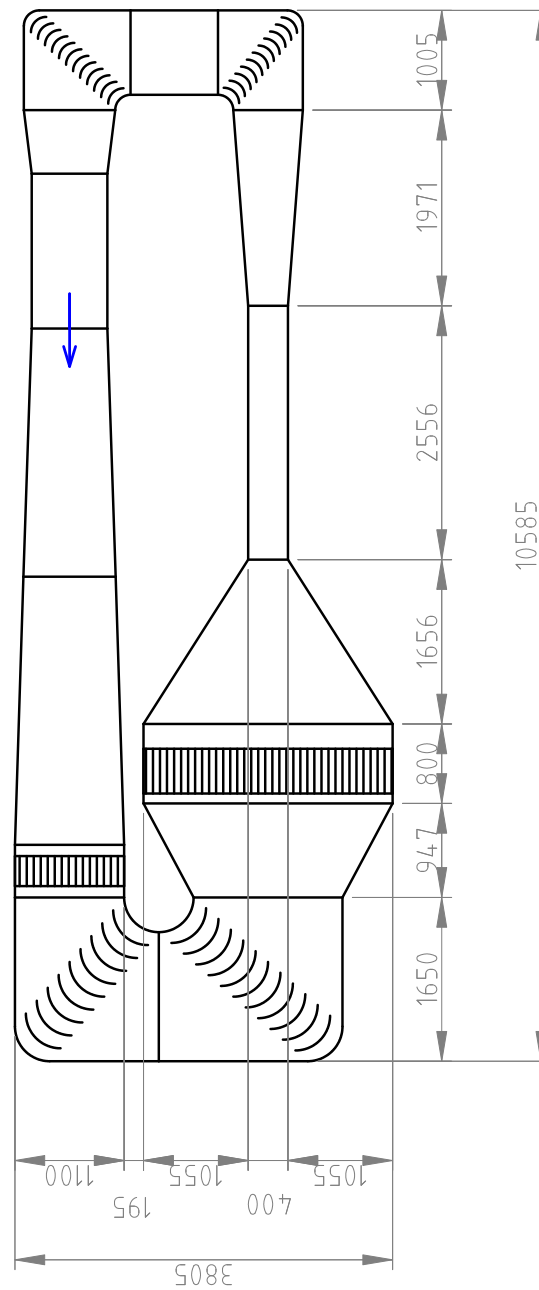


Figure E.1: Former closed circuit wind tunnel David from the Applied Physics Department, dimensions in mm.

E.0.2 Open circuit ABL wind tunnel from the KU Leuven

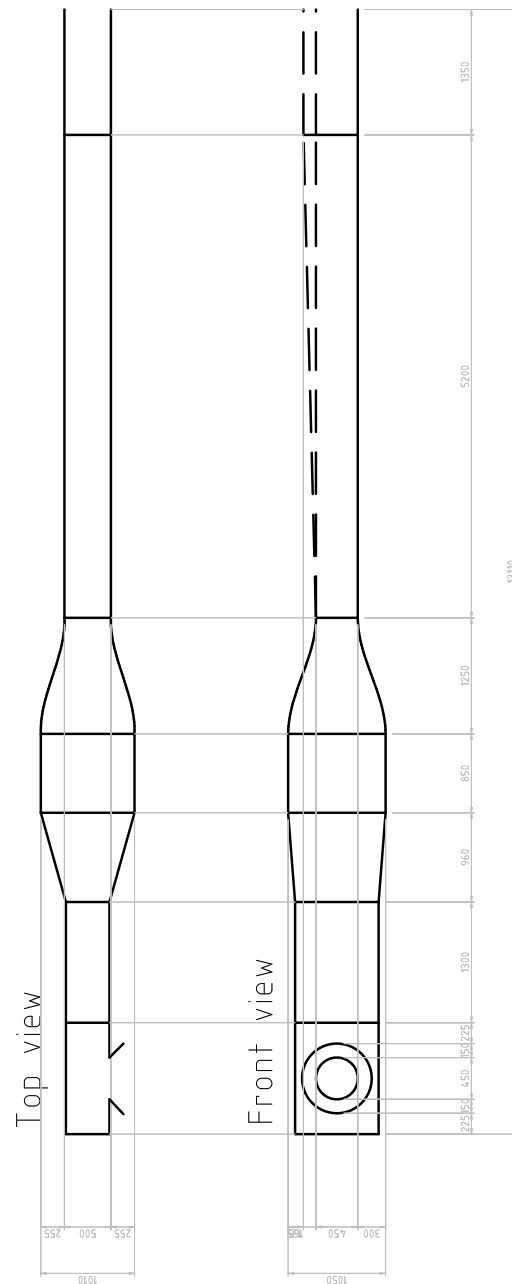


Figure E.2: Former open circuit ABL wind tunnel from the KU Leuven, dimensions in mm.

Appendix F

Fan information and performance curves

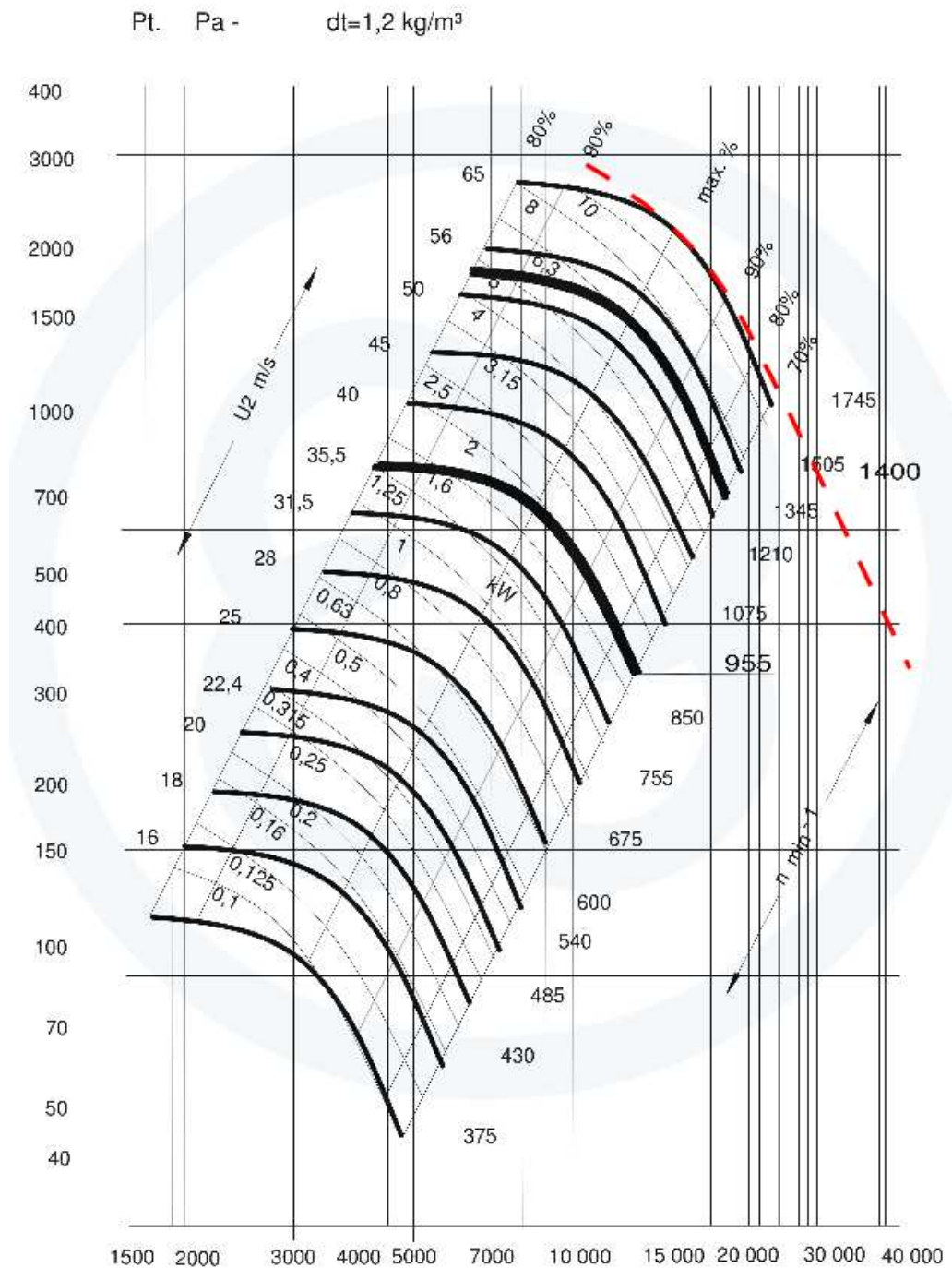


Figure F.1: Fan performance curve from the centrifugal fan from KU Leuven. The installed motor has a maximum power of 11.5 kW in combination with 1745 rotation / min. In red the approximate performance curve of the existing centrifugal fan at the TU/e.



Rucon Systemair BV
van Leeuwenhoekstraat 2
3846 CB Harderwijk

Telephone	+31 (0)341 439100
Fax	+31 (0)341 439190
E-Mail	martin.verweij@ruconsystemair.nl
web	www.systemair.nl
Item	1
Reference	EFC 2004/H (G)
Quantity	1

Date	12-4-2013
Offer No.	-
Customer	-
Contact	-
Project	Untitled Project

Belt Driven Axial Fan

Temperature	15°C
Altitude	0 m
Density	1,226 kg/m ³
Capacity	216.000 m ³ /h
Pressure	230 Pa Static Suction

Fan model	EFC 2004/H (G)
Capacity	216.000 m ³ /h (100%)
Static pressure	230 Pa @ 1,226 kg/m ³
Dynamic pressure	224 Pa @ 1,226 kg/m ³
Total pressure	454 Pa @ 1,226 kg/m ³
Total pressure	454 Pa @ 1,226 kg/m ³
Velocity	19,10 m/s
Impeller speed	574 rpm
Pitch angle	28 °
Absorbed power	33,5 kW @ 1,226 kg/m ³
Absorbed power	33,5 kW @ 1,226 kg/m ³
Service Factor	22 %
Power installed	45,0 kW (D280S 6 pole)
Total Efficiency	81 %
Static Efficiency	41 %
Arrangement	1
Class	-
Maximum speed	950 rpm
Moment of inertia	27,5 (J) kgm ²
Weight	900 kg
Axial thrust	2127 N
Discharge thrust	-
Starting time	0,7 s
Sound power level	111 dB Ambient (Density=1,226 kg/m ³)
Sound pressure level	93 dB Ambient (Density=1,226 kg/m ³ - Distance=1m)
Sound pressure level	87 dBA Ambient (Density=1,226 kg/m ³ - Distance=1m)
Frequency	63 125 250 500 1k 2k 4k 8k Hz
Sound power in-duct	94 98 111 104 102 95 90 82 dB inlet
Sound power in-duct	98 102 115 108 106 99 94 86 dB outlet

12-4-2013 17:00:22
V:3.0.19.0 D: T - 7 : C - 12

Figure F.2: Fan information from Rucon EFC 2004/H(G) axial fan, with external motor and changeable frequency

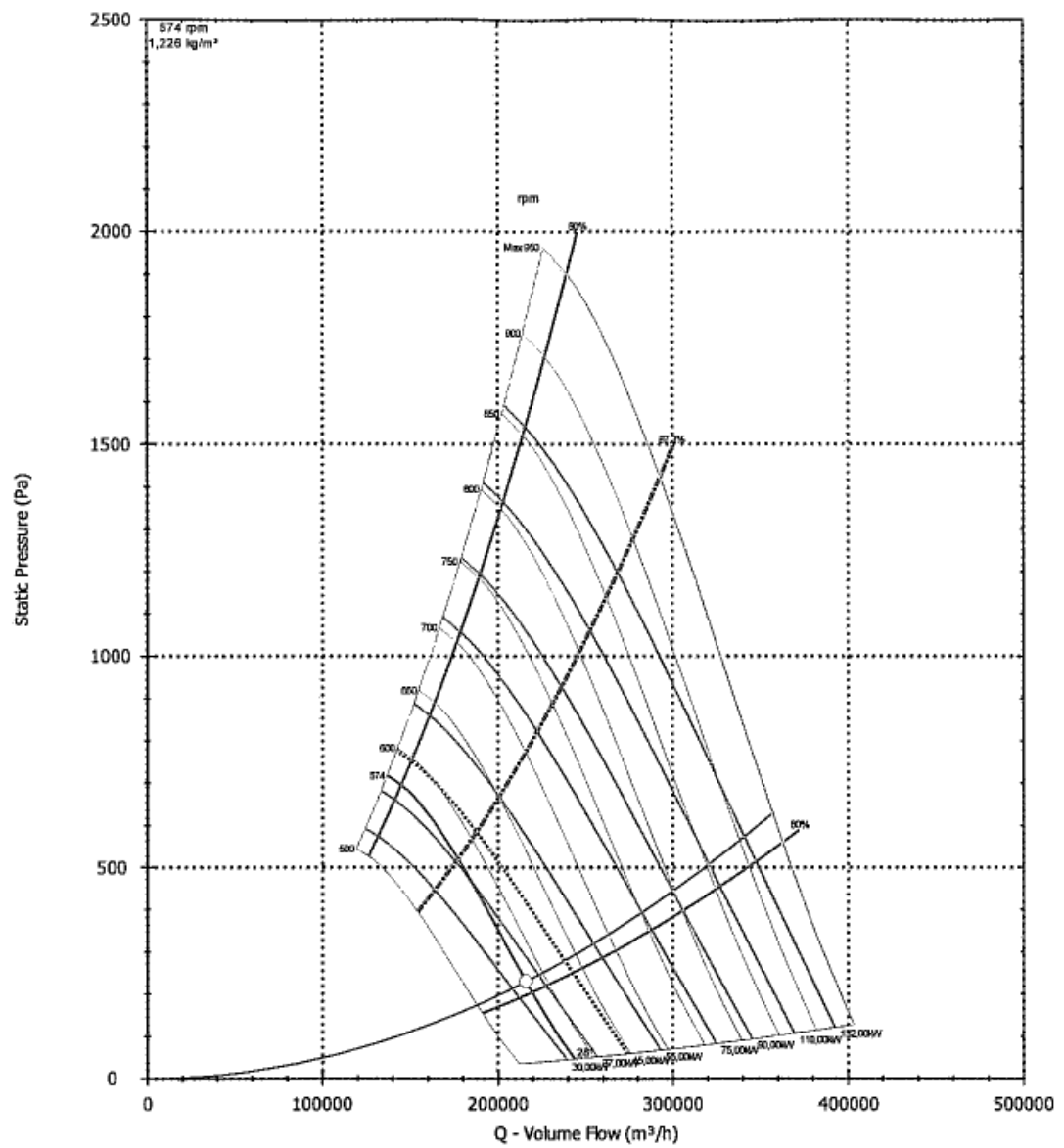


Figure F.3: Fan performance curve from Rucon EFC 2004/H(G) axial fan, with external motor and changeable frequency

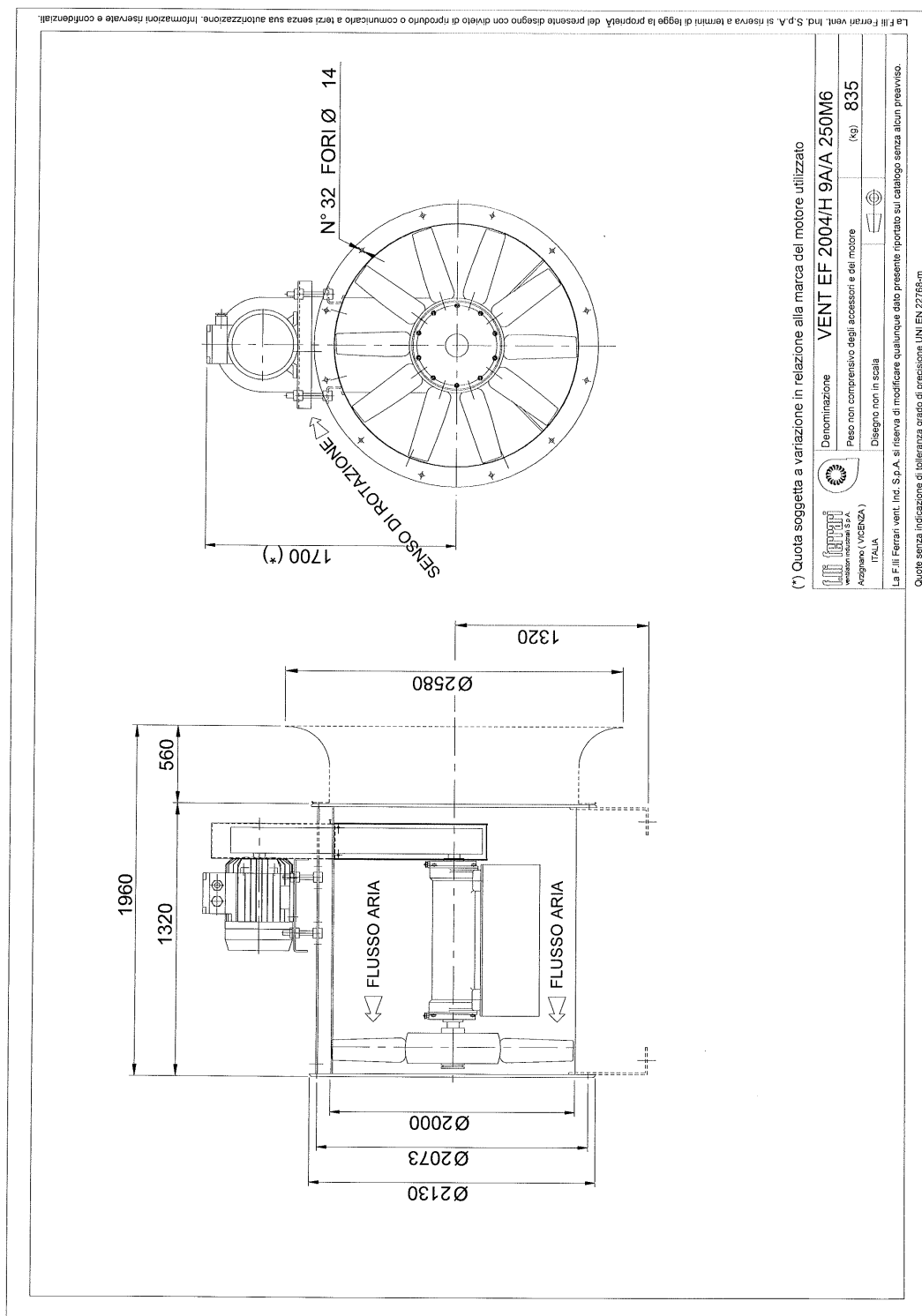


Figure F.4: Drawing from Rucon EFC 2004/H(G) axial fan, with external motor and changeable frequency



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Project -

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Item -
Reference -
Quantity -

Direct Driven Axial Fan

Temperature 15°C
Altitude 0 m
Density 1,226 kg/m³
Capacity 216.000 m³/h
Pressure 230 Pa Static Suction

Fan model EF 1806/H (G)
Capacity 219.794 m³/h (102%)
Static pressure 238 Pa @ 1,226 kg/m³
Dynamic pressure 353 Pa @ 1,226 kg/m³
Total pressure 591 Pa @ 1,226 kg/m³
Total pressure 591 Pa @ 1,226 kg/m³
Velocity 23,99 m/s
Impeller speed 960 rpm
Pitch angle 18 °
Absorbed power 40,3 kW @ 1,226 kg/m³
Absorbed power 40,3 kW @ 1,226 kg/m³
Service Factor 12 %
Power installed 45,0 kW (D280S 6 pole)
Total Efficiency 89 %
Static Efficiency 36 %
Arrangement 4
Class -
Maximum speed -
Moment of inertia 19,3 (J) kgm²
Weight -
Axial thrust 2402 N
Discharge thrust -
Starting time 1,3 s

Sound power level 122 dB Ambient (Density=1,226 kg/m³)
Sound pressure level 105 dB Ambient (Density=1,226 kg/m³ - Distance=1m)
Sound pressure level 102 dBA Ambient (Density=1,226 kg/m³ - Distance=1m)
Frequency 63 125 250 500 1k 2k 4k 8k Hz
Sound power in-duct - - - - - - - - dB inlet
Sound power in-duct 113 116 120 119 114 117 105 96 dB outlet

Figure F.5: Information from Rucon EFC 1806/H(G) axial fan, with internal motor and changeable frequency

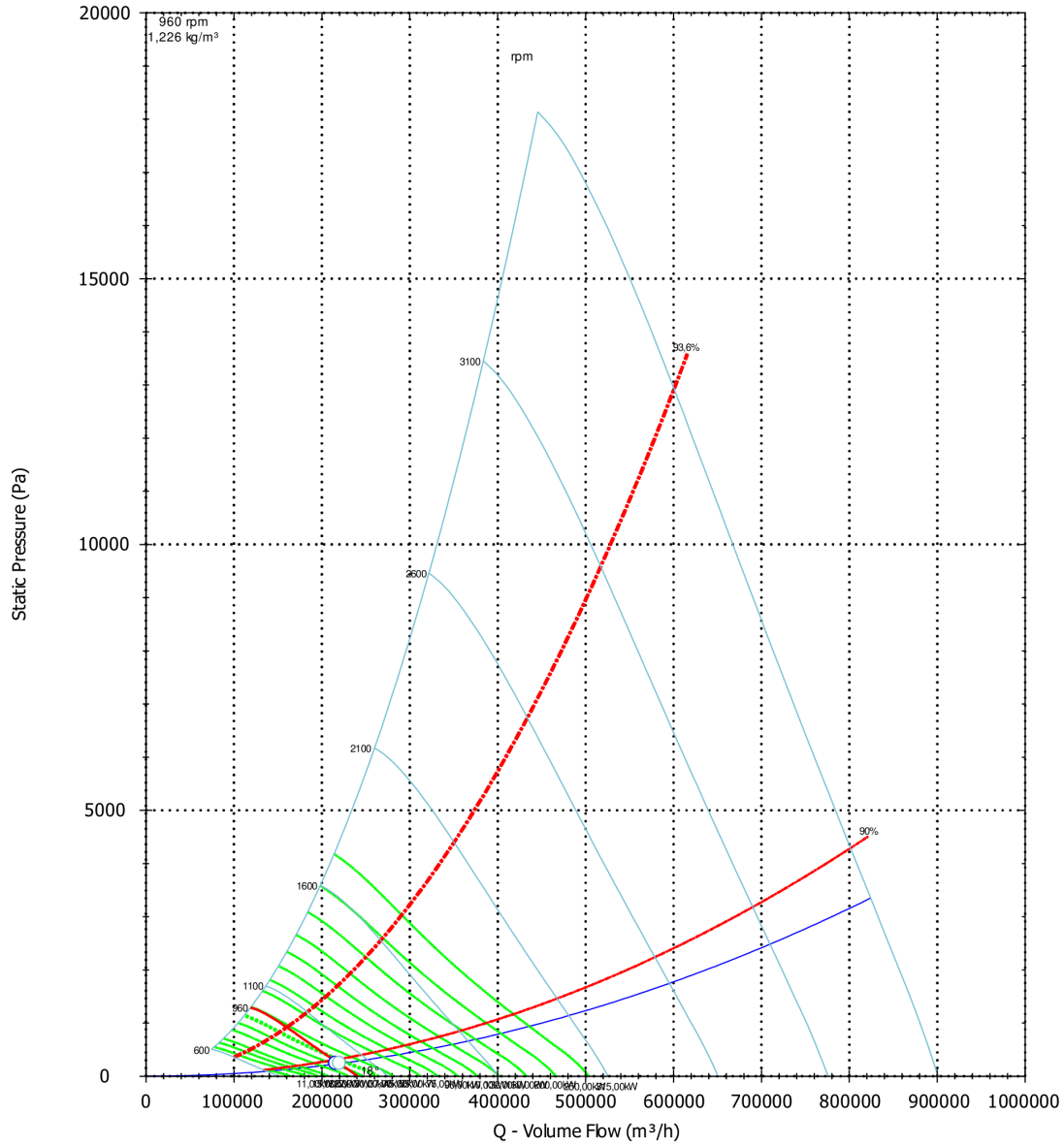


Figure F.6: Fan performance curve from Rucon EFC 1806/H(G) axial fan, with internal motor and changeable frequency

Fan type : A-M8L5/V1.4/2000/GD/10

Total press. pt, static press. pst, dynamic press. pd2 [Pa]

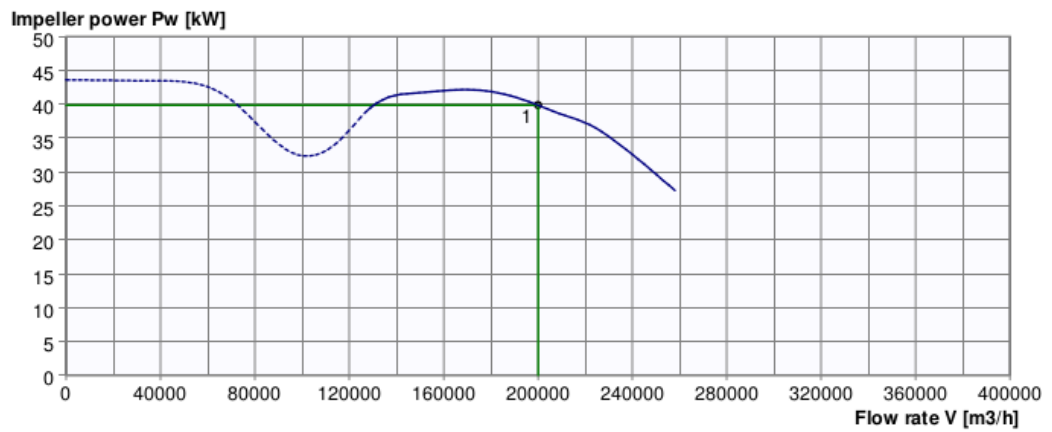
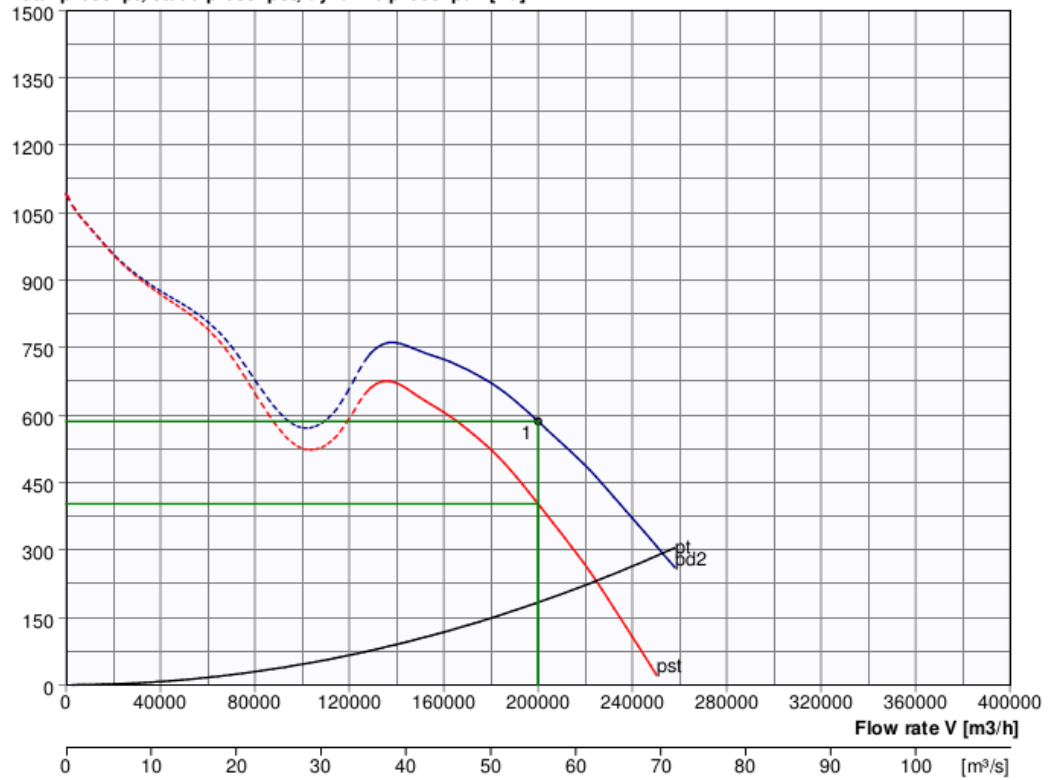


Figure F.7: Fan performance curve from Witt & Sohs 41kW fan



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